Nonintegral Laplacian spectra of ideal intersection graphs

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Abstract

We prove that the ideal intersection graph of $\mathbb{Z}_{p^a q^b r^c}$ is Laplacian nonintegral for any three distinct primes and every positive exponent triple. Adjacency means nonzero intersection of distinct nonzero proper ideals. For squarefree composite integers, the integral case is exactly two prime factors.

A six-support complement quotient reduces the three-prime problem to a real symmetric matrix. Its uniform positive root below three forces an integer spectrum onto one of two endpoint surfaces. On endpoint two, a written large-minimum argument supplies another root in $(4, 5)$; the remaining geometric domain has a complete 8 658-triple certificate. On endpoint one, an exponent-sum bound and written small-gap brackets reduce the final obstruction to the largest quotient root. For real $2 \le a \le b \le c$, $b - a \ge 3$ and $c \ge a^2 - a$, we prove

$$bc + a + b + c < \lambda_{\max} < bc + a + b + c + 1.$$

Positive polynomial identities, a slab lower bound and a Schur complement establish this unit interval throughout its unbounded domain. It closes the remaining integer spectra without enumerating the endpoint-one exponent points or solving the residual quartic's integer-root equation.

The global classification combines these written arguments with the established repeated-exponent and small-minimum results, including a complete 27 562-triple minimum-through-seven certificate. We retain further spectral gaps, arithmetic obstructions and local equality-curve splitting results. Exact verification uses integer and rational arithmetic; no Lean formalization is claimed. For any number of prime factors at least three, a unit exponent gives a noninteger eigenvalue in $(V - 1, V)$ by an explicit complement Rayleigh quotient below one, without a finite base. For a distinguished exponent $m \ge 2$, the same conclusion holds whenever $A \ge (m^2 - 1)B - (m - 1)$, including equality, where A is the product of the other exponents and $B = \prod_{i<t}(k_i + 1) - 1 - A$. In particular, every other exponent at least $m^2(t - 1)$ suffices. For four prime factors, a four-dimensional antisymmetric subspace also settles every positive exponent vector with two entries equal to two. Its smallest restricted eigenvalue lies in $(1, 3)$; the only possible integer endpoint pair has an irreducible residual cubic modulo 17. The analogous family with two exponents equal to three is nonintegral as well: tensor positivity and a complete endpoint reduction exclude the sole possible integer without an exceptional exponent pair. The nonsquarefree higher-prime classification outside these families remains open.

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1 Introduction

The ideal intersection graph associates a graph to the intersection structure of the nonzero proper ideals of a ring [8]. For $\mathbb{Z}_n$, its isomorphism type depends on the exponents in the prime factorization

of n. We ask which of these graphs have integer Laplacian eigenvalues.

Graphs defined by ring multiplication provide related constructions: Beck's coloring graph and the zero-divisor graph [3, 2], and the annihilating-ideal graph [4, 5, 1]. Their multiplication-based adjacency differs from the nonzero-intersection convention used here.

The Laplacian viewpoint originates in algebraic connectivity [10] and its spectral development [13, 15]; general graph-spectral background is given in [9, 6]. Equitable partitions and generalized joins [12, 7] reduce the present graphs to small invariant matrices.

Nikandish [16] develops a generalized-join formula for the Laplacian spectrum and a Schur–Weyl reduction for equal prime exponents. His results include nonintegrality for $(pqr)^k$ for every $k \geq 1$. His Question (Q3) asks for the full integrality characterization; the present arguments complete its three-prime case: every positive exponent triple is nonintegral (Theorem 17.3). Theorem 9.2 also settles every positive exponent vector with a unit entry and at least three prime factors. An explicit zero-sum function on the connected complement has Rayleigh quotient below one; reflection and the universal-vertex shift give a graph eigenvalue in $(V-1, V)$, without a finite base. Theorem 9.3 extends this obstruction to a distinguished exponent $m \geq 2$ under the exact support-weight condition $A \geq (m^2-1)B - (m-1)$, including equality. A rational trial function factors the energy-minus-norm sign; its boundary value is not a full eigenvector, giving a strict bound there as well. Corollary 9.4 covers every vector whose other exponents are at least $m^2(t-1)$. Theorem 9.5 settles the four-prime vectors $(2, 2, b, c)$ by an antisymmetric invariant subspace. Positive principal minors place its smallest eigenvalue between one and three; five symbolic endpoint polynomials leave only $\{b, c\} = \{5, 8\}$, whose residual cubic is irreducible modulo 17. This includes the balanced vector $(2, 2, 2, 2)$ outside the preceding sufficient condition. Theorem 9.6 also settles every $(3, 3, b, c)$, using tensor normalization for a general repeated-pair lower bound. A second positive tail, eight nonsquare discriminants and two rational factorizations exclude the integer endpoint without an exceptional pair. The remaining higher-prime case has every exponent at least two and fails the support-weight criterion for each distinguished coordinate; for four primes it also has at most one exponent equal to two and at most one equal to three.

Two further complete residue classes are excluded by Proposition 14.6: every permutation of $(1, 1, 2)$ or $(1, 4, 4)$ modulo five. An irreducible quadratic or cubic in the reduced complement quintic supplies the obstruction. On the equality curve, the cubic case has square discriminant modulo five, showing why the root equation gives information beyond discriminant congruences. Proposition 14.7 resolves a different degeneracy on that curve: the reduction $x(x+1)^2$ has zero discriminant, but its normalized discriminant is 13 times a square modulo every odd prime dividing b on the branch $ac \equiv -1$. This excludes integer spectra when 13 is a nonresidue, in particular whenever 7 or 11 divides b, and reduces the six compatible modulo-five pairs to four. Proposition 14.8 completes the corresponding local splitting test, except the negative branch at 13. In particular, the two remaining $b \equiv 0 \pmod 5$ rows split modulo every power of five, by simple-root Hensel lifting. The global largest-root obstruction proved below excludes integral spectra even in those locally split rows. At the exceptional prime 13, Proposition 14.9 uses the next discriminant term to exclude the negative branch unless $b \equiv 104 \pmod{169}$. Its further local analysis is independent of the completed global three-prime nonintegrality theorem.

We prove a complete classification for squarefree composite integers (Theorem 4.1), an explicit quadratic obstruction for $(1, 1, k)$ (Theorem 5.1), and nonintegrality for every repeated-exponent triple (a, a, b), $a, b \geq 1$ (Theorem 8.1). The last result combines an arithmetic restriction on the antisymmetric block with a consecutive-integer sign interval for the symmetric cubic. The boundary family $b = (a-1)(2a-1)$ is proved in Theorem 6.1. An exact divisor criterion isolates every possible square discriminant for fixed a. A second divisor criterion fixes the smaller-factor index instead, giving a finite reduction over all a (Proposition 7.6). The first three nonboundary factor levels have

complete finite modular certificates. Every later level lies strictly below the uniform sign threshold, closing the whole repeated-exponent family.

For a unit exponent, the complement of the general six-support quotient has a nonzero eigenvalue strictly between 0 and 1, except for three small pairs already covered by $(1, 1, k)$. This proves nonintegrality for every $(1, b, c)$ (Theorem 9.1), including fully unequal triples. Positive-coefficient expansions at a minimum exponent of two, together with one exceptional rational sign interval, then settle every $(2, b, c)$ (Theorem 10.1). Thus every three-prime exponent vector with minimum entry at most two is covered. The same endpoint polynomial gives a uniform cutoff $c \geq 4a^2 - 2a$ for ordered $2 \leq a \leq b \leq c$ (Proposition 11.1). For a minimum exponent of three, this leaves exactly 325 pairs. An exhaustive integer sign certificate, positive second-endpoint expansions and one rational exception settle all of them (Theorem 11.2). Thus every triple with minimum entry at most three is nonintegral.

For nearby unequal exponents, low roots can occur in pairs without an endpoint sign change. A symmetric Schur complement gives an inertia criterion for two roots in a specified interval (Proposition 12.1). At the upper endpoint three, four bounded-gap families satisfy the criterion for every $a \geq 4$. Positive first-endpoint factors and root-free second-endpoint factors exclude both possible integers. This proves Theorem 12.2, including the full family $(a, a + 1, a + 2)$ and every triple whose maximum minus minimum is at most three.

The requested diagnostic for minimum entries four through seven exhausts the finite region left by the uniform cutoff. All 27 562 quintics have positive nonsquare discriminants, verified by independent integer Sylvester determinants. Together with the written tail and repeated-exponent proofs, this yields Corollary 12.4: every triple with minimum entry at most seven is nonintegral. The dependence on the complete finite certificate is explicit.

The Schur complement also admits a uniform comparison that covers mixed diagonal signs. For every positive real parameter triple, the complement quotient has a positive eigenvalue below three (Theorem 13.2). If $(a - 2)(a + b + c - 2) > 2bc$, that eigenvalue lies in $(2, 3)$ (Corollary 13.3). This settles the growing balanced region $a > \rho + 4 + \sqrt{3}(\rho + 2)$, where ρ is the exponent span (Corollary 13.4). The universal bound also reduces any possible integral triple to the endpoint-zero condition $h_C(1)h_C(2) = 0$.

An integer-scaling argument then settles every triple with common exponent divisor at least three (Theorem 14.1). Finite-field factorization settles every all-odd triple and explicit common-residue classes, including residues $1, 3, 4$ modulo five (Theorem 14.2). These arithmetic arguments allow unbounded exponent differences and do not require excluding all integer quotient roots. Scaling also settles every common 2-adic valuation (Corollary 14.4). Theorem 14.5 collects the mixed-parity exclusions and necessary conditions: six classes modulo eight, the full $(3, 3, 2)$ and $(3, 0, 0)$ classes modulo four, and the sum, derivative and binary-cubic restrictions for the $(1, 3, 2)$ role. Its low-root consequences confine $(1, 3, 2)$ and $(3, 3, 0)$ to the endpoint-two surface inside a linear region. Theorem 15.1 combines that linear bound, the endpoint divisor constraints, the complete middle-exponent-through-fifteen result and two further modulo-three exclusions. One worked shifted-root example is given in the main text; all case calculations and endpoint details are grouped in Appendices B and C.

Theorem 15.8 then completes the endpoint-two alternative. Positive identities give $h_C(4) > 0 > h_C(5)$ on this surface at minimum at least forty. The geometric bounds $b < 2a - 2$ and $c < 9a/4 - 8$ leave exactly 8 658 fully distinct integer triples at minima eight through thirty-nine; every endpoint value is independently checked by integer Horner and Bareiss determinant evaluation. The middle bound holds already for real $a > (7 + \sqrt{13})/3$, with its unique boundary equality given in the rectangle lemma. Together with the earlier repeated and small-minimum results, this proves nonintegrality for every all-even triple and the full $(1, 3, 2)$ and $(3, 3, 0)$ classes modulo four

(Corollary 15.9). The complete positive identities and finite-base counts are in Appendix D. The final low-root alternative is endpoint one, completed below.

Theorem 15.2 proves the global gap $\lambda_3 < a+b < a+b+c \le \lambda_4$ for real $2 \le a \le b \le c$, using principal interlacing and a determinant-sign count. On endpoint one it also proves a sharp exponent-sum lower bound for $a \ge 2$. For $a > 2+\sqrt{3}$ it places the smallest residual root in $(a, \min\{c, a+b\})$, with the other three above $s = a+b+c$. Equality in the global lower bound occurs only at $(2,2,2)$. The resulting constant-term divisor window in Corollary 15.3 focuses the remaining integer-splitting question. Its proof uses a quadratic with three positive coefficients and a determinant-sign count, with no finite base. Proposition 15.6 further identifies b as the simple third quotient root on $c = b(b+2) - a$ for all real $2 \le a \le b$, reducing the equality splitting problem to a cubic whose roots all exceed s. Corollary 15.3 also sharpens the smallest-root window using the exact coefficients and a convexity bound for the three larger roots' reciprocal sum.

The graph and lifting arguments are given explicitly, so the proofs do not require a complete spectral decomposition. Appendix A describes the exact computations supporting the displayed formulas and finite certificates. Theorem 17.1 finally places the largest quotient root in $(bc+s, bc+s+1)$ for real $a \ge 2$ when $b - a \ge 3$ and $c \ge a^2 - a$. On endpoint one, the exponent-sum bound supplies the latter inequality; written unit brackets handle gaps one and two for every real $a \ge 2$. This proves the complete three-prime nonintegrality theorem. The new real interval needs no finite base; the combined classification retains the earlier 27 562- and 8 658-triple certificates. The remaining higher-prime classification concerns vectors with every exponent at least two.

2 The graph and its invariant subspaces

Write $n = \prod_{i=1}^{t} p_i^{k_i}$, where the primes are distinct and $k_i \ge 1$. Denote by G_n the graph whose vertices are the nonzero proper ideals of $\mathbb{Z}_n$, with distinct ideals adjacent if their intersection is nonzero. We use the combinatorial Laplacian $L = D - A$; the graph is *Laplacian integral* when every eigenvalue of L is an integer, including when the graph is disconnected.

The ideal $(\prod_i p_i^{\alpha_i})$ is encoded by $0 \le \alpha_i \le k_i$. We exclude $(0, \dots, 0)$, which encodes the whole ring, and $(k_1, \dots, k_t)$, which encodes the zero ideal. Ideal intersection is generated by the least common multiple. Therefore two distinct vertices α, β are adjacent exactly when

$$\max(\alpha_i, \beta_i) < k_i \quad \text{for some } i.$$

Equivalently, putting $S(\alpha) = \{i : \alpha_i < k_i\}$, adjacency is $S(\alpha) \cap S(\beta) \ne \varnothing$.

For nonempty $S \subseteq [t]$, the support class $\mathcal{C}_S$ is a clique of size $\prod_{i \in S} k_i$, except that the full support has size $u = \prod_i k_i - 1$. Edges between distinct support classes are complete or absent according as their supports intersect or are disjoint. The full support consists of universal vertices. Remove them and call the resulting graph H; thus $G_n = K_u \vee H$, with the evident interpretation for $u = 0$.

Lemma 2.1 (Lifting across the universal vertices)**.** *If $L(H)v = \lambda v$ and $\lambda \ne 0$, extension of v by zero on the u universal vertices is an eigenvector of $L(G_n)$ with eigenvalue $\lambda + u$.*

Proof. Since $L(H)$ is symmetric with zero row sums, $\lambda \ne 0$ implies $\sum v = 0$. At a vertex of H, its degree increases by u and all new neighbor values are zero. At a universal vertex the Laplacian action is $-\sum v = 0$. These are exactly the claimed eigenvector equations. □

For $(k_1, k_2, k_3) = (a, a, b)$, H has six support classes of sizes a, a, b, a^2, ab, ab on $1, 2, 3, 12, 13, 23$, respectively. The partition into

$$X = \mathcal{C}_1 \cup \mathcal{C}_2, \quad Y = \mathcal{C}_3, \quad Z = \mathcal{C}_{12}, \quad W = \mathcal{C}_{13} \cup \mathcal{C}_{23}$$

is equitable, and on functions constant on these cells the Laplacian acts by

$$B = \begin{pmatrix} a^2+ab & 0 & -a^2 & -ab \\ 0 & 2ab & 0 & -2ab \\ -2a & 0 & 2a+2ab & -2ab \\ -a & -b & -a^2 & a^2+a+b \end{pmatrix}. \tag{1}$$

For example, an X vertex has a^2 neighbors in Z and ab in W, while a W vertex has a, b, a^2 neighbors in X, Y, Z. Internal neighbors cancel for cell-constant functions. The cell sizes are $2a, b, a^2, 2ab$, and multiplying B on the left by their diagonal matrix gives a symmetric matrix. Thus B is similar to a real symmetric matrix.

The functions taking values $s, -s$ on $\mathcal{C}_1, \mathcal{C}_2$, $t, -t$ on $\mathcal{C}_{13}, \mathcal{C}_{23}$, and zero elsewhere form a second invariant subspace, with matrix

$$B_- = \begin{pmatrix} a^2+ab & -ab \\ -a & a^2+a+b+2ab \end{pmatrix}. \tag{2}$$

These functions have sum zero. Their eigenvalues therefore also lift by $u = a^2b - 1$, directly by the proof of Lemma 2.1.

Expanding (1) gives $\det(xI - B) = x f_{a,b}(x)$, where

$$\begin{aligned} f_{a,b}(x) &= x^3 - E_1x^2 + E_2x - E_3, \\ E_1 &= 2a^2 + 5ab + 3a + b, \\ E_2 &= a^4 + 7a^3b + 3a^3 + 8a^2b^2 + 11a^2b + 2a^2 + 3ab^2 + 2ab, \\ E_3 &= 2a^2b(a+b+1)(a^2+2ab+2a+b). \end{aligned} \tag{3}$$

Its roots are real and nonzero: reality follows from the similarity above, and $E_3 > 0$. Every root λ gives an eigenvalue $\lambda + u$ of G_n. A noninteger root of $f_{a,b}$ therefore proves nonintegrality.

3 Structure of the proof

The complement quotient explains the endpoint alternatives in the three-prime classification. Its low-root bound localizes a possible integer spectrum to two endpoint surfaces, whose geometric bounds lead to different spectral obstructions. We summarize this mechanism before giving the component proofs.

3.1 Localization and the sharp threshold

Let C be the complement Laplacian quotient on the six proper nonempty supports, of weights a, b, c, ab, ac, bc, and put $h_C(x) = \det(xI - C)/x$. The off-diagonal entry from S to T is $-w_T$ when $S \cap T$ is empty, and zero otherwise; the diagonal is the sum of these neighbor weights. The weighted symmetric quotient is positive semidefinite with a simple zero eigenvalue, since the disjointness graph on these supports is connected.

For positive integer exponents, every nonzero complement quotient root must be an integer if G_n is Laplacian integral: reflection across the complement and the universal-vertex lift transfer every noninteger root to a noninteger graph eigenvalue. Theorem 13.2 gives a positive quotient root below three for every ordered positive real triple. Hence the smallest positive root also lies in $(0, 3)$, and an integral graph spectrum requires

$$h_C(1) = 0 \qquad \text{or} \qquad h_C(2) = 0.$$

These are necessary conditions; the following arguments exclude integer spectra on each surface by a further root or an exponent-root bracket.

Proposition 3.1. *The uniform threshold three is optimal even for integer exponents. For $a = b = c = m \geq 1$, the smallest positive quotient root satisfies*

$$\mu_- = \frac{m^2 + 4m - m\sqrt{(m+4)^2 - 12}}{2} \in \left(\frac{3m}{m+4}, 3\right), \qquad \lim_{m\to\infty} \mu_- = 3.$$

Proof. Direct substitution in the quotient gives

$$h_{m,m,m}(x) = (x - m^2 - m)\big[x^2 - (m^2 + 4m)x + 3m^2\big]^2.$$

For the quadratic q_m, we have $q_m(0) > 0$ and $q_m(m) = -m^3 < 0$. Its smaller root therefore lies in $(0, m)$ and is below the other three positive roots, counted with multiplicity. Also $q_m(3) = 9 - 12m < 0$, so $\mu_- < 3$. Rationalization gives

$$\mu_- = \frac{6m}{m + 4 + \sqrt{(m+4)^2 - 12}} > \frac{3m}{m+4}.$$

Dividing the denominator by m proves the limit, including along positive integers. Thus no smaller constant is a uniform upper bound. □

3.2 The endpoint surfaces and their scales

After repeated exponents and minima at most seven have been settled, the remaining classification domain is $8 \leq a < b < c$. In this domain, Lemma 15.7 shows that

$$h_C(2) = 0 \quad \Longrightarrow \quad b < 2a - 2, \qquad c < \frac{9a}{4} - 8.$$

Both larger exponents are bounded multiples of the minimum. The rectangle is finite at each fixed integer a, but a is unbounded. Theorem 15.8 supplies a root in $(4, 5)$ for $a \geq 40$; the complete 8 658-triple certificate handles $8 \leq a \leq 39$. The real tail argument and the finite base together close this alternative.

Endpoint one has a different necessary scale. For real $2 \leq a \leq b \leq c$, Theorem 15.2 gives

$$h_C(1) = 0 \quad \Longrightarrow \quad b + c \geq 2a^2 - 2a + 1, \qquad c \geq a^2 - a + \tfrac{1}{2}.$$

The second inequality follows from $b \leq c$. This quadratic lower bound supplies the hypothesis of the largest-root interval in Theorem 17.1 whenever $b - a \geq 3$. For gaps one and two, Lemma 17.2 instead places the only possible exponent-root above b between consecutive integers. This route settles the spectrum without enumerating integer points on the endpoint-one surface.

3.3 The fixed-gap endpoint-one expansion

The small-gap anchors also have an asymptotic interpretation. The following result concerns the exponent-root c, with the gap fixed; it is supplementary to the exact brackets used in the classification.

Proposition 3.2. *Fix a real $d \geq 1$ and put $b = a + d$. For all sufficiently large real a, the equation $h_C(1) = 0$, regarded as a cubic in c, has exactly one root c^* above b. As $a \to \infty$ with d fixed,*

$$c^* = 2a^2 + (2d-3)a - \frac{(d+2)(2d-1)}{2} + \frac{d^2}{2a} + \frac{d^2(d+1)(2d-3)}{4a^2} + O_d(a^{-3}). \tag{4}$$

The remainder constant and the required minimum may depend on d.

Proof. Write $Q_{a,d}(c) = h_C(1)$ at $b = a + d$. The cubic coefficients are displayed in Lemma 17.2; their identity holds for arbitrary d. Set $t = 1/a$, $y = t^2 c$ and

$$F_d(t, y) = t^8 Q_{1/t,d}(y/t^2).$$

This is a polynomial in t, y, with

$$F_d(0, y) = 2y^2(y-2), \qquad \partial_y F_d(0, 2) = 8.$$

The analytic implicit-function theorem gives a unique real analytic branch $y_d(t)$ near $t = 0$ with $y_d(0) = 2$. Substitution and comparison of successive powers of t give

$$y_d(t) = 2 + (2d-3)t - \frac{(d+2)(2d-1)}{2}t^2 + \frac{d^2}{2}t^3 + \frac{d^2(d+1)(2d-3)}{4}t^4 + O_d(t^5).$$

Multiplication by t^{-2} proves (4). For fixed d and sufficiently large a, the cubic has positive leading coefficient and $Q_{a,d}(0) > 0$, whereas

$$Q_{a,d}(a + d) = -4a^6 + O_d(a^5) < 0.$$

It therefore has a root in each of $(-\infty, 0)$ and $(0, b)$. The analytic branch has $c^* \sim 2a^2 > b$ and supplies the third root. These distinct roots exhaust the cubic degree, proving uniqueness above b. □

For $d = 1, 2$, the polynomial parts of this expansion are respectively $2a^2 - a - 2$ and $2a^2 + a - 6$, followed by a positive term of order a^{-1}. The exact unit brackets in Lemma 17.2 hold for all real $a \geq 2$. The expansion here is for fixed d; it supplies no uniform error bound when d grows with a. The largest-root argument handles every $d \geq 3$ in the classification using the quadratic lower bound on c.

4 Squarefree integers

For squarefree integers the support classes have size one, so the graph is described directly by subsets of the prime factors. Alternating tensors provide invariant subspaces for this subset action [11] and give the complete squarefree classification.

Theorem 4.1. *For squarefree composite n, the graph G_n is Laplacian integral if and only if n has exactly two prime factors.*

Proof. Let t be the number of prime factors. For $t = 2$ the graph consists of the two ideals (p) and (q), with no edge, so $L = 0$. Assume $t \geq 3$. The vertices are the nonempty proper subsets of $[t]$, adjacent when they intersect. On the tensor space encoding all subsets put

$$E = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad c = 2^t - 1.$$

The principal submatrix of $\Phi = cI - E^{\otimes t} + F^{\otimes t} - J^{\otimes t}$ on the graph vertices is $L(G_n)$: the diagonal entry at S is $2^t - 2 - 2^{t-|S|}$, and the off-diagonal entry is -1 for intersecting subsets and zero otherwise.

With e_0, e_1 denoting the absence and presence basis, set $w = e_0 \otimes e_1 - e_1 \otimes e_0$. For every 2×2 matrix T, one has $T^{\otimes 2} w = (\det T) w$. A paired factor w thus contributes $2, -1, 0$ under E, F, J, respectively. It vanishes at the empty and full subsets. Each subspace used below is invariant for Φ and is supported on graph vertices, so its restriction is also a restriction of L.

If $t = 2r + 1$, $r \geq 1$, use $w^{\otimes r} \otimes \operatorname{span}\{e_0, e_1\}$. Putting $a = 2^r$ and $s = (-1)^r$, the restricted matrix is

$$\begin{pmatrix} c - 2a + s & s \\ s & c - a \end{pmatrix},$$

with discriminant $(a - s)^2 + 4$. The integer $h = a - s$ is at least 3, so $h^2 < h^2 + 4 < (h + 1)^2$. Both eigenvalues are irrational.

If $t = 2r$, $r \geq 2$, use $r - 1$ paired factors and the symmetric subspace on the last two coordinates, with basis $e_0e_0, (e_0e_1 + e_1e_0)/\sqrt{2}, e_1e_1$. Put $a = 2^{r-1}$ and $s = (-1)^{r-1}$. The restriction is $cI - K$, where

$$K = \begin{pmatrix} 4a - s & -s\sqrt{2} & -s \\ -s\sqrt{2} & 2a - s & 0 \\ -s & 0 & a \end{pmatrix}.$$

The leading principal minors of $K - (a - 1)I$ are

$$3a + 1 - s, \quad (3a + 1 - s)(a + 1 - s) - 2, \quad (3a - s)(a + 1 - s) - 2.$$

They are positive for $a \geq 2$ and $s = \pm 1$: the latter two are at least 10 and 8, respectively. Sylvester's criterion gives $\lambda_{\min}(K) > a - 1$. The principal submatrix of $K - aI$ on coordinates $1, 3$ is $\left(\begin{smallmatrix} 3a - s & -s \\ -s & 0 \end{smallmatrix}\right)$, with determinant -1. A vector supported on those coordinates has a negative Rayleigh quotient, giving $\lambda_{\min}(K) < a$. Hence $c - \lambda_{\min}(K)$ is an eigenvalue of L strictly between the consecutive integers $c - a$ and $c - a + 1$. □

5 The family pqr^k

With two unit exponents, exchanging their coordinates isolates a two-dimensional invariant subspace. Its discriminant gives explicit irrational eigenvalues for every positive third exponent.

Theorem 5.1. *For distinct primes p, q, r and every $k \geq 1$, the graph G_{pqr^k} has the irrational eigenvalues*

$$\frac{6k + 1 \pm \sqrt{(2k + 2)^2 - 3}}{2}.$$

In particular, every permutation of the exponent vector $(1, 1, k)$ gives a nonintegral Laplacian spectrum.

Proof. The support classes $1, 2$ have size 1, and $13, 23$ have size k. Consider functions with values $s, -s$ on the first two classes, $t, -t$ on the latter two, and zero elsewhere. A vertex of support 1 has $2k$ neighbors and Laplacian value $2ks - kt$. A vertex of support 13 has degree $4k$; after subtracting the same-class and opposite-class values its Laplacian value is $-s + (4k+1)t$. The swapped classes give their negatives, and all other classes give zero by cancellation. The restricted matrix of the full graph is therefore

$$B_k = \begin{pmatrix} 2k & -k \\ -1 & 4k+1 \end{pmatrix}.$$

Its characteristic polynomial is $x^2 - (6k+1)x + 8k^2 + k$, with discriminant $(2k+2)^2 - 3$. Since

$$(2k+1)^2 < (2k+2)^2 - 3 < (2k+2)^2$$

for $k \geq 1$, the two roots are irrational. Relabeling prime coordinates preserves the graph. □

6 A boundary family with integral antisymmetric block

The following argument was supplied by Reza Nikandish in the public collaboration discussion. We include it because it is the boundary case needed for the linear cutoff in Section 7.

Theorem 6.1. *For every $a \geq 2$, put $C = (a-1)(2a-1)$. For distinct primes p, q, r, the graph $G_{p^a q^a r^C}$ is not Laplacian integral, although the antisymmetric block* (2) *has integer eigenvalues.*

Proof. Specialize $f_{a,b}$ from (3) to $b = C$. For $a = 2$ it is $x^3 - 47x^2 + 702x - 3312$. Its values at $0, 1, 2, 3, 4$ modulo 5 are $3, 4, 2, 3, 3$. Thus it has no root in $\mathbb{F}_5$ and is irreducible over $\mathbb{Q}$; its real roots are noninteger. Lemma 2.1 gives noninteger graph eigenvalues.

For $a \geq 3$ put $m = 2a^3 - 2a^2 + 2a - 1$. Expansion gives

$$f_{a,C}(m) = 2(5a^4 - 10a^3 + 7a^2 - 1),$$
$$f_{a,C}(m-1) = -2(a-1)(a+2)(2a^4 - 7a^3 + 7a^2 - a - 3).$$

The first value is positive because its parenthesis is $5a^3(a-2)+7a^2-1$. Writing $z = a-3 \geq 0$, the last factor in the second value is $2z^4+17z^3+52z^2+68z+30 > 0$. Consequently $f_{a,C}(m-1) < 0 < f_{a,C}(m)$, so a real root lies strictly between $m-1$ and m. Its integer shift by $u = a^2C - 1$ is noninteger.

The characteristic discriminant of B_- is $D = (a+1)^2b^2 + 2a(3a+1)b + a^2$. At $b = C$ it equals $(2a^3 - a^2 + a - 1)^2$, and the two eigenvalues are $4a^3 - 3a^2 + a$ and $2a^3 - 2a^2 + 1$. Thus the obstruction in this family comes from the symmetric cubic. □

7 Finite reduction and a linear cutoff

The repeated-exponent decomposition leaves two possible obstructions: a nonsquare quadratic discriminant and a noninteger root of the symmetric cubic. Exact factor pairs reduce the first exceptional set, while sign intervals settle the large-exponent tail.

Fix $a, b \geq 1$, and put

$$h = a + 1, \quad C = (a-1)(2a-1), \quad M = a^3(2a+1), \quad A = h^2b + a(3a+1).$$

The characteristic discriminant of (2) is

$$D = h^2b^2 + 2a(3a+1)b + a^2. \tag{5}$$

If D is not a square, this block has irrational eigenvalues, which remain irrational after the integer shift $a^2b - 1$.

Proposition 7.1 (Exact divisor reduction)**.** *For fixed $a \geq 1$, the positive integers b for which D is a square are exactly those obtained from positive factor pairs $d \leq e$ of M such that*

$$b = \frac{d + e - a(3a+1)}{h^2} \in \mathbb{Z}_{\geq 1}, \qquad s = \frac{e-d}{h} \in \mathbb{Z}_{\geq 0}. \tag{6}$$

In that case $D = s^2$.

Proof. The polynomial identity

$$A^2 - h^2 D = 4M \tag{7}$$

holds for arbitrary a, b. If $D = s^2$ with $s \geq 0$, then $A - hs$ and $A + hs$ are positive: their product is positive and their sum is $2A > 0$. They have the same parity; since their product is divisible by 4, both are even. Write $A - hs = 2d$ and $A + hs = 2e$. Then $de = M$, $d + e = A$ and $e - d = hs$, giving (6). Conversely those conditions yield $A = d + e$ and $hs = e - d$. Using $(d+e)^2 - (e-d)^2 = 4de = 4M$ in (7) gives $D = s^2$. □

The divisor set is finite for every fixed a; this criterion has no truncation in b. It gives first a quadratic bound.

Corollary 7.2. *If D is a square then $b \leq C$. Consequently every $b > C$ gives a nonintegral Laplacian spectrum. For $a = 1$, every $b \geq 1$ is covered.*

Proof. Since $(d-1)(e-1) \geq 0$, $A = d + e \leq M + 1$. The identity

$$M + 1 - a(3a+1) = h^2 C$$

then gives $b \leq C$. The nonsquare case is covered by the antisymmetric block. At $a = 1$, $C = 0$. □

Lemma 7.3. *Every admissible factor pair in Proposition 7.1 satisfies $d \equiv e \equiv 1 \pmod{h}$.*

Proof. From $e - d = hs$ and $d + e = A$, reduction modulo h gives $e \equiv d$ and $2d \equiv 2$, since $a(3a+1) \equiv 2 \pmod{h}$. Also

$$(d-1)(e-1) = M - A + 1 = h^2(C - b). \tag{8}$$

If h is odd, the congruence $2(d-1) \equiv 0$ gives the assertion. If h is even, the alternative to $d \equiv 1$ is $d - 1 \equiv h/2 \pmod{h}$. Then $d - 1$ and $e - 1$ would both equal $h/2$ times an odd integer. Their product divided by h^2 would be an odd integer divided by 4, contradicting (8). This excludes the alternative for either parity of a. □

Theorem 7.4 (Unified linear cutoff)**.** *Let $a \geq 2$ and*

$$R(a) = \frac{2(a-1)(a-2)}{a+2} = 2a - 10 + \frac{24}{a+2}.$$

For every integer $b > R(a)$ and distinct primes p, q, r, the graph $G_{p^a q^a r^b}$ is not Laplacian integral. Every unresolved repeated-exponent pair therefore has square D and $1 \leq b \leq \lfloor R(a) \rfloor$.

Proof. If D is nonsquare the antisymmetric block suffices. Otherwise use Proposition 7.1. When $d = 1$, $e = M$ and $b = C$; Theorem 6.1 settles this case. If $d > 1$, Lemma 7.3 gives $d \geq T = a + 2$. Since $e \geq d \geq T$,

$$\frac{M}{T} + T - (d + e) = (d - T)\left(\frac{e}{T} - 1\right) \geq 0.$$

Hence

$$b \leq \frac{M/T + T - a(3a+1)}{h^2} = \frac{2(a-1)(a-2)}{a+2} = R(a).$$

Thus $b > R(a)$ leaves either a nonsquare discriminant or the already settled boundary case. □

At $(a, b) = (4, 2)$ the square-discriminant pair has $d = 6 = a + 2$, so equality in the factor estimate is attained. The graph is still nonintegral, as shown in the next section.

7.1 Settling the first nonboundary factor

The first possible value after $d = 1$ is $d = a + 2$. Since d divides M, the identity

$$M = (a + 2)(2a^3 - 3a^2 + 6a - 12) + 24$$

forces $a+2 \mid 24$. With $a \geq 2$ this leaves $a = 2, 4, 6, 10, 22$. Substitution into (6) gives $b = 0, 2, 5, 12, 35$, respectively. Excluding $b = 0$, the complete positive-b list and its cubic certificates are:

(a, b)	$f_{a,b}(x)$	Root-free prime
$(4, 2)$	$x^3 - 86x^2 + 2304x - 18816$	5
$(6, 5)$	$x^3 - 245x^2 + 19266x - 488160$	7
$(10, 12)$	$x^3 - 842x^2 + 230160x - 20534400$	13
$(22, 35)$	$x^3 - 4919x^2 + 7887858x - 4132479120$	17

Each polynomial has no root modulo its indicated prime. Its monic cubic is therefore irreducible over $\mathbb{Q}$, and its real nonzero roots lift to noninteger graph eigenvalues. Thus every possible $d = a+2$ pair is settled, over all a, by this finite divisor argument.

Theorem 7.5 (Second linear cutoff)**.** *For $a \geq 2$ put*

$$S(a) = \frac{(a - 3)(2a - 3)}{2a + 3} = a - 6 + \frac{27}{2a + 3}.$$

For every positive integer $b > S(a)$ and distinct primes p, q, r, $G_{p^a q^a r^b}$ is not Laplacian integral. Every remaining square-discriminant case has $d \geq 2a + 3$ and $b \leq \lfloor S(a) \rfloor$.

Proof. Nonsquare D is covered by the antisymmetric block. If D is square, the cases $d = 1$ and $d = a + 2$ are already settled. Every other admissible factor satisfies $d \geq T = 2a + 3$ by Lemma 7.3. Since $e \geq d \geq T$, the same factor inequality gives

$$b \leq \frac{M/T + T - a(3a + 1)}{(a + 1)^2} = \frac{(a - 3)(2a - 3)}{2a + 3} = S(a).$$

This proves the claim for $b > S(a)$. □

The new bound is strictly smaller than $R(a)$ for all $a \geq 2$, since

$$R(a) - S(a) = \frac{2a^3 - a^2 - a - 6}{(a + 2)(2a + 3)} > 0.$$

Writing $z = a - 2 \geq 0$, its numerator is $2z^3 + 11z^2 + 19z + 4$.

7.2 A finite reduction for each factor index

For a nonboundary square-discriminant pair, write $d = 1 + hj$, $j \geq 1$. The following criterion fixes j instead of a.

Proposition 7.6. *Fix $j \geq 1$ and set $K_j = (j+1)^3(j+2)$. Every admissible square-discriminant pair of index j is obtained from a positive divisor d of K_j with $a = (d-1)/j - 1 \in \mathbb{Z}_{\geq 2}$. Set $e = M/d$, $v = (e-1)/(a+1)$, and retain exactly those candidates with $d \leq e$ and $b = C - jv \geq 1$. Then e, v, b are integers and $D = (v-j)^2$. Conversely every retained candidate is admissible.*

Proof. Modulo $d = j(a+1)+1$ we have $ja \equiv -(j+1)$, whence

$$j^4 M = (ja)^3 j(2a+1) \equiv (j+1)^3(j+2) = K_j \pmod{d}.$$

Since $\gcd(d, j) = 1$, we have $d \mid M$ if and only if $d \mid K_j$. Thus e is integral. Since $M \equiv d \equiv 1 \pmod{h}$, also $e \equiv 1 \pmod{h}$, so v is integral. The identity

$$M = 1 + h(3a-2) + h^2 C = (1+hj)(1+hv)$$

gives $j + v + hjv = 3a - 2 + hC$. Defining $b = C - jv$, we obtain $d + e = h^2 b + a(3a+1)$ and $e - d = h(v-j)$. The ordering condition gives $v - j \geq 0$; Proposition 7.1 now proves admissibility and $D = (v-j)^2$. The reverse direction follows by applying these identities to any admissible index-j pair. □

This is a complete finite problem for each fixed j, without a truncation in a. Section 8 combines the first three such problems with a uniform argument for all later indices. For $j = 2$, $K_2 = 108$ and $d = 2a+3$ is an odd divisor of 108 with $d \geq 7$. The only candidates are $d = 9, 27$, giving $a = 3, 12$. The first has $e = 21, v = 5, b = 0$ and is excluded. The second has $e = 1600, v = 123, b = 7$ and $\sqrt{D} = 121$. Thus $(12, 7)$ is the only positive-b case with $j = 2$.

Its symmetric cubic is $f_{12,7}(x) = x^3 - 751x^2 + 180348x - 13829760$. Its values at all residues modulo 13 are $4, 7, 9, 3, 8, 4, 10, 6, 11, 5, 7, 10, 7$. As a monic cubic with no root modulo 13, it is irreducible over $\mathbb{Q}$. Its real nonzero roots lift to noninteger graph eigenvalues, settling the entire index-2 case.

Theorem 7.7 (Third linear cutoff)**.** *For every $a \geq 2$, put $T(a) = 2(a-4)(a-2)/(3a+4)$. If $b \geq 1$ and $b > T(a)$, then $G_{p^a q^{a+r^b}}$ is not Laplacian integral. Every unresolved square-discriminant pair has $j \geq 3$, $d \geq 3a+4$, and $b \leq \lfloor T(a) \rfloor$.*

Proof. Nonsquare D gives irrational antisymmetric eigenvalues. For square D, the cases $j = 0, 1, 2$ are all settled above. Every other pair has $d \geq Q = 3a+4$, and $e \geq d \geq Q$ gives

$$b \leq \frac{M/Q + Q - a(3a+1)}{(a+1)^2} = \frac{2(a-4)(a-2)}{3a+4} = T(a).$$

This proves the claimed implication. □

For $a \geq 3$ the new bound is strictly below $S(a)$, since

$$S(a) - T(a) = \frac{2a^3 - a^2 - 5a - 12}{(2a+3)(3a+4)} > 0.$$

With $z = a - 3 \geq 0$, the numerator is $2z^3 + 17z^2 + 43z + 18$. The exact expression $T(a) = 2a/3 - 44/9 + 320/(9(3a+4))$ shows its leading term.

8 Completion of the repeated-exponent family

The preceding factor-index reduction makes the repeated-exponent problem finite at its first few levels. A uniform cubic sign interval handles every later level, completing the family without a bound on either exponent.

Theorem 8.1. *For distinct primes p, q, r and all integers $a, b \geq 1$, the graph $G_{p^a q^a r^b}$ is not Laplacian integral. The same conclusion holds for every permutation of the exponent vector (a, a, b).*

The arithmetic reduction and a uniform sign obstruction for the cubic combine to exhaust the family. We first settle one last finite factor level.

Lemma 8.2. *Every square-discriminant pair with index $j = 3$ gives a nonintegral Laplacian spectrum.*

Proof. Proposition 7.6 gives $d \mid K_3 = 320$ and $d = 3a + 4$. For $a \geq 2$, the complete divisor list of this shape is $10, 16, 40, 64, 160$. The resulting values are

d	a	e	v	b	Outcome
10	2	4	1	0	$d > e$
16	4	36	7	0	$b = 0$
40	12	1080	83	4	Root-free modulo 5
64	20	5125	244	9	Root-free modulo 7
160	52	92274	1741	30	Root-free modulo 7

The three positive-b cubics are

$$\begin{aligned} f_{12,4}(x) &= x^3 - 568x^2 + 100032x - 5248512, \\ f_{20,9}(x) &= x^3 - 1769x^2 + 992820x - 174744000, \\ f_{52,30}(x) &= x^3 - 13394x^2 + 57771168x - 80229951360. \end{aligned}$$

Their values at all residues of the indicated primes are, respectively, $(3, 3, 3, 4, 2)$, $(3, 2, 4, 1, 6, 4, 1)$ and $(1, 6, 4, 1, 3, 2, 4)$. Each monic cubic is therefore irreducible over $\mathbb{Q}$ and has noninteger real roots. Section 2 supplies the lift to noninteger graph eigenvalues. □

Lemma 8.3 (A uniform consecutive-integer interval)**.** *For $a, b \geq 1$, put*

$$L(a) = \frac{a^3 + 2a^2 + 2a + 1}{a(2a + 1)}.$$

If $b < L(a)$, then $f_{a,b}$ has a root strictly between $2ab - 1$ and $2ab$.

Proof. Substitution in (3) gives

$$\begin{aligned} f_{a,b}(2ab) &= 2a^2b^2 > 0, \\ f_{a,b}(2ab - 1) &= -(a + b + 1)\big(a^3 + 2a^2 + 2a + 1 - a(2a + 1)b\big). \end{aligned}$$

The second value is strictly negative when $b < L(a)$. The intermediate value theorem gives the asserted noninteger, nonzero root. □

Proof of Theorem 8.1. For $a = 1$, Corollary 7.2 settles every positive b. For $a \geq 2$, nonsquare D gives irrational antisymmetric eigenvalues. When D is square, index $j = 0$ is Theorem 6.1, and indices $j = 1, 2$ are settled in Section 7. Lemma 8.2 settles $j = 3$.

It remains to consider $j \geq 4$. Then $d \geq Q = 4a + 5$, and $e \geq d \geq Q$ implies

$$\frac{M}{Q} + Q - (d + e) = (d - Q)\left(\frac{e}{Q} - 1\right) \geq 0.$$

Since $d + e = (a + 1)^2 b + a(3a + 1)$, we obtain

$$b \leq B_4(a) = \frac{M/Q + Q - a(3a + 1)}{(a + 1)^2} = \frac{(a - 5)(2a - 5)}{4a + 5}.$$

For every $a \geq 2$,

$$L(a) - B_4(a) = \frac{41a^3 - 17a^2 - 11a + 5}{a(2a + 1)(4a + 5)} > 0.$$

Indeed, with $z = a-2 \geq 0$, the numerator becomes $41z^3+229z^2+413z+243$. Thus $b \leq B_4(a) < L(a)$, and Lemma 8.3 gives a noninteger cubic root, which lifts across the universal vertices. These cases exhaust all positive a, b. Permuting the prime factors preserves the graph's isomorphism type. □

The preceding cutoffs and fixed-exponent theorems are intermediate consequences of the arithmetic reduction. The uniform interval now settles the infinite range left by those cutoffs, without a parameter scan.

9 Small exponents and complement Rayleigh bounds

The complement quotient replaces coordinate symmetry when the other two exponents are arbitrary. A root between zero and one then settles the entire unit-exponent family after the small exceptions are identified. For arbitrary prime-factor count, a rational trial function also gives an explicit sufficient condition with a distinguished exponent $m \geq 2$.

Theorem 9.1. *For distinct primes p, q, r and every $b, c \geq 1$, the graph $G_{pq^b r^c}$ is not Laplacian integral. The same holds for permutations of the exponent vector $(1, b, c)$.*

The equal-exponent decomposition is replaced by a complement identity on all six support classes. For arbitrary $a, b, c \geq 1$, remove the $u = abc - 1$ universal vertices and denote the remaining graph by H. In the order $1, 2, 3, 12, 13, 23$, its class-size vector is $w = (a, b, c, ab, ac, bc)^T$ and its vertex count is $N = a + b + c + ab + ac + bc$. Disjoint supports are adjacent in the complement $\overline{H}$, whose cell-constant Laplacian quotient is

$$C = \begin{pmatrix} b + c + bc & -b & -c & 0 & 0 & -bc \\ -a & a + c + ac & -c & 0 & -ac & 0 \\ -a & -b & a + b + ab & -ab & 0 & 0 \\ 0 & 0 & -c & c & 0 & 0 \\ 0 & -b & 0 & 0 & b & 0 \\ -a & 0 & 0 & 0 & 0 & a \end{pmatrix}.$$

Multiplication on the left by $\mathrm{diag}(w)$ makes C symmetric. Moreover $C\mathbf{1} = 0$ and $w^T C = 0$. If B is the quotient of H, the complement relation gives

$$B + C = NI - \mathbf{1}w^T. \tag{9}$$

Thus an eigenvector $Cv = \mu v$ with $\mu \neq 0$ satisfies $w^T v = 0$ and $Bv = (N - \mu)v$. When $N - \mu \neq 0$, Lemma 2.1 lifts it to an eigenvalue $N + u - \mu$ of G_n.

Proof of Theorem 9.1. Write $\det(xI - C) = x h_{a,b,c}(x)$. Direct determinant expansion yields

$$h_{a,b,c}(0) = -abc(a + b + c)N,$$
$$h_{a,b,c}(a) = -a^2 b^2 c^2 (a^2 + 2a - b - c).$$

At $a = 1$, the first value is strictly negative and

$$h_{1,b,c}(1) = b^2 c^2 (b + c - 3).$$

When $b + c > 3$, a real root μ lies strictly between 0 and 1. Equation (9) and the universal-vertex lift produce a Laplacian eigenvalue $V - \mu \in (V - 1, V)$, where $V = N + u$ is the integer vertex count. This eigenvalue is noninteger. When $b + c \leq 3$, the positive pairs are $(1, 1), (1, 2), (2, 1)$; Theorem 5.1 settles the corresponding permutations of $(1, 1, k)$. Permuting the prime factors preserves the graph's isomorphism type. □

The unit-exponent obstruction extends to any number of prime factors by a variational argument on the complement graph.

Theorem 9.2. *Let $n = \prod_{i=1}^{t} p_i^{k_i}$ with $t \geq 3$, distinct primes p_i and positive integer exponents k_i. If some $k_i = 1$, then G_n has a Laplacian eigenvalue in $(V - 1, V)$, where $V = \prod_{i=1}^{t}(k_i + 1) - 2$. In particular, G_n is not Laplacian integral.*

Proof. Permute the factors so that $k_t = 1$, and put

$$A = \prod_{i<t} k_i, \qquad B = \prod_{i<t}(k_i + 1) - 1 - A.$$

The quantity B is the sum of the weights of all nonempty proper supports on the first $t - 1$ coordinates. Since $t - 1 \geq 2$, it is positive. Remove the $A - 1$ full-support universal vertices and let $F = \overline{H}$. Then $N = |H| = A + 2B + 1$ and $V = N + A - 1 = 2A + 2B$. The graph F is connected: its singleton support classes form a connected complete multipartite graph, and every proper support has a singleton outside it.

The support $\{t\}$ consists of a single vertex z. Each of the A vertices supported on $[t - 1]$ has z as its sole neighbor in F. The remaining $2B$ vertices form two collections of weight B: nonempty proper supports $S \subset [t - 1]$, and their counterparts $S \cup \{t\}$. The vertex z is adjacent to all vertices of the first collection and none of the second.

Give a function f the value $2B$ on the A leaves, zero at z, and $-A$ on the other $2B$ vertices. It is nonzero and orthogonal to the constant vector. Its Laplacian energy and norm are

$$f^T L(F) f = A(2B)^2 + BA^2 = AB(4B + A), \qquad f^T f = A(2B)^2 + 2BA^2 = 2AB(2B + A).$$

Indeed, only edges incident to z have unequal endpoint values. Connectedness and the variational characterization therefore give

$$0 < \lambda_2(F) \leq \frac{4B + A}{4B + 2A} < 1.$$

For a corresponding eigenvector, the complement identity $L(H) + L(F) = NI - J$ gives the nonzero eigenvalue $N - \lambda_2(F)$ of H. Lemma 2.1 lifts it to $V - \lambda_2(F) \in (V - 1, V)$, proving the claim. □

The restriction $t \geq 3$ is sharp for this conclusion. If $t = 2$ and $k_2 = 1$, then $B = 0$ and $F = K_{1,A}$, so the complement and universal-vertex identities give an integral spectrum for G_n. Theorem 9.2 also gives a proof of Theorems 4.1 for $t \geq 3$ and 9.1 without their separate spectral decompositions or small exceptions.

The same support partition gives an obstruction when the distinguished exponent is larger than one and the other support weights are sufficiently concentrated on their full support.

Theorem 9.3. *Let $n = (\prod_{i<t} p_i^{k_i}) p_t^m$ have $t \geq 3$ distinct prime factors and positive integer exponents. Put*

$$A = \prod_{i<t} k_i, \qquad B = \prod_{i<t}(k_i + 1) - 1 - A.$$

If $A \geq (m^2 - 1)B - (m - 1)$, then G_n has a Laplacian eigenvalue in $(V - 1, V)$, where $V = (m + 1)\prod_{i<t}(k_i + 1) - 2$. In particular, it is not Laplacian integral. The condition includes equality and can be applied after any permutation of the factors.

Proof. Remove the $mA - 1$ universal vertices and let $F = \overline{H}$. As in Theorem 9.2, F is connected and $B > 0$. Its vertices split into three collections: L supported on $[t - 1]$, Z supported on $\{t\}$, and the remaining collection R. Their sizes are $A, m, (m + 1)B$, respectively. There are mA edges between L and Z, none between L and R, and mB between Z and R. Among the vertices of R, those whose support contains t have total weight mB and no neighbor in Z; the others have total weight B and are adjacent to every vertex of Z.

Give a trial function f the values

$$x = m \text{ on } L, \qquad y = m - 1 \text{ on } Z, \qquad z = -\frac{m(A + m - 1)}{(m + 1)B} \text{ on } R.$$

It is nonzero and $Ax + my + (m + 1)Bz = 0$. Its energy E and squared norm D are

$$E = mA(x - y)^2 + mB(y - z)^2, \qquad D = Ax^2 + my^2 + (m + 1)Bz^2.$$

Direct expansion gives

$$E - D = \frac{m\big((m^2 - 1)B - A - (m - 1)\big)\big(mA + (m^2 - 1)B + m(m - 1)\big)}{B(m + 1)^2}.$$

The second factor is positive. Thus $E/D < 1$ when the hypothesis is strict, and the variational principle gives $0 < \lambda_2(F) < 1$.

At equality, necessarily $m \geq 2$, and $z = -m(m - 1)$, with $E/D = 1$. At every vertex of R whose support contains t, the full Laplacian action on f is zero: its neighbors are in R and have the same trial value. But $f = -m(m - 1) \neq 0$ there, so f is not an eigenvector at one. Equality in the Rayleigh minimum would force f to be such an eigenvector. Hence $\lambda_2(F) < 1$ also at the boundary.

In both cases, the complement identity reflects this positive eigenvalue to $N - \lambda_2(F)$, where $N = A + m + (m + 1)B$. Lemma 2.1 then shifts it by $mA - 1$, giving $V - \lambda_2(F) \in (V - 1, V)$. □

The exact product condition has a convenient reciprocal sufficient condition, giving unbounded families with every exponent at least two.

Corollary 9.4. *Let n have $t \geq 3$ distinct prime factors, one exponent $m \geq 2$ and other positive integer exponents $k_1, \ldots, k_{t-1}$. Suppose*

$$\sum_{i<t} \frac{1}{k_i} \leq \frac{1}{m^2}.$$

Then G_n is not Laplacian integral. In particular, every $k_i \geq m^2(t - 1)$ suffices.

Proof. Put $u_i = 1/k_i$ and $s = \sum u_i < 1$. Induction gives $\prod(1-u_i) \geq 1-s$, while $1+u_i < (1-u_i)^{-1}$. Therefore

$$\frac{B}{A} = \prod_{i<t}(1+u_i) - 1 - \frac{1}{A} < \frac{1}{1-s} - 1 \leq \frac{1}{m^2-1}.$$

Thus $A > (m^2-1)B$, and Theorem 9.3 applies. The stated coordinate bound implies the reciprocal bound. □

For example, $(10,10,10,2)$ has $A = 1000, B = 330$ and satisfies the strict condition. The boundary vector $(4,19,74,2)$ has $A = 5624, B = 1875$ and $A = 3B-1$; the non-eigenvector argument in the proof supplies strictness. The criterion does not apply to $(2,2,2,2)$, for which $A = 8, B = 18$. It is a sufficient obstruction, and the higher-prime classification outside it remains open.

The general six-support reduction also gives an exact diagnostic for fully distinct triples. For $1 \leq a < b < c \leq 7$, all thirty-five quintics have a nonlinear irreducible factor over $\mathbb{Q}$. The quintics at $(1,2,7)$ and $(2,3,5)$ have factor degrees $(1,4)$, and that at $(2,4,6)$ has factor degrees $(2,3)$; irreducibility of the entire quintic is therefore not a uniform obstruction. Theorem 9.1 covers every triple beginning with 1 without a range restriction; the thirty-five-case calculation itself remains finite evidence.

9.1 Four prime factors and two exponents equal to two

A repeated pair also produces a small invariant subspace in four coordinates. This settles a family including balanced vectors outside the preceding Rayleigh criterion.

Theorem 9.5. *For distinct primes p_1, p_2, p_3, p_4 and every $b, c \geq 2$, the graph $G_{p_1^2 p_2^2 p_3^b p_4^c}$ is not Laplacian integral. Together with Theorem 9.2, this covers every positive four-prime exponent vector with at least two entries equal to two.*

Proof. Remove the $4bc-1$ universal vertices and let $F = \overline{H}$. For each $T \subseteq \{3,4\}$, assign a function the values $h(T)$ on support $\{1\} \cup T$, $-h(T)$ on $\{2\} \cup T$, and zero elsewhere. The paired classes have equal weight, so the function has weighted sum zero. Classes containing neither or both of coordinates $1,2$ see equal opposite contributions. At support $\{1\} \cup T$, the complement degree is $3\prod_{i\in\{3,4\}\setminus T}(k_i+1) - 1$ and its nonzero-valued neighbors are precisely the classes $\{2\} \cup T'$ with $T' \cap T = \varnothing$, of weight $2\prod_{i\in T'} k_i$. Thus the four-dimensional space is invariant. In the order $T = \varnothing, \{4\}, \{3\}, \{3,4\}$, its complement action is $K - I$, where

$$K = \begin{pmatrix} 3(b+1)(c+1)+2 & 2c & 2b & 2bc \\ 2 & 3(b+1) & 2b & 0 \\ 2 & 2c & 3(c+1) & 0 \\ 2 & 0 & 0 & 3 \end{pmatrix}.$$

The matrix $\operatorname{diag}(1,c,b,bc)K$ is symmetric. If κ is an eigenvalue of K, complement reflection and the zero-sum lifting calculation in the proof of Lemma 2.1 give the graph eigenvalue $V+1-\kappa$, with $V = 9(b+1)(c+1) - 2$.

The leading principal minors of $K - I$ are

$$\begin{aligned}
D_1 &= 3bc + 3b + 3c + 4,\\
D_2 &= 9b^2c + 9b^2 + 15bc + 18b + 2c + 8,\\
D_3 &= 15b^2c^2 + 33b^2c + 6b^2 + 33bc^2 + 84bc\\
&\qquad + 28b + 6c^2 + 28c + 16,\\
D_4 &= 2\big(5b^2c^2 + 21b^2c + 6b^2 + 21bc^2 + 76bc\\
&\qquad + 28b + 6c^2 + 28c + 16\big).
\end{aligned}$$

All coefficients are positive. The symmetric similar matrix has the same leading minors, so Sylvester's criterion gives $\kappa_{\min} > 1$. The principal submatrix of $K - 3I$ on coordinates $1, 4$ has determinant $-4bc$, hence a negative Rayleigh quotient in its symmetric similar matrix. Thus $\kappa_{\min} < 3$. Unless $\det(K - 2I) = 0$, this smallest eigenvalue is noninteger and so is its graph lift.

Direct expansion gives

$$\begin{aligned}
P(b,c) := \det(K - 2I) = &-5b^2c^2 + 12b^2c - 3b^2 + 12bc^2\\
&+ 48bc + 8b - 3c^2 + 8c + 3.
\end{aligned}$$

Order $b \le c$. For $b = 7 + u, c = 7 + v$, with $u, v \ge 0$,

$$\begin{aligned}
-P(7+u, 7+v) = &\,5u^2v^2 + 58u^2v + 164u^2 + 58uv^2 + 596uv\\
&+ 1364u + 164v^2 + 1364v + 1600 > 0.
\end{aligned}$$

For the remaining five values of b, the polynomial is

b	$P(b,c)$
2	$c^2 + 152c + 7$
3	$4c(65 - 3c)$
4	$-35c^2 + 392c - 13$
5	$-4(c - 8)(17c - 1)$
6	$-111c^2 + 728c - 57$

The first two rows have no positive integer zero. The discriminants of rows $4, 6$ are $151844 \in (389^2, 390^2)$ and $504676 \in (710^2, 711^2)$, so neither has a rational zero. The only positive integer endpoint pair is therefore $(b, c) = (5, 8)$, up to exchange.

At this pair,

$$\det(xI - K) = (x - 2)(x^3 - 210x^2 + 7701x - 53240).$$

The cubic reduces to $x^3 + 11x^2 + 4$ modulo 17. At the residues $0, \dots, 16$ its values, in order, are

$$\begin{aligned}
&4, 16, 5, 11, 6, 13, 4, 2, 13,\\
&9, 13, 14, 1, 14, 8, 6, 14.
\end{aligned}$$

None vanishes, so the cubic is irreducible over $\mathbb{F}_{17}$ and over $\mathbb{Q}$. Its roots are real eigenvalues of K and irrational; their graph lifts settle the exceptional pair as well. The proof uses the five symbolic rows and this 17-residue certificate, with no finite exponent base. □

In particular, $(2, 2, 2, 2)$ is nonintegral, although it fails Theorem 9.3's sufficient condition. The present argument uses an actual invariant subspace, whereas the preceding three-collection compression supplies trial functions.

9.2 Two exponents equal to three

Tensor normalization supplies a general lower bound for the repeated-pair restriction. At exponent three it leaves an endpoint equation with no integer solutions.

Theorem 9.6. *For distinct primes p_1, p_2, p_3, p_4 and every $b, c \geq 2$, the graph $G_{p_1^3 p_2^3 p_3^b p_4^c}$ has a noninteger Laplacian eigenvalue in $(V-2, V)$, different from $V-1$, where $V = 16(b+1)(c+1)-2$. Together with Theorem 9.2, this proves nonintegrality for every positive four-prime vector with two entries equal to three.*

Proof. For the general vector (a, a, b, c), use the paired-support functions of Theorem 9.5. Their weighted sum is zero. The degree on $\{1\} \cup T$ is now $(a+1)\prod_{i\in\{3,4\}\setminus T}(k_i+1)-1$, while the opposite neighbor classes have weights $a\prod_{i\in T'} k_i$. Thus the complement restriction is $K_a - I$, with

$$K_a = (a+1)E_b \otimes E_c + aF_b \otimes F_c, \qquad E_k = \begin{pmatrix} k+1 & 0 \\ 0 & 1 \end{pmatrix}, \quad F_k = \begin{pmatrix} 1 & k \\ 1 & 0 \end{pmatrix}.$$

The symmetrizer is $\mathrm{diag}(1, c, b, bc)$, and an eigenvalue κ lifts to $V + 1 - \kappa$, where $V = (a+1)^2(b+1)(c+1) - 2$, by the zero-sum calculation above.

In the weighted symmetric basis, F_k becomes $S_k = \begin{pmatrix} 1 & \sqrt{k} \\ \sqrt{k} & 0 \end{pmatrix}$. Put $T_k = E_k^{-1/2} S_k E_k^{-1/2}$. Its characteristic polynomial is $(x-1)(x + k/(k+1))$, so its eigenvalues are $1, -k/(k+1)$. Every eigenvalue of $T_b \otimes T_c$ is strictly greater than -1. With $E = E_b \otimes E_c$, the symmetric similar matrix of K_a is

$$E^{1/2}\big((a+1)I + aT_b \otimes T_c\big)E^{1/2}.$$

The bracket is strictly greater than I, so this matrix is strictly greater than $E \geq I$. Hence $\kappa_{\min}(K_a) > 1$ for all $a, b, c \geq 1$.

Set $a = 3$ and write $K = K_3$. Direct expansion gives

$$\begin{aligned} \det(K - 3I) = &-35b^2c^2 + 8b^2c - 20b^2 + 8bc^2 \\ &+ 109bc + 11b - 20c^2 + 11c + 4. \end{aligned}$$

For $b = 2 + u, c = 2 + v$, its negative is

$$\begin{aligned} &35u^2v^2 + 132u^2v + 144u^2 + 132uv^2 + 387uv \\ &\quad + 315u + 144v^2 + 315v + 108 > 0. \end{aligned}$$

The symmetric similar matrix $K - 3I$ therefore has a negative eigenvalue, giving $1 < \kappa_{\min}(K) < 3$. It suffices to exclude two.

Write $P(b, c) = \det(K - 2I)$. Its exact polynomial is

$$\begin{aligned} P(b,c) = &-7b^2c^2 + 48b^2c - 8b^2 + 48bc^2 \\ &+ 302bc + 76b - 8c^2 + 76c + 40. \end{aligned}$$

Order $b \leq c$. For $b = 17 + u, c = 17 + v$, with $u, v \geq 0$,

$$\begin{aligned} -P(17+u, 17+v) = &\,7u^2v^2 + 190u^2v + 1215u^2 + 190uv^2 + 4526uv \\ &+ 22228u + 1215v^2 + 22228v + 27721 > 0. \end{aligned}$$

For $2 \le b \le 16$, write $P = A_b c^2 + B_b c + C_b$, where

$$A_b = -7b^2 + 48b - 8, \quad B_b = 48b^2 + 302b + 76, \quad C_b = -8b^2 + 76b + 40.$$

For $b = 2, \dots, 6$, the A_b values are $(60, 73, 72, 57, 28)$ and the C_b values are $(160, 196, 216, 220, 208)$; also $B_b > 0$. These rows have no positive zero. Two further rows factor as

$$P(10, c) = -12c(19c - 658), \qquad P(14, c) = -4(3c - 58)(59c - 2).$$

Their positive roots $658/19, 58/3, 2/59$ are not integers. For the remaining eight rows, the discriminant $D_b = B_b^2 - 4A_bC_b$ lies strictly between the consecutive squares shown below:

b	D_b	h_b with $h_b^2 < D_b < (h_b+1)^2$
7	20640564	4543
8	30997264	5567
9	44692596	6685
11	84630100	9199
12	112262544	10595
13	146014164	12083
15	235204596	15336
16	292433040	17100

Thus none of these quadratics has a rational zero. The complete symbolic reduction proves $P(b, c) \neq 0$ for all integers $b, c \ge 2$, with no exceptional pair. Therefore $\kappa_{\min}$ is noninteger in $(1, 3)$, and its graph lift $V + 1 - \kappa_{\min}$ has the stated location. The fixed arithmetic consists of the eight discriminants and two factorizations; no finite two-parameter exponent base is used. □

10 A minimum exponent of two

Specializing the complement quintic to a minimum exponent of two reduces its endpoint signs to positive polynomial expansions. One remaining pair is handled by a rational interval, completing this minimum uniformly.

Theorem 10.1. *For distinct primes p, q, r and all $b, c \ge 1$, the graph $G_{p^2q^br^c}$ is not Laplacian integral. Consequently every three-prime exponent vector with minimum entry at most two gives a nonintegral graph.*

Proof. Use the complement quotient and quintic from Section 9. For $a = 2$, $h_{2,b,c}(0) < 0$. Putting $s = b + c$ and $t = bc$ gives

$$h_{2,b,c}(1) = (s - 13)t^2 - (2s + 6)t + 3s^3 + s^2 - 3s - 1.$$

By symmetry order $b \le c$. A unit exponent is covered by Theorem 9.1; if $b = 2$ or $b = c$, use Theorem 8.1. It remains to consider $3 \le b < c$.

When $b = 3$ and $c = 5 + v$, $v \ge 0$, the exact expansion is

$$h_{2,3,5+v}(1) = 12v^3 + 112v^2 + 268v + 120 > 0.$$

When $b \ge 4$ and $c > b$, write $b = 4 + u$ and $c = 5 + u + v$, with $u, v \ge 0$. Then

$$\begin{aligned} h_{2,4+u,5+u+v}(1) = {} & (u^2 + 8u + 19)v^3 \\ & + (4u^3 + 38u^2 + 128u + 170)v^2 \\ & + (5u^4 + 62u^3 + 271u^2 + 502u + 368)v \\ & + 2u^5 + 32u^4 + 190u^3 + 504u^2 + 552u + 160 > 0. \end{aligned}$$

All coefficients are positive. In either region the opposite signs at 0 and 1 give a root $\mu \in (0,1)$ and, by (9) and the universal-vertex lift, a noninteger graph eigenvalue in $(V-1, V)$.

The only remaining pair is $(b,c) = (3,4)$. Here

$$h_{2,3,4}(x) = x^5 - 53x^4 + 953x^3 - 6523x^2 + 13134x - 7560,$$

and $h_{2,3,4}(1) = -48$ while $h_{2,3,4}(3/2) = 13437/32 > 0$. A root lies in $(1, 3/2)$. Since $V = 58$, its lifted graph eigenvalue lies in $(113/2, 57)$, an interval containing no integer. This exhausts all positive b, c. Combining with Theorem 9.1 proves the minimum-entry assertion. □

11 A uniform cutoff and a minimum exponent of three

The first endpoint polynomial gives a cutoff depending only on the minimum exponent. At minimum three, the finite region below that cutoff is exhausted by an exact certificate, so the whole minimum-three family follows.

Proposition 11.1. *For distinct primes p, q, r and positive exponents ordered as $2 \le a \le b \le c$, the graph $G_{p^a q^b r^c}$ is not Laplacian integral whenever $c \ge 4a^2 - 2a$.*

Proof. Use the complement quintic $h = h_{a,b,c}$ of Section 9. Its first endpoint is the cubic

$$h(1) = Lc^3 + Qc^2 + Rc + S, \tag{10}$$

where

$$\begin{aligned} L &= a^2 + b^2 - 1, \\ Q &= a^3 - 4a^2b^2 + 2a^2b - a^2 + 2ab^2 + ab - 3a + b^3 - b^2 - 3b + 3, \\ R &= (a-1)(b-1)(2ab + 3a + 3b - 3), \\ S &= (a-1)(b-1)(a+b-1)(ab + a + b - 1). \end{aligned}$$

Here $L, R, S > 0$. In fact the sharper condition $Lc + Q \ge 0$ already gives $h(1) > 0$. Put $T(a) = 4a^2 - 2a$. Then

$$Q + T(a)L = b^2(b-1) + (2a^2 + a - 3)b + K(a),$$

where $K(a) = 4a^4 - a^3 - 5a^2 - a + 3$. For $a = 2 + u$, $u \ge 0$,

$$K(2+u) = 4u^4 + 31u^3 + 85u^2 + 95u + 37 > 0.$$

Consequently $Lc + Q > 0$ for $c \ge T(a)$. Since $h(0) < 0$, a root lies in $(0,1)$, producing a noninteger graph eigenvalue in $(V-1, V)$ by (9) and Lemma 2.1. □

Theorem 11.2. *For distinct primes p, q, r and all $b, c \ge 1$, the graph $G_{p^3 q^b r^c}$ is not Laplacian integral. Consequently every three-prime exponent vector with minimum entry at most three gives a nonintegral graph.*

Proof. Theorems 9.1, 10.1 and 8.1 leave $4 \le b < c$. Proposition 11.1 settles $c \ge 30$. At $a = 3$, equation (10) is

$$\begin{aligned} h_{3,b,c}(1) = (b^2+8)c^3 &+ (b-1)(b^2 - 30b - 12)c^2 \\ &+ 6(b-1)(3b+2)c + 4(b-1)(b+2)(2b+1). \end{aligned}$$

The remaining domain $4 \le b < c < 30$ has exactly $\binom{26}{2} = 325$ pairs. Exhaustive integer evaluation of this cubic gives 257 positive and 68 negative values, with no zero. Table 1 lists every negative range; all other pairs in this finite domain have positive values. The complete polynomial and separate Horner sign certificates are in `results/distinct-tail.json`; their construction is described in Section A. Thus a positive value gives a root in $(0, 1)$.

b	4	5	6	7	8	9	10	11	12	13
$c_{\min}$	5	6	7	8	9	10	11	12	13	14
$c_{\max}$	13	15	16	17	17	16	16	15	14	14

Table 1: Every negative $h_{3,b,c}(1)$ in $4 \le b < c < 30$; each column includes all integers between its two endpoints.

For the negative values, the second endpoint is positive except at $(b, c) = (4, 5)$. Indeed, for $v \ge 0$,

$$h_{3,4,6+v}(2) = 104v^3 + 1238v^2 + 4018v + 2344 > 0.$$

For $b \ge 5$, write $b = 5 + u$, $c = 6 + u + v$, with $u, v \ge 0$. Direct expansion gives

$$\begin{aligned} h_{3,5+u,6+u+v}(2) = {} & (5u^2 + 54u + 153)v^3 \\ & + (20u^3 + 262u^2 + 1179u + 1863)v^2 \\ & + (25u^4 + 426u^3 + 2642u^2 + 7011u + 6624)v \\ & + 10u^5 + 218u^4 + 1828u^3 + 7200u^2 + 12600u + 6480 > 0. \end{aligned}$$

These regions exhaust $4 \le b < c$ except $(4, 5)$. Whenever $h(1) < 0 < h(2)$, a root lies in $(1, 2)$, giving a noninteger graph eigenvalue in $(V - 2, V - 1)$.

Finally

$$h_{3,4,5}(x) = x^5 - 83x^4 + 2327x^3 - 24133x^2 + 60576x - 42480.$$

Here $h(1) = -3792$ and $h(3/2) = 48825/32 > 0$. The root in $(1, 3/2)$ lifts to a graph eigenvalue in $(233/2, 117)$, since $V = 118$. Every interval used contains no integer. The finite nonvanishing certificate and the written infinite-range arguments exhaust all b, c; Theorems 9.1 and 10.1 give the final assertion. □

12 An inertia criterion and four families with bounded exponent gaps

An endpoint sign change need not detect two roots in the same interval. For example, the complement quintic at equal exponents factors as

$$h_{a,a,a}(x) = (x - a^2 - a)\big(x^2 - (a^2 + 4a)x + 3a^2\big)^2.$$

We instead count low eigenvalues by a symmetric Schur complement.

Proposition 12.1. *Order the exponents as $a \le b \le c$, and let m be an integer with $2 \le m < a$. If*

$$(c - m)(a + b + c - m) < mab, \tag{11}$$

then the six-support complement quotient has exactly two nonzero eigenvalues in $(0, m)$*. If, in addition,*

$$(a - m + 1)(a + b + c - m + 1) > (m - 1)bc, \tag{12}$$

both lie in $(m - 1, m)$*, so* $G_{p^a q^b r^c}$ *is not Laplacian integral.*

Proof. Put $s = a+b+c$ and $P = abc$. Symmetrize C by the square roots of its cell sizes and reorder the classes as $1, 2, 3, 23, 13, 12$. The resulting real symmetric matrix is

$$\mathcal{C} = \begin{pmatrix} A & -\sqrt{P}I \\ -\sqrt{P}I & D \end{pmatrix}, \quad D = \operatorname{diag}(a, b, c),$$

where A has diagonal $(b + c + bc, a + c + ac, a + b + ab)$ and off-diagonal entries $A_{ij} = -\sqrt{a_i a_j}$, with $(a_1, a_2, a_3) = (a, b, c)$. For $0 < x < a$, the block $D - xI$ is positive definite, and Sylvester's law of inertia gives

$$\operatorname{inertia}(\mathcal{C} - xI) = \operatorname{inertia}(D - xI) + \operatorname{inertia}(K(x)),$$

where

$$K(x) = \operatorname{diag}(k_a(x), k_b(x), k_c(x)) - vv^T, \quad v = (\sqrt{a}, \sqrt{b}, \sqrt{c})^T, \quad k_t(x) = s - x - \frac{xP}{t(t - x)}.$$

For fixed $x > 0$, the function $k_t(x)$ increases with $t > x$, since

$$\frac{\partial k_t(x)}{\partial t} = \frac{xP(2t - x)}{t^2(t - x)^2} > 0.$$

Condition (11) is exactly $k_c(m) < 0$. Thus all three diagonal values are negative, so $K(m)$ is negative definite. The matrix $\mathcal{C} - mI$ has three negative and three positive eigenvalues. The complement support graph is a triangle with one pendant cell at each vertex; it is connected, so C has a simple zero eigenvalue and all other eigenvalues are positive. Exactly two nonzero eigenvalues therefore lie in $(0, m)$.

For $x = m - 1$, condition (12) says $k_a(x) > 0$, so all three $k_t(x)$ are positive. Congruence by the inverse square roots of these values turns $K(x)$ into $I - zz^T$, where

$$\|z\|^2 = \frac{a}{k_a(x)} + \frac{b}{k_b(x)} + \frac{c}{k_c(x)} > 1,$$

because each $k_t(x) < s$. Consequently $K(x)$ has one negative and two positive eigenvalues, and the only eigenvalue of C below x is zero. Both low positive eigenvalues lie in $(m - 1, m)$. They are less than $a < N$, so the complement identity and universal-vertex lift apply. Their graph eigenvalues lie in $(V - m, V - m + 1)$, which contains no integer. □

Theorem 12.2. *For every* $a \geq 1$*, each of the exponent triples*

$$(a, a + 1, a + 2), \quad (a, a + 1, a + 3), \quad (a, a + 2, a + 3), \quad (a, a + 3, a + 4)$$

gives a nonintegral ideal intersection graph on three distinct primes.

Proof. For $a \leq 3$, use Theorem 11.2. Assume $a \geq 4$ and write the triple as $(a, a + d, a + e)$, where $(d, e) = (1, 2), (1, 3), (2, 3), (3, 4)$. The left side minus the right side of (11), at $m = 3$, is respectively

$$-6a, \quad -2a, \quad -4a, \quad 4 - 2a.$$

(d,e)	$M_{d,e}(a)$	Descending coefficients of $P_{d,e}(4+u)$
$(1,2)$	$a(a+1)$	$(1,18,116,314,294)$
$(1,3)$	$a(a+1)(a+3)$	$(1,13,51,57)$
$(2,3)$	$a+2$	$(1,25,242,1126,2484,2022)$
$(3,4)$	$a+3$	$(1,28,302,1550,3699,3126)$

Table 2: Exact first-endpoint identities and positive shifted factors.

These values are negative. Proposition 12.1 therefore places exactly two nonzero complement eigenvalues in $(0,3)$. It remains to exclude the integers 1 and 2.

The first endpoints have the identities $h_{a,a+d,a+e}(1) = -4M_{d,e}(a)P_{d,e}(a)$ from Table 2. Every multiplier is positive for $a \geq 4$, and every coefficient of $P_{d,e}(4+u)$ is positive. Thus $h(1) < 0$ in each family.

The second endpoints factor as

$$\begin{aligned}
h_{a,a+1,a+2}(2) &= -aF_{1,2}(a),\\
h_{a,a+1,a+3}(2) &= -(a-2)(a+1)F_{1,3}(a),\\
h_{a,a+2,a+3}(2) &= -aF_{2,3}(a),\\
h_{a,a+3,a+4}(2) &= -F_{3,4}(a),
\end{aligned}$$

where

$$\begin{aligned}
F_{1,2}(a) &= a^5 - 9a^4 - 14a^3 + 93a^2 + 67a + 6,\\
F_{1,3}(a) &= a^4 - 6a^3 - 42a^2 - 39a - 10,\\
F_{2,3}(a) &= a^5 - 5a^4 - 48a^3 - 59a^2 - 39a - 54,\\
F_{3,4}(a) &= a^6 - a^5 - 74a^4 - 375a^3 - 967a^2 - 1220a - 340.
\end{aligned}$$

Table 3 evaluates each factor at every residue of one prime. No value is zero, so no factor has an integer root. The linear multipliers are nonzero for $a \geq 4$, hence $h(2) \neq 0$. Both positive eigenvalues in $(0,3)$ are therefore noninteger, and their graph lifts are noninteger as well.

Factor	Prime	Values at residues $0,\ldots,p-1$
$F_{1,2}$	7	$(6,4,1,5,6,5,1)$
$F_{1,3}$	11	$(1,3,9,8,2,6,4,4,5,8,5)$
$F_{2,3}$	7	$(2,6,5,3,5,4,3)$
$F_{3,4}$	13	$(11,1,3,4,3,1,9,9,11,11,6,1,8)$

Table 3: Complete root-free modular certificates for the second endpoints.

□

Corollary 12.3. *Every positive three-prime exponent triple with maximum minus minimum at most three gives a nonintegral ideal intersection graph.*

Proof. Repeated entries are covered by Theorem 8.1. After sorting distinct entries, the only possibilities have gaps $(d,e) = (1,2),(1,3),(2,3)$, all covered by Theorem 12.2. □

12.1 The complete finite region for minimum exponents four through seven

The uniform cutoff makes the requested diagnostic range complete for each of the minimum exponents $4, 5, 6, 7$. Here the relevant quintic is $g_B(x) = \det(xI - B)/x$ for H, whereas the preceding endpoint arguments use the complement quintic $h_C(x) = -g_B(N - x)$.

Corollary 12.4. *Every positive three-prime exponent triple with minimum entry at most seven gives a nonintegral ideal intersection graph.*

Proof. Theorem 11.2 handles minimum entries at most three. For a minimum entry $a \in \{4, 5, 6, 7\}$, repeated entries are covered by Theorem 8.1, and Proposition 11.1 settles $c \geq 4a^2 - 2a$. The remaining ordered triples lie in the exact finite domain $a < b < c < 4a^2 - 2a$.

For all 27 562 triples in this domain, the integer discriminant $\Delta_{a,b,c}$ of g_B is positive and strictly between two consecutive integer squares. Table 4 gives the complete counts. The certificate `results/open-region-discriminants.csv` records every triple, discriminant and integer q satisfying $q^2 < \Delta_{a,b,c} < (q+1)^2$. Each discriminant is independently recomputed as a 9×9 integer Sylvester determinant by `scripts/check_open_region_certificate.py`, using fraction-free elimination and exact divisions. The certificate exhausts the domain without omitted or duplicate triples.

If all roots of the monic quintic g_B were integers, its discriminant, the product of the squared pairwise root differences, would be an integer square. Thus every certified quintic has a noninteger root. The quotient is similar to a real symmetric Laplacian, so the root is real and nonzero; Lemma 2.1 gives a noninteger graph eigenvalue. This complete finite certificate and the written cutoff exhaust all triples with minimum entry at most seven. □

Minimum a	4	5	6	7	Total
Exclusive cutoff	56	90	132	182	—
Complete pairs	1275	3486	7750	15051	27562
Nonsquare discriminants	1275	3486	7750	15051	27562

Table 4: Exhaustive discriminant certificates in the region derived from the uniform cutoff.

The signs of $(g_B(1), g_B(2), g_B(3))$ are $(-,-,-)$ throughout this finite range. The complement signs instead have three patterns, $(-,+,-), (-,+,+), (+,+,+)$, recorded separately in the diagnostic. Different endpoint patterns do not rule out a uniform proof, and equal signs can miss two roots in one interval. Proposition 12.1 addresses that issue without extrapolating from these finite counts.

13 A uniform low eigenvalue and a balanced region

The Schur complement of Proposition 12.1 also counts roots when its diagonal has mixed signs. A comparison at three gives a low eigenvalue for every positive real parameter triple.

Lemma 13.1. *For $0 < x < a$, let q and z count the negative and zero values among the three $k_t(x)$. If $z = 0$, put*

$$F(x) = 1 - \frac{a}{k_a(x)} - \frac{b}{k_b(x)} - \frac{c}{k_c(x)}.$$

Then $n_-(K(x)) = q + \mathbf{1}_{\{F(x)<0\}}$ and nullity $K(x) = \mathbf{1}_{\{F(x)=0\}}$. *If $z > 0$, then*

$$n_-(K(x)) = q + 1, \qquad \text{nullity } K(x) = z - 1.$$

The number of positive complement quotient eigenvalues strictly below x is $n_-(K(x)) - 1$, and the multiplicity at x is nullity $K(x)$.

Proof. For $z = 0$, the bordered matrix

$$\begin{pmatrix} \operatorname{diag}(k_t(x)) & v \\ v^T & 1 \end{pmatrix}$$

is congruent both to $K(x) \oplus [1]$ and to $\operatorname{diag}(k_t(x)) \oplus [F(x)]$. Sylvester's law of inertia gives the formulas. If $k_j(x) = 0$, change coordinates to $w_j = v^T y$ and $w_i = y_i$ for $i \neq j$. Since $v_j > 0$, this is invertible, and

$$y^T K(x) y = -w_j^2 + \sum_{i \neq j} k_i(x) w_i^2.$$

The singular formulas follow without reciprocals. The block $D - xI$ is positive definite, and C has a simple zero; congruence and subtracting this eigenvalue give the quotient counts and multiplicity. □

A direct comparison at three supplies a uniform low positive root without choosing a diagonal sign pattern.

Theorem 13.2. *For every ordered real triple $0 < a \leq b \leq c$, the six-support complement quotient has a nonzero eigenvalue in $(0, 3)$.*

Proof. Use $s = a + b + c$, $P = abc$ and $v = (\sqrt{a}, \sqrt{b}, \sqrt{c})^T$ from Proposition 12.1, and define

$$L = \operatorname{diag}\left(s - \frac{3P}{a^2}, s - \frac{3P}{b^2}, s - \frac{3P}{c^2}\right) - vv^T.$$

Expansion of this diagonal-minus-rank-one determinant gives

$$\det L = 6\big(c(a-b)^2 + b(a-c)^2 + a(b-c)^2\big),$$
$$v^T L v = -3P\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) < 0.$$

The determinant and Rayleigh value give exactly two negative eigenvalues at unequal parameters. At equal parameters $L = -vv^T$; thus L is nonpositive on a two-dimensional subspace in both cases.

For $a > 3$, the identity

$$K(3) = L - \operatorname{diag}_{t \in \{a,b,c\}}\left(3 + \frac{9P}{t^2(t-3)}\right)$$

subtracts a positive definite matrix. Since $D - 3I > 0$, this gives at least two negative eigenvalues of $\mathcal{C} - 3I$.

If $b \le 3$, restrict $\mathcal{C} - 3I$ to the first two singleton supports and their paired supports. Its blocks are $\left[\begin{smallmatrix} A_0 & -\sqrt{P}I_2 \\ -\sqrt{P}I_2 & D_0 \end{smallmatrix}\right]$ with $D_0 \le 0$. On vectors (z, tz), the quadratic form is at most $z^T(A_0 - 2t\sqrt{P}I_2)z$, which is negative definite for large t.

It remains to consider $a \le 3 < b$. Eliminate the positive paired diagonals $b-3, c-3$. On singleton supports $2, 3$, the vector $(\sqrt{b}, \sqrt{c})$ has quadratic value

$$q = (a-3)(b+c) - 3P\left(\frac{1}{b-3} + \frac{1}{c-3}\right) < 0.$$

Compress the remaining matrix to singleton 1, its paired support, and this vector. The resulting matrix is

$$\begin{pmatrix} M & -\sqrt{P} & r \\ -\sqrt{P} & a-3 & 0 \\ r & 0 & q \end{pmatrix}, \quad M = bc + b + c - 3 > 0, \quad r = -\sqrt{a}(b+c).$$

Its determinant $M(a-3)q - Pq - (a-3)r^2$ is positive. Its positive and negative diagonal entries therefore force exactly two negative eigenvalues. This includes the singular boundary $a = 3$.

In every case $\mathcal{C} - 3I$ has at least two negative eigenvalues. Since $\mathcal{C}$ is positive semidefinite with a simple zero, at least one positive eigenvalue lies strictly below three. □

Corollary 13.3. *For $4 \le a \le b \le c$, if*

$$(a-2)(a+b+c-2) > 2bc, \tag{13}$$

then C has an eigenvalue in $(2,3)$, and $G_{p^a q^b r^c}$ is not Laplacian integral. Nonzero values of both $h_C(1)$ and $h_C(2)$ also imply nonintegrality. Hence $h_C(1)h_C(2) = 0$ is necessary for an integral graph.

Proof. Condition (13) is exactly $k_a(2) > 0$, so all three diagonal values are positive. As in Proposition 12.1, congruence turns $K(2)$ into $I - zz^T$, with $\|z\|^2 = \sum_t t/k_t(2) > 1$, since every $k_t(2) < s$. It has one negative and two positive eigenvalues and no zero eigenvalue. Consequently the only quotient eigenvalue at or below two is zero. The positive root below three from Theorem 13.2 lies in $(2,3)$ and lifts to $(V-3, V-2)$. For the endpoint assertion, a positive root in $(0,3)$ can be an integer only if it equals one or two, giving the necessary condition. □

Corollary 13.4. *Let a be the minimum exponent and ρ the maximum minus minimum. Every positive three-prime exponent triple with*

$$a > \rho + 4 + \sqrt{3}(\rho + 2)$$

gives a nonintegral ideal intersection graph. In particular, $a \ge 3\rho + 8$ still suffices.

Proof. Write the ordered triple as $(a, a+d, a+\rho)$ with $0 \le d \le \rho$. The left side minus the right side of (13) is

$$a^2 - 2a\rho - 8a - 2\rho^2 - 4\rho + 4 + (\rho - d)(a + 2\rho + 2).$$

The last term is nonnegative, while the first part factors as

$$\big(a - \rho - 4 - \sqrt{3}(\rho+2)\big)\big(a - \rho - 4 + \sqrt{3}(\rho+2)\big).$$

Both factors are positive under the stated bound, which also gives $a > 4 + 2\sqrt{3} > 4$. Corollary 13.3 applies. The old linear bound is larger, since $3\rho + 8 - [\rho + 4 + \sqrt{3}(\rho+2)] = (2 - \sqrt{3})(\rho + 2) > 0$. □

The new threshold is exact for guaranteeing the strict diagonal criterion (13) from the minimum and span alone: at fixed a, ρ its left side minus right side is minimized by $b = c = a + \rho$, and it vanishes at the displayed threshold. This optimality concerns that sufficient criterion, not the entire spectral region where a root lies in $(2, 3)$. For example, $(35, 45, 45)$ and $(281, 381, 381)$ satisfy the new bound but not $a \geq 3\rho + 8$.

For example, $(20, 21, 24)$, $(38, 43, 48)$ and $(308, 350, 408)$ satisfy the span criterion and lie outside the four fixed-gap families of Theorem 12.2. The argument allows an unbounded span and uses no exponent scan or modular endpoint certificate. The necessary endpoint-zero condition is not sufficient for integrality: those triples may still have other noninteger quotient eigenvalues.

14 Arithmetic obstructions from the complement quotient

The uniform low-root bound rules out common exponent divisors greater than two. Reduction of the characteristic polynomial gives further infinite congruence classes, even when another quotient root is an integer.

Theorem 14.1. *Every positive three-prime exponent triple with* $\gcd(a, b, c) \geq 3$ *gives a nonintegral ideal intersection graph.*

Proof. Put $d = \gcd(a, b, c)$. All entries of the six-support complement quotient C are integer combinations of the exponents and their pairwise products, so C/d is an integer matrix. Every rational eigenvalue of an integer matrix is an integer, by the rational-root theorem for its monic characteristic polynomial. Therefore every integer eigenvalue of C is a multiple of d.

Order $a \leq b \leq c$. If $a = 3$, Theorem 11.2 applies. Otherwise $a \geq 4$, and Theorem 13.2 supplies an eigenvalue $0 < \mu < 3$. No positive multiple of $d \geq 3$ lies in this interval. Hence μ and its graph lift $V - \mu$ are noninteger. □

In particular, if m has exactly three distinct prime factors, then G_{m^d} is not Laplacian integral for every $d \geq 3$.

Theorem 14.2. *A positive three-prime exponent triple gives a nonintegral ideal intersection graph under either of the following conditions:*

1. *all three exponents are odd;*
2. *there is an odd prime ℓ and a nonzero residue t such that $a \equiv b \equiv c \equiv t \pmod{\ell}$ and $t^2 + 8t + 4$ is a quadratic nonresidue modulo ℓ.*

In particular, common residues $1, 3, 4$ *modulo* 5*, or* $1, 2, 4, 5$ *modulo* 7*, imply nonintegrality.*

Proof. The monic integer quintic $h_C(x) = \det(xI - C)/x$ would split into linear factors modulo every prime if all its roots were integers. A nonlinear irreducible factor in one reduction therefore supplies a noninteger real complement root and a noninteger graph lift.

If the exponents are odd, reduce the general quotient modulo two:

$$h_C(x) \equiv x(x^2 + x + 1)^2 \pmod{2}.$$

The quadratic has no root in $\mathbb{F}_2$, proving the first assertion. For a common residue t, the equal-exponent identity gives

$$h_C(x) \equiv (x - t^2 - t)(x^2 - (t^2 + 4t)x + 3t^2)^2 \pmod{\ell}.$$

The quadratic discriminant is $t^2(t^2+8t+4)$. For odd ℓ and nonzero t, the assumed nonresidue makes the quadratic irreducible. Modulo five the discriminant factors for $t = 1, 3, 4$ are $3, 2, 2$, respectively; the squares are $0, 1, 4$. Modulo seven the corresponding values for $t = 1, 2, 4, 5$ are $6, 3, 3, 6$, outside the square residues. □

These conditions allow unequal exponents with arbitrarily large ratios or differences. For example, $(11, 16, 21)$, $(13, 18, 23)$ and $(14, 19, 24)$ have gcd one and mixed parity, but meet the modulo-five criterion. The other parity reductions are x^5 for three even exponents and $x^2(x+1)^3$ for one or two odd exponents. Their complete splitting does not establish integrality.

The arithmetic arguments do not assume that the smallest positive quotient root is noninteger. Scaling also yields an obstruction for even exponents whose binary reductions split completely.

Theorem 14.3. *Every positive three-prime exponent triple with $a \equiv b \equiv c \equiv 2 \pmod 4$ gives a nonintegral ideal intersection graph.*

Proof. Write $(a, b, c) = (4u+2, 4v+2, 4w+2)$, set $Q = C/2$, and put $f(x) = \det(xI - Q)/x$. The matrix Q is integral. If all eigenvalues of C were integers, those of Q would be rational and hence integers by the rational-root theorem.

Modulo two, Q is block lower triangular. Its upper block has zero diagonal and all off-diagonal entries one, and its lower block is I_3. Consequently

$$\det(xI - Q) \equiv x(x+1)^5 \pmod 2, \qquad f(x) \equiv (x+1)^5 \pmod 2.$$

Thus all five hypothetical integer roots of f are odd. For every odd integer t, their five factors $t - \lambda_i$ would give $32 \mid f(t)$. On the other hand, expansion of the general scaled quotient gives the coefficientwise polynomial congruence

$$f(2y+1) \equiv 8(u+v+w+y-1) \pmod{32}.$$

In particular $f(3) - f(1) \equiv 8 \pmod{32}$, contradicting the required divisibility of both values. A noninteger complement eigenvalue and its noninteger graph lift follow. □

Corollary 14.4. *If the three positive exponents have the same 2-adic valuation, their ideal intersection graph is not Laplacian integral.*

Proof. At valuation zero, Theorem 14.2 applies. At valuation one, Theorem 14.3 applies. At valuation at least two, their common divisor is at least four, so Theorem 14.1 applies. □

For example, $(10, 14, 18)$, $(14, 22, 34)$ and $(18, 26, 42)$ have gcd two and lie below the linear even-tail cutoff of the following section. The new obstruction applies regardless of the minimum exponent or ratios. The necessary endpoint-zero condition of Corollary 13.3 remains insufficient: the already settled boundary triple $(9, 9, 136)$ has $h_C(1) = 0$ and other noninteger quotient roots. The remaining three-prime problem has gcd at most two, at least one even exponent, and lies outside the congruence classes above as well as the earlier proved regions.

14.1 Mixed-parity residue classes

Shifted reductions of the complement quintic organize the mixed-parity restrictions in one statement. The exponent letters in the residue rows identify roles, independently of their increasing order.

Theorem 14.5. *For positive three-prime exponents, the following conclusions hold for every permutation of the indicated residue roles.*

1. *Each of the classes*

$$(1,3,2),\ (1,7,2),\ (3,7,2),\ (3,5,6),\ (3,7,6),\ (5,7,6) \pmod 8$$

is nonintegral. The classes $(3,3,2)$ *and* $(3,0,0)$ *modulo four are also nonintegral. These exclusions have no size or ratio bound.*

2. *For roles* $a \equiv 1$, $b \equiv 3$, $c \equiv 2 \pmod 4$, *an integral graph requires*

$$a + c \equiv b(b+2) \pmod{128}, \qquad b \equiv 11 \text{ or } 15 \pmod{16}.$$

With $u = (a-1)/4$ *and* $t = (a + c - b(b+2))/128$, *it further requires*

$$\begin{cases} t + u + v \equiv 1 \pmod 2, & b = 16v + 11, \\ t + v \equiv 1 \pmod 2, & b = 16v + 15. \end{cases}$$

3. *For either* $(1,3,2)$ *or* $(3,3,0)$ *modulo four, if* $m = \min\{a,b,c\} \geq 4$ *and* $M = \max\{a,b,c\}$, *an integral graph requires*

$$h_C(2) = 0, \qquad M < 7m - 16 + \frac{40}{m+2}.$$

Every violation of a necessary condition proves nonintegrality.

Proof. The case table and complete shifted-polynomial arguments are in Appendix B. The common mechanism is that an integer-root factorization of h_C forces its shifted binary cubics to split into linear factors; their residue and derivative conditions give the first two parts. For the last part, the odd roots would be three modulo four, excluding one. The positive root below three from Theorem 13.2 must then equal two, and Theorem 15.1 gives the maximum bound. □

For one worked example, take the residue roles $(a,b,c) = (19,27,10)$, so $(a,b,c) = (4u+3, 4v+3, 4w+2)$ with $(u,v,w) = (4,6,2)$. The shifted quintic satisfies

$$h_C(2y+1)/8 \in \mathbb{Z}[y], \qquad h_C(2y+1)/8 \equiv y^3 \pmod 2.$$

Integer roots would therefore force all three odd roots to be one modulo four and $16 \mid h_C'(1)$. But $h_C'(1) \equiv 8(u+v+1) \equiv 8 \pmod{16}$, a contradiction. The same identity proves the whole $(3,3,2)$ class in Appendix B. The arithmetic conditions in this theorem do not assert integral examples. Corollary 15.9 below completes the remaining $(1,3,2)$ and $(3,3,0)$ classes modulo four.

14.2 Two further congruence classes modulo five

Proposition 14.6. *Every positive exponent triple that is a permutation of* $(1,1,2)$ *or* $(1,4,4)$ *modulo five gives a nonintegral ideal intersection graph.*

Proof. The monic integer quintic $h_C(x) = \det(xI - C)/x$ is symmetric in the exponents. Reducing the six-support quotient gives the complete identities

$$\begin{aligned} (1,1,2): &\quad h_C(x) \equiv (x-1)^2(x-3)(x^2+3x+4), \\ (1,4,4): &\quad h_C(x) \equiv (x-1)^2(x^3+x^2+4x+3) \pmod 5. \end{aligned}$$

The quadratic discriminant is 3, a nonsquare modulo five. The cubic takes the values $3,4,3,1,4$ at $0,1,2,3,4$, respectively, and hence is irreducible over $\mathbb{F}_5$. An integer quotient spectrum would make h_C split into linear factors modulo every prime, contradicting either reduction. The complement and universal-class lifts give the graph conclusion. No size, parity or endpoint-zero hypothesis is needed. □

On the equality subfamily $c = b(b+2) - a$, this gives a splitting obstruction that a square-discriminant congruence can miss. If $b \equiv 1 \pmod 5$, put $q = ac \equiv a(3-a)$. The curve equation in Remark 15.5 becomes $G \equiv 2(q-1)(q-2) \pmod 5$. Thus $G = 0$ permits precisely $a \equiv 1, 2, 4 \pmod 5$, giving one of the two excluded exponent classes. For $q \equiv 1$, the irreducible cubic has discriminant 1 modulo five. Consequently an integral equality spectrum requires $b \not\equiv 1 \pmod 5$, even though the discriminant-square test alone permits one excluded row.

14.3 A local discriminant obstruction on the equality curve

Proposition 14.7. *Let $a \le b \le c$ be positive integers with $c = b(b+2) - a$ and $G(a,b) = 0$, where G is the equality equation in Remark 15.5. If an odd prime p divides b, $a^2 \equiv 1 \pmod p$, and 13 is a quadratic nonresidue modulo p, then the ideal intersection graph is Laplacian nonintegral. In particular, an integral equality spectrum requires $7 \nmid b$ and $11 \nmid b$.*

Proof. Put $q = ac$ and let $\Delta(q,b)$ be the discriminant of the explicit cubic P_3 in Section 18. On $G = 0$, $h_C = (x-1)(x-b)P_3$, so an integer quotient spectrum would make Δ an integer square. For $v = -1 + 2b + 3b^2$, expansion of the displayed coefficient formulas gives identities in $\mathbb{Z}[b,q]$:

$$\begin{aligned} G(v,b) &= b^3(-b^4 - 8b^3 + 8b^2 + 20b - 7), \\ \Delta(v,b) &= 13b^2 + b^3 R(b), \qquad R \in \mathbb{Z}[b], \\ G(q,b) - G(v,b) &= (q-v)\big[(3b-1)(q+v) + b(b^2+4b-1)\big]. \end{aligned}$$

The assumption $a^2 \equiv 1$ gives $q \equiv -1 \pmod p$. Write $e = v_p(b) \ge 1$. The bracket in the third identity is 2 modulo p, so it is a unit. Since $G(q,b) = 0$, the first and third identities imply $q \equiv v \pmod{p^{3e}}$. The discriminant is an integer polynomial, and therefore

$$\Delta(q,b) \equiv 13b^2 \pmod{p^{3e}}.$$

Because 13 is a nonresidue, $p \neq 13$. Hence $v_p(\Delta) = 2e$, and the unit Δ/p^{2e} is congruent to $13(b/p^e)^2$ modulo p. This is a nonresidue, contradicting square discriminant. The complement and universal-class lifts give the graph conclusion.

For $p = 7$ or 11, reducing the curve at $p \mid b$ gives $q^2 \equiv 1$ and hence $a^4 \equiv 1$. Since -1 is a nonresidue at both primes, $a^2 \equiv 1$ follows. Moreover 13 reduces to the nonresidues 6 and 2, respectively, so the first assertion applies. □

The obstruction resolves a repeated-root reduction: on the branch $q \equiv -1$, one has $P_3 \equiv x(x+1)^2 \pmod p$ and $\Delta \equiv 0 \pmod p$. Normalizing by $p^{2v_p(b)}$ reveals the nonsquare unit that the reduction modulo p misses. At $p = 5$, this excludes $(a,b) \equiv (1,0), (4,0)$ among the previously compatible equality rows. The argument is uniform in p and its valuation and requires no finite exponent base.

Proposition 14.8. *At an integer equality point $c = b(b+2) - a$, $q = ac$, $G = 0$, let p be an odd prime dividing b. If $q \equiv 1 \pmod p$, then P_3 has three distinct roots in $\mathbb{Z}_p$. If $q \equiv -1 \pmod p$ and $p \neq 13$, then P_3 splits into three distinct $\mathbb{Z}_p$ roots exactly when 13 is a quadratic residue modulo p. If 13 is a nonresidue in this negative branch, it has exactly one $\mathbb{Q}_p$ root. In the split negative branch, the two roots reducing to -1 differ by valuation $v_p(b)$. Moreover, for every fixed $b \in p\mathbb{Z}_p$, each residue a_0 with $a_0^4 \equiv 1 \pmod p$ lifts uniquely to a curve point $a \in a_0 + p\mathbb{Z}_p$ on $G = 0$.*

Proof. At $p \mid b$, the curve gives $q^2 \equiv 1$ and $q \equiv -a^2$. For fixed $b \in p\mathbb{Z}_p$, the polynomial $G(a[b(b+2) - a], b)$ reduces to $1 - a^4$ and has derivative $-4a^3$ modulo p. At $a_0^4 \equiv 1$, this derivative is a unit; Hensel's lemma gives the asserted unique local curve point. On the positive branch, $a^2 \equiv -1$

and $P_3 \equiv x(x^2+1)$. Its roots $0, a, -a$ have derivatives $1, -2, -2$ modulo p, respectively. They are distinct simple roots, so Hensel's lemma lifts all three to $\mathbb{Z}_p$.

On the negative branch, $P_3 \equiv x(x+1)^2$, and the simple root 0 lifts uniquely to $d_0 \in p\mathbb{Z}_p$. Polynomial division gives

$$\begin{aligned} P_3(x) &= (x - d_0)Q(x), \\ Q(x) &= x^2 + (d_0 - A_3)x + d_0^2 - A_3 d_0 + B_3, \\ \delta_Q &= A_3^2 + 2A_3 d_0 - 3d_0^2 - 4B_3. \end{aligned}$$

Here $P_3'(d_0) \equiv 1 \pmod{p}$, $Q \equiv (x+1)^2$, and $\Delta = P_3'(d_0)^2 \delta_Q$. Proposition 14.7 and its anchor identities give $v_p(\Delta) = 2e$ and normalized unit $13(b/p^e)^2 \pmod{p}$, where $e = v_p(b)$ and $p \neq 13$. An odd-prime-adic unit is a square if and only if its reduction is a nonzero square, by Hensel's lemma for $z^2 - u$. Thus δ_Q is a square in $\mathbb{Q}_p$ if and only if 13 is a residue modulo p. Since 2 is a unit, the quadratic formula then supplies two integral roots; otherwise Q is irreducible over $\mathbb{Q}_p$. In the split case the roots reduce to -1, and their difference is a square root of δ_Q, of valuation e. Their differences from d_0 are units. All three roots are distinct because $\Delta \neq 0$. □

The surviving $(a, b) \equiv (2, 0), (3, 0) \pmod{5}$ rows lie on the positive branch. At every actual equality point in either row, the cubic splits over $\mathbb{Z}_5$ and modulo every power of five. Higher five-power splitting tables cannot exclude these rows. For every fixed $b \in 5\mathbb{Z}_5$, both residues $a_0 = 2, 3$ also lift to local equality points with split cubics. Thus simultaneous curve and splitting solutions exist in both rows modulo every power of five. These local points leave integer equality and global integer splitting as separate questions. Only the negative branch at $p = 13$ is left outside the odd-prime-divisor classification.

Proposition 14.9. *At an integer equality point $c = b(b+2) - a$, $q = ac$, $G = 0$, suppose $13 \mid b$ and $q \equiv -1 \pmod{13}$, equivalently $a^2 \equiv 1 \pmod{13}$. If $b \not\equiv 104 \pmod{169}$, then P_3 does not split over $\mathbb{Q}_{13}$ and the ideal intersection graph is Laplacian nonintegral. Thus an integral equality spectrum on this negative branch requires $b \equiv 104 \pmod{169}$, in particular $13^2 \nmid b$.*

Proof. Let $e = v_{13}(b) \geq 1$ and refine the previous anchor to $w = -1 + 2b + 3b^2 + \frac{7}{2}b^3$. Substitution into the explicit curve and cubic-discriminant formulas gives identities in $\mathbb{Z}[1/2][b]$:

$$\begin{aligned} G(w, b) &= b^4 \left(\frac{143}{4} b^3 + \frac{185}{4} b^2 + 43b - \frac{37}{2} \right), \\ \Delta(w, b) &= 13b^2 - 18b^3 + b^4 T(b), \qquad T \in \mathbb{Z}[1/2][b]. \end{aligned}$$

The curve difference identity of Proposition 14.7 holds with w in place of v. Since $q \equiv w \equiv -1 \pmod{13}$, its bracket is 2 modulo 13. The coefficients are integral at 13 and $G(q, b) = 0$, so $q \equiv w \pmod{13^{4e}}$. Polynomial divisibility of the discriminant difference then gives

$$\Delta(q, b) \equiv 13b^2 - 18b^3 \pmod{13^{4e}}.$$

If $e \geq 2$, the displayed terms have valuations $2e + 1$ and $3e$, and the remainder has valuation at least $4e$. Hence $v_{13}(\Delta) = 2e + 1$, which is odd, and Δ is nonsquare.

If $e = 1$, write $b = 13u$ with u a unit. The congruence becomes

$$\Delta \equiv 13^3 u^2 (1 - 18u) \pmod{13^4}.$$

Unless $u \equiv 8 \pmod{13}$, the factor $1 - 18u$ is a unit and $v_{13}(\Delta) = 3$, again odd. The remaining residue is precisely $b \equiv 104 \pmod{169}$. A split cubic has square discriminant, so the nonsquare cases exclude splitting over $\mathbb{Q}_{13}$ and over the integers. The equality factorization and complement and universal-class lifts give the graph conclusion. □

On the remaining negative-branch class $b \equiv 104 \pmod{169}$, the discriminant has valuation at least four; its square status remains undecided by this congruence. The positive branch at 13 retains the three simple local roots of Proposition 14.8.

15 Endpoint surfaces and their reductions

The positive quotient root in $(0, 3)$ reduces possible integer spectra to the two surfaces $h_C(1) = 0$ and $h_C(2) = 0$. Their size, divisor and finite-certificate restrictions are collected together.

Theorem 15.1. *For positive ordered exponents $a \le b \le c$, put $R(a) = 7a - 16 + 40/(a + 2)$. Then:*

1. *If $a \ge 4$, then $c \ge R(a)$ gives $h_C(2) > 0$, so $h_C(2) = 0$ requires $c < R(a)$. In this range, all-even triples and permutations of $(1, 3, 2)$ or $(3, 3, 0)$ modulo four are nonintegral when $c \ge R(a)$. For $a \ge 8$, $c \ge 7a - 12$ suffices.*

2. *For $a, b \ge 4$, an endpoint zero requires, respectively,*

$$h_C(1) = 0 \implies c \mid (a-1)(b-1)(a+b-1)(ab+a+b-1),$$
$$h_C(2) = 0 \implies c \mid 2(a-2)(b-2)(a+b-2)(ab+a+b-2).$$

3. *Every triple with $b \le 15$ is nonintegral. Every permutation of $(2, 2, 2)$ or $(1, 2, 2)$ modulo three is also nonintegral.*

Proof. The full endpoint polynomial and positivity proof are in Appendix C. Its constant terms give the divisor conditions. For the remaining ordered pairs $4 \le a < b \le 15$, the complete certificate checks all 132 endpoint cubics over 66 pairs: 8 527 positive constant-term divisors, including every one of the 7 585 divisors above b, give no endpoint zero. Repeated exponents and minima at most three were already settled. Theorem 13.2 then proves the middle-exponent claim. The two modulo-three classes have both endpoint values nonzero, as shown by the reductions in Appendix C. □

Finitely many endpoint-value congruences alone do not eliminate every candidate, even with the listed gcd, valuation and geometric restrictions: Appendix C gives explicit CRT constructions below both maximum bounds and outside the sufficient inertia exclusion. This limits these endpoint tests; another-root obstruction or further arithmetic information can still exclude the candidates. The following arguments sharpen endpoint one and complete the endpoint-two alternative.

15.1 The spectrum on the endpoint-one surface

Principal interlacing separates the five positive quotient roots into two below $a + b$ and three above $a + b + c$, apart from the boundary $(2, 2, 2)$. On endpoint one, this gap isolates the smallest residual root from the other three.

Theorem 15.2. *Let $2 \le a \le b \le c$ be real and put $s = a + b + c$. The ordered eigenvalues of the six-support complement quotient, counted with multiplicity, satisfy*

$$0 = \lambda_1 < \lambda_2 \le \lambda_3 < a + b < s \le \lambda_4 \le \lambda_5 \le \lambda_6.$$

Equality $\lambda_4 = s$ holds exactly at $(a, b, c) = (2, 2, 2)$. If also $h_C(1) = 0$, then

$$b + c \ge 2a^2 - 2a + 1,$$

with equality exactly when $b = a$ and $c = (a-1)(2a-1)$. If in addition $a > 2+\sqrt{3}$, root one is simple and is the smallest positive quotient root; all four other quotient roots are strictly greater than a. Their ordered roots satisfy

$$a < \mu_1 < \min\{c, a+b\}, \qquad s < \mu_2 \le \mu_3 \le \mu_4.$$

Proof. Let $S = W^{1/2}CW^{-1/2}$, with $W = \operatorname{diag}(a, b, c, ab, ac, bc)$. It is symmetric positive semidefinite with a one-dimensional zero kernel. The principal block on supports $3, 12, 13, 23$ is

$$\begin{pmatrix} ab+a+b & -\sqrt{abc} \\ -\sqrt{abc} & c \end{pmatrix} \oplus (b) \oplus (a).$$

The two-dimensional block has polynomial $Q(x) = x^2 - (ab+a+b+c)x + c(a+b)$, with $Q(a+b) = -ab(a+b) < 0$. Exactly three eigenvalues of this four-dimensional block lie below $a+b$; interlacing gives $\lambda_3 < a+b$.

The principal block on singleton supports $1, 2$, shifted by s, is

$$\begin{pmatrix} bc-a & -\sqrt{ab} \\ -\sqrt{ab} & ac-b \end{pmatrix}.$$

Its first leading minor is $bc - a \ge a(a-1) > 0$. Its determinant is $c(abc - a^2 - b^2)$, where

$$abc - a^2 - b^2 = ab(c-b) + (a-1)(b-a)(b+a) + (a-2)a^2.$$

This is positive except at $(2,2,2)$. Away from that point, interlacing gives $\lambda_5, \lambda_6 > s$. Put $p = ab+ac+bc$, $r = abc$. Direct expansion gives $h_C(s) = -rsV$, where

$$V = r(s+3) - sp = a^2(bc-b-c) + b^2(ac-a-c) + c^2(ab-a-b) > 0.$$

The three brackets are nonnegative and vanish simultaneously only at $(2,2,2)$. Thus no eigenvalue equals s. At least three lie below it and at most four. The negative determinant $sh_C(s)$ requires an odd number below it, counting multiplicity; hence exactly three lie below it and $\lambda_4 > s$. At the boundary the characteristic polynomial is $x(x-6)(x^2-12x+12)^2$, giving $\lambda_4 = s = 6$.

Now suppose $h_C(1) = 0$. Put $q = b+c$, $z = bc$ and $K = 4a^2 - 2a + 1$. The generic determinant is

$$\begin{aligned} h_C(1) &= (q-K)z^2 - a(a-1)(q+2a-1)z + P_a(q), \\ P_a(q) &= (a^2-1)q^3 + (a^3-a^2-3a+3)q^2 - 3(a-1)^2 q - (a-1)^3. \end{aligned}$$

Since $(b-a)(c-a) \ge 0$, we have $z \ge z_0 = a(q-a)$. When $q \le Q := 2a^2 - 2a + 1$, the coefficient $q - K \le -2a^2$ is negative, and the difference between the displayed polynomial at z and at z_0 is

$$(z-z_0)\big[(q-K)(z+z_0) - a(a-1)(q+2a-1)\big] \le 0,$$

strictly unless $z = z_0$. Set $t = q - a \ge a$. At z_0 the value factors as

$$\big[(a-1)(2a-1) - t\big]B(a,t),$$

where $B = a^3 - 2a^2t^2 + a^2t + a^2 + 3at - 3a + t^2 - 2t + 1$. For $t = a+u$, with $u \ge 0$, the identity

$$\begin{aligned} -B(a, a+u) = (2a^2-1)u^2 &\\ + \big[(a-2)(4a^2+7a+9) + 20\big]u &\\ + (a-2)\big[2a^3 + a(2a-1) + 3\big] + 5 & \end{aligned}$$

has strictly positive coefficients for every $a \geq 2$, so $B < 0$. Thus $q < Q$ gives $h_C(1) < 0$. At $q = Q$, equality requires $z = z_0$, hence $b = a$ by ordering, and $c = (a-1)(2a-1)$. The factorization also verifies this equality case.

Now assume also $a > 2 + \sqrt{3}$. The pair-support principal block is $\mathrm{diag}(c, b, a)$. The sum bound implies $b + c - (a^2 + 2a) \geq a^2 - 4a + 1 > 0$, so

$$h_C(a) = -a^2b^2c^2(a^2 + 2a - b - c) > 0.$$

No eigenvalue equals a. The positive determinant $\det(aI - S) = ah_C(a)$ forces an even number of eigenvalues below a, counted with multiplicity. There are at least two, zero and one, and at most three by interlacing. There are therefore exactly two: zero and a simple one. All four remaining roots exceed a.

The pair-support block gives $\lambda_3 \leq c$. The identity $h_C(c) = -a^2b^2c^2(c^2 + 2c - a - b) < 0$ excludes equality. Together with the global total-sum gap, this proves the residual bounds.

The real cutoff is sharp. On the ordered endpoint-one family $b = a$, $c = (a-1)(2a-1)$, we have $h_C(a) = a^4(a-1)^2(2a-1)^2(a^2-4a+1)$. At $a = 2+\sqrt{3}$ a residual root equals a. For $2 \leq a < 2+\sqrt{3}$, the determinant at a is negative, so the same interlacing and parity argument gives three roots below a; a residual root lies below a. □

Corollary 15.3. *For integer exponents $4 \leq a \leq b \leq c$ on $h_C(1) = 0$, define $F(x) = h_C(x)/(x-1)$ and $s = a+b+c$, $p = ab+ac+bc$, $r = abc$. Then F is a monic integer quartic with $F(0) = rs(p+s) > 0$. Its ordered real roots satisfy*

$$a < \mu_1 < U := \min\{c, a+b\}, \qquad s < \mu_2 \leq \mu_3 \leq \mu_4.$$

Writing $F = x^4 - Ax^3 + Bx^2 - Dx + N$, with $N = rs(p+s)$ and $A = p + 3s - 1$, put $R = 2/s + 1/(A - U - 2s)$. Then

$$\frac{N}{D} < \mu_1 < \frac{N}{D - NR}, \qquad D - NR > 0.$$

An integer spectrum therefore requires a positive divisor d of $rs(p+s)$ with

$$\max\{a, N/D\} < d < \min\{U, N/(D - NR)\}, \qquad F(d) = 0.$$

The other three integer roots would each be at least $s + 1$.

Proof. The determinant gives $h_C(0) = -rs(p+s)$, so division by $x - 1$ retains the stated nonzero constant. Apply the rational-root theorem to the smallest quartic root and apply Theorem 15.2 to all four roots.

For three numbers $x_i > s$ of sum T, convexity of $1/(s+y)$ on the simplex $y_i \geq 0$, $\sum y_i = T - 3s$, gives $\sum 1/x_i < 2/s + 1/(T - 2s)$: the closed-simplex maximum has two $y_i = 0$, whereas all three are positive here. For the larger roots, $T = A - \mu_1 > A - U$, so their reciprocal sum lies in $(0, R)$. Vieta gives $D/N = \sum_{i=1}^{4} 1/\mu_i$, yielding the stated bounds.

To check positivity before inverting, $A - U - 2s = p + s - 1 - U > s$, since $p \geq 4c + ab > U + 1$. Thus $R < 3/s$. The determinant coefficients give $D = r^2 + (s^2 - s + 1)r + s^2p + p^2 + (s-1)^3$ and

$$D - 3N/s = r(r - 4s) + r(s^2 - 3p + 1) + s^2p + p^2 + (s-1)^3 > 0.$$

Here $r/s \geq 16/3 > 4$ and $s^2 - 3p = ((a-b)^2 + (a-c)^2 + (b-c)^2)/2 \geq 0$. Hence $D - NR > D - 3N/s > 0$. □

All four quartic roots exceed a; the upper window in the corollary applies to their smallest root. For example, $(8, 8, 105)$ has the quartic factor

$$(x - 968)(x^3 - 1138x^2 + 177145x - 1566600),$$

so another root can exceed both c and $a + b$. Testing the divisor window is finite for each fixed triple. A surviving candidate still requires checking the other factors, and the exponent domain is unbounded.

Remark 15.4. The generic residual $[h_C(x) - xh_C(1)]/(x-1)$ is positive on $[0, 4]$ for all real $a, b, c \geq 8$, including off the endpoint-one surface. A separate complete 350-term positive identity after $a = 8+m$, $b = a + t$, $c = b + u$, $x = 4v/(1 + v)$ verifies that statement, including $x = 4$. The shorter proof above suffices for the endpoint-one spectral theorem; the generic residual calculation is retained in the verification note.

Remark 15.5 (The equality curve). On $c = b(b + 2) - a$, put $t = b(b + 2)$ and $q = a(t - a) = ac$. Then $h_C(1) = -(b - 1)G(a, b)$, where the plane septic is

$$G = (3b - 1)q^2 + b(b^2 + 4b - 1)q - (b^2 + 2b - 1)(b^2 + 3b - 1)(b^3 + 3b^2 + 3b - 1).$$

Its smooth projective model is geometrically irreducible of genus ten. Indeed, the involution $a \mapsto t - a$ has quotient $y^2 = D(b)$, where D is the squarefree degree-eight discriminant of the displayed quadratic in q, so the quotient has genus three. Recovering a adjoins $\sqrt{t^2 - 4q}$. The norm of $t^2 - 4q$ has nine distinct simple zeros away from the quotient branch points and one simple pole at $b = 1/3$. This nonsquare norm makes the cover connected. Above these points the cover has ten simple branches; at the two quotient points at infinity, $t^2 - 4q$ has even pole order four. Riemann–Hurwitz gives $2g-2 = 2(2\cdot3-2)+10 = 18$. The explicit norm and squarefreeness identities are retained in the curve verification note. By Siegel's theorem [14, 18], this affine equality curve has finitely many integer points. Their effective determination and the classification of admissible exponent triples remain open.

Proposition 15.6. *For real $2 \leq a \leq b$ and $c = b(b + 2) - a$, put $s = a + b + c$. The ordered eigenvalues of the six-support complement quotient satisfy*

$$0 = \lambda_1 < \lambda_2 < \lambda_3 = b < a + b < s < \lambda_4 \leq \lambda_5 \leq \lambda_6,$$

and b is simple. If also $h_C(1) = 0$, then $\lambda_2 = 1$ and $F(x) = (x - b)P_3(x)$, where the monic cubic P_3 has all three roots strictly greater than $s = a + b + c$.

Proof. Use the symmetric similarity S from Theorem 15.2. Here $c - b = b^2 + (b - a) > 0$, so the exponents are ordered and the exceptional global-gap boundary $(2, 2, 2)$ is absent. Every proper principal block is positive definite: S is positive semidefinite with a one-dimensional zero kernel whose vector has every coordinate nonzero. Delete singleton support 2. The five-support block B is the direct sum of the scalar b on support 13 and the block T on $1, 3, 12, 23$. Its characteristic polynomial is

$$\begin{aligned}\det(xI - T) &= (x - b)K(x),\\ K(x) &= x^3 - b(b^2 + 4b + 5)x^2\\ &\quad + \big[ac(b^2 + 1) + b^2(b + 2)(b^2 + 3b + 3)\big]x - abc(b + 3).\end{aligned}$$

The three roots of K are positive and real. Moreover,

$$K(b) = b(b + 1)\big[a(b - a)(b - 2) + ab(b - 2)(b + 1) + b^2(b + 1)(b + 2)\big] > 0.$$

The first two bracketed terms are nonnegative for $b \geq a \geq 2$, and the third is strictly positive, including at $a = b = 2$. The pair-support block on $12, 23$ is $\mathrm{diag}(c, a)$, so the largest eigenvalue of T is at least $c > b$. A positive monic cubic value at b requires one or three roots below b; the largest root rules out three. Thus B has $\beta_2 = \beta_3 = b$, and interlacing gives $\beta_2 \leq \lambda_3 \leq \beta_3$, hence $\lambda_3 = b$.

The kernel of $B - bI$ has dimension two. Its support-13 unit vector couples to the deleted singleton by $-\sqrt{abc} \neq 0$. Projecting the full eigenvector equation onto that vector forces the deleted coordinate to vanish. The remaining eigenvectors lie in $\ker(B - bI)$ and satisfy one nonzero coupling condition. Their space is one-dimensional, proving simplicity and $\lambda_2 < b < \lambda_4$.

The global gap in Theorem 15.2 gives $\lambda_4 > s$. If $h_C(1) = 0$, then $1 < b = \lambda_3$ and all later roots exceed s, so the root 1 must be λ_2. It is simple because $\lambda_1 = 0 < 1 < b = \lambda_3$. Division by the simple roots $1, b$ yields the stated cubic. □

This proposition narrows the equality splitting problem: for integer endpoint-one equality points, all three integer roots of P_3 would have to exceed s. The global spectral verification note records the determinant identities and exact controls.

15.2 Completion of the endpoint-two alternative

Two geometric bounds confine the endpoint-two surface to a rectangle where positive polynomial identities give a second noninteger root at large minima. A complete finite base handles the smaller distinct triples.

Lemma 15.7. *Put $\alpha = (7 + \sqrt{13})/3$. For ordered real exponents $\alpha \leq a \leq b \leq c$, the equation $h_C(2) = 0$ requires $b \leq 2a - 2$, with equality exactly at $a = \alpha$, $b = c = 2\alpha - 2$. In particular, $b < 2a - 2$ if $a > \alpha$. If $a \geq 8$, it also requires $c < 9a/4 - 8$.*

Proof. First put $b = 2a - 2 + u$ and $c = b + v$, where $u, v \geq 0$. Define

$$\begin{aligned}
A_0(u) &= 2u^2 + 7(a-2)u + 2(3a^2 - 14a + 12),\\
B_0(u) &= (a+2)u^3 + (4a^2 + 10a - 12)u^2\\
&\quad + (4a^3 + 25a^2 - 48a + 12)u + 2(13a^3 - 28a^2 + 8a + 8),\\
D_0(u) &= 2(a-2)(9a^3 + 24a^2 - 56a + 16)\\
&\quad + (33a^3 + 20a^2 - 208a + 136)u + (20a^2 + 23a - 62)u^2 + 4(a+2)u^3,\\
E_0(u) &= 2(3a^3 - a^2 - 8a + 4) + (5a^2 + 3a - 10)u + (a+2)u^2.
\end{aligned}$$

Direct expansion of the quotient determinant gives

$$h_C(2) = A_0 B_0 + \tfrac{1}{2}(A_0 B_0)' v + D_0 v^2 + E_0 v^3,$$

where the derivative is in u. At $a = 3 + m$, all coefficients of B_0, D_0, E_0 are positive for $m \geq 0$. For $a \geq \alpha$, $A_0(0) = 6(a - \alpha)(a - (7 - \sqrt{13})/3) \geq 0$ and $A_0'(u) = 4u + 7(a-2) > 0$. Hence $A_0 \geq 0$, vanishing only at $a = \alpha, u = 0$, and $(A_0 B_0)' > 0$. The identity is positive unless $a = \alpha, u = v = 0$, where its zero proves the equality case and sharpness.

For $a \geq 8$ and $a \leq b < 2a - 2$, parameterize the proposed maximum tail by

$$a = 8 + m, \qquad b = \frac{a + (2a-2)t}{1+t}, \qquad c = \frac{9a}{4} - 8 + v, \qquad m, t, v \geq 0.$$

The inverse $t = (b-a)/(2a-2-b)$ has positive denominator. The exact identity is

$$64(1+t)^3 h_C(2) = 9m^6(1+2t)\big[14(1-t)^2 + 5t + t^2\big] + 308m^5 t(t-1)^2 + P(m,t,v).$$

The first two terms are nonnegative. The complete polynomial P, listed in Appendix D, has 84 strictly positive coefficients and constant 3 014 656. Hence the right side is positive, and so is $h_C(2)$. Together with the first bound, this proves $c < 9a/4 - 8$ on the zero surface. □

Theorem 15.8. *For ordered real $40 \le a \le b \le c$ with $h_C(2) = 0$,*

$$h_C(4) > 0 > h_C(5).$$

Consequently the quotient has a noninteger root in $(4,5)$. For every positive integer exponent triple with $h_C(2) = 0$, the quotient spectrum is nonintegral. The integer conclusion uses the complete finite base specified below and the earlier small-minimum certificates.

Proof. Write $s = a+b+c$, $p = ab+ac+bc$ and $r = abc$, and set

$$D_4 = h_C(4) - 3h_C(2), \qquad D_5 = 4h_C(2) - h_C(5).$$

The full identities in Appendix D give the following positive expansions. For $a = 8+m, b = 8+t, c = 8+u$, D_5 has 44 strictly positive coefficients and constant 4 629 843. Thus $D_5 > 0$ for arbitrary $a, b, c \ge 8$.

Lemma 15.7 confines an endpoint-two zero at $a \ge 40$ to $a \le b < 2a$ and $a \le c < 9a/4$. Put

$$a = 40 + m, \qquad b = \frac{a(1+2t)}{1+t}, \qquad c = \frac{a(4+9u)}{4(1+u)}, \qquad m, t, u \ge 0.$$

The inverse substitutions are $t = (b-a)/(2a-b)$ and $u = 4(c-a)/(9a-4c)$, with positive denominators. The polynomial $64(1+t)^3(1+u)^3 D_4$ has 112 strictly positive coefficients and constant 611 305 472, all displayed in the appendix. Therefore $D_4 > 0$. At $h_C(2) = 0$ these two signs give $h_C(4) > 0 > h_C(5)$; the intermediate value theorem supplies the root in $(4,5)$.

For positive integer triples, repeated exponents are nonintegral by Theorem 8.1, and minima at most seven by Corollary 12.4. The only remaining base is

$$8 \le a \le 39, \qquad a < b < 2a - 2, \qquad b < c < \frac{9a}{4} - 8.$$

The largest permitted integer c is $\lfloor (9a-33)/4 \rfloor$. This domain contains exactly 8 658 triples; its per-minimum counts are listed in Appendix D. Every endpoint value is nonzero, with smallest absolute value 8 208. The complete integer certificate records $h_C(2)$ by integer Horner evaluation and $\det(2I - C) = 2h_C(2)$ by an independent fraction-free 6×6 Bareiss determinant for every triple. Thus no fully distinct endpoint-two zero occurs in this necessary base. The real tail covers all larger minima, completing the integer conclusion. The proof does not require emptiness of the unbounded integer zero surface. □

Corollary 15.9. *Every all-even exponent triple and every permutation of $(1,3,2)$ or $(3,3,0)$ modulo four has a nonintegral ideal intersection graph, without a size, gap or ratio bound.*

Proof. Repeated triples and minima at most seven are already excluded. For the others, Theorem 13.2 gives a quotient root in $(0, 3)$. If all exponents are even, $C/2$ is an integer matrix, so an integer spectrum of C would have only even roots by the rational-root theorem. In the two mixed classes, the shifted-root argument in Theorem 14.5 forces hypothetical odd integer roots to be three modulo four. In either case the low root cannot be one, so it must be two. Theorem 15.8 contradicts integrality. The complement and universal-class lifts transfer the result to the ideal intersection graph. Exclusion of one in the mixed classes is conditional on an integer spectrum. □

16 Complete fixed-exponent families and exact diagnostics

These fixed-exponent diagnostics illustrate how the divisor reductions and the symmetric cubic complement one another. Their exact certificates also distinguish a successful cubic test from a complete spectral obstruction.

Theorem 16.1. *For every $a \in \{2, 3, 4\}$, every $b \geq 1$ and distinct primes p, q, r, the graph $G_{p^a q^a r^b}$ is not Laplacian integral.*

Proof. The cutoff values are $R(2) = 0$, $R(3) = 4/5$, and $R(4) = 2$. Theorem 7.4 settles all b for $a = 2, 3$, and all $b > 2$ for $a = 4$. At $(a, b) = (4, 1)$, $D = 145$ is nonsquare. At $(4, 2)$ the symmetric cubic is

$$f_{4,2}(x) = x^3 - 86x^2 + 2304x - 18816.$$

Its values at $0, 1, 2, 3, 4$ modulo 5 are $4, 3, 1, 4, 3$, so it has no root modulo 5. A cubic over a field without a root is irreducible. Since this cubic is monic, reduction modulo 5 proves irreducibility over $\mathbb{Q}$. Its real nonzero roots lift to noninteger graph eigenvalues. □

The second cutoff reaches the next two fixed exponents with only one remaining discriminant check in each case.

Corollary 16.2. *The same all-b conclusion holds for $a = 5, 6$. Thus it holds for every $a \in \{2, 3, 4, 5, 6\}$.*

Proof. The second cutoff gives $S(5) = 14/13$ and $S(6) = 9/5$, leaving only $b = 1$ in each case. The discriminants are 221 and 313, respectively. Since $14^2 < 221 < 15^2$ and $17^2 < 313 < 18^2$, both give irrational antisymmetric eigenvalues. Every larger b is covered by Theorem 7.5. □

Corollary 16.3. *Every positive b also gives a nonintegral spectrum for $a = 7, 8, 9$. Thus every repeated exponent $a \in \{2, 3, 4, 5, 6, 7, 8, 9\}$ is settled.*

Proof. The third cutoff gives $T(7) = 6/5$, $T(8) = 12/7$, $T(9) = 70/31$. Only four pairs remain under these bounds:

$$(a, b) = (7, 1), (8, 1), (9, 1), (9, 2).$$

Their discriminants are $421, 545, 685, 1489$, respectively, between the consecutive squares with lower square roots $20, 23, 26, 38$. Each therefore gives irrational antisymmetric eigenvalues, and Theorem 7.7 covers every larger b. □

For comparison, the complete divisor certificates before applying the linear cutoff are given in Table 5. The constants factor as $M = 40 = 2^3 \cdot 5$, $189 = 3^3 \cdot 7$, and $576 = 2^6 \cdot 3^2$. Testing every divisor $d \leq \sqrt{M}$ in (6) gives exactly the displayed admissible pairs. The corresponding cubics in Table 6 have no root modulo the listed primes. These finite certificates supply a second proof

a	M	(d,e)	b	$\sqrt{D}$
2	40	$(1,40)$	3	13
3	189	$(1,189)$	10	47
4	576	$(6,96)$	2	18
4	576	$(1,576)$	21	115

Table 5: Every admissible factor pair for $a = 2, 3, 4$.

(a,b)	$f_{a,b}(x)$	Witness prime
$(2,3)$	$x^3 - 47x^2 + 702x - 3312$	5
$(3,10)$	$x^3 - 187x^2 + 11220x - 214200$	13
$(4,2)$	$x^3 - 86x^2 + 2304x - 18816$	5
$(4,21)$	$x^3 - 485x^2 + 75492x - 3721536$	5

Table 6: Root-free modular reductions for the exceptional cubics.

using the complete exceptional list. Three pairs are in the boundary family; $(4,2)$ is the additional symmetric-block obstruction.

The separate diagnostic range $a = 2, 3, 4$, $1 \le b \le 30$ contains 90 cubics, each with a noninteger root certified in an interval between consecutive integers. The reducible cases in that range are $b = 2, 6$ for $a = 2$, $b = 3, 12$ for $a = 3$, and $b = 4, 20$ for $a = 4$. Each has a linear and an irreducible quadratic factor. These observations do not require whole-cubic irreducibility. Indeed, for every $a \ge 1$,

$$f_{a,a}(x) = (x - 2a(a+1))(x^2 - (5a^2 + 2a)x + 6a^4 + 3a^3),$$
$$f_{a,a(a+1)}(x) = (x - 2a(a+1)^2)(x^2 - (3a^3 + 4a^2 + 2a)x + 2a^6 + 6a^5 + 7a^4 + 3a^3).$$

Consequently no prime can make every diagnostic cubic irreducible: the monic factors in the diagonal family retain positive degree modulo every prime.

For the second family the quadratic discriminant is $a^2(a^4 + 4a + 4)$. For $a \ge 3$ the radicand is between a^4 and $(a^2+1)^2$, since $2a^2 - 4a - 3 = 2(a-3)^2 + 8(a-3) + 3 > 0$. At $a = 2$ it is 28, between 5^2 and 6^2. Thus the quadratic is irreducible for every $a \ge 2$.

Remark 16.4. At $a = 1, b = 2$ the cubic is $(x-3)(x-6)(x-8)$, although the full graph has irrational antisymmetric eigenvalues $(13 \pm \sqrt{33})/2$. A direct construction of its 10 vertices gives

$$\det(xI - L) = x(x-10)(x-9)^3(x-7)^2(x-4)(x^2 - 13x + 34).$$

Thus a noninteger cubic root is a sufficient condition for graph nonintegrality; the converse cannot be used without the other blocks.

17 Completion of the three-prime classification

The largest quotient root supplies the final obstruction. Its open unit interval completes endpoint one without deciding the quartic's smallest root or enumerating integer exponent points on the endpoint surface.

Theorem 17.1. *Let $2 \le a \le b \le c$ be real, with $b - a \ge 3$ and $c \ge a^2 - a$. Put $s = a + b + c$ and $M = bc + s$. Then*

$$M < \lambda_{\max}(C) < M + 1.$$

For integer exponents in this domain, the ideal intersection graph is Laplacian nonintegral. The real interval has a written proof with no finite exponent base.

Proof. The quotient determinant gives

$$h_C(M) = -a^2bcT(a, b, c),$$

where

$$T = (2b^2 - ab + b - a)c^2 - [(a - 1)b^2 + a^2]c - ab(a + b).$$

The quadratic coefficient of T in c is positive, and

$$T(a, b, b) = 2b[(b - a + 1)b^2 - ab - a^2] \ge 2b^3(b - a - 1) > 0, \qquad \partial_c T(a, b, b) \ge b^3 > 0.$$

The derivative bound follows from $\partial_c T(a, b, b) = b^2(4b-3a+3)-2ab-a^2$ and $a \le b$. Thus $T > 0$ for $c \ge b$. Put $P(a, b, c) = h_C(M + 1)$. For $a \ge 3$, under $a = 3 + m$, $b = a + 3 + u$, $c = a^2 - a + v$, with $m, u, v \ge 0$, the complete identities in Appendix E express T and P as polynomials with respectively 36 and 170 strictly positive coefficients, including constants 1404 and 513220. For $2 \le a \le 3$, write $a = 2 + m$, $b = a + 3 + u$, $c = b + v$, where $0 \le m \le 1$ and $u, v \ge 0$. The same appendix gives $P \ge B(u, v) > 0$, with all 35 coefficients of B displayed. Thus $h_C(M) < 0 < h_C(M + 1)$, giving a root in $(M, M + 1)$.

To identify it as the largest root, use the symmetric similarity S from Theorem 15.2. For $x > c$, the pair-support block of $xI - S$ is positive definite. Its Schur complement is

$$H(x) = \operatorname{diag}(\ell_a, \ell_b, \ell_c) + zz^T, \qquad z = (\sqrt{a}, \sqrt{b}, \sqrt{c})^T, \qquad \ell_t = x - s - \frac{xabc}{t(x - t)}.$$

At $x = M + 1$, both $x - b$ and $x - c$ exceed bc. Hence

$$\ell_b = c(b - a) + 1 - \frac{abc}{x - b} > c(b - a) + 1 - a > 0,$$
$$\ell_c = b(c - a) + 1 - \frac{abc}{x - c} > b(c - a) + 1 - a > 0.$$

Here $c \ge b \ge a + 3$. The b, c principal block of H is positive definite, and $\det(xI - S) = xP > 0$ gives $\det H > 0$. Its remaining scalar Schur complement is therefore positive. Consequently $xI - S$ is positive definite, and every eigenvalue is less than $M+1$. Since $\det(MI-S) = Mh_C(M) < 0$, not all eigenvalues can lie below M, so the largest lies in the stated interval. For integer exponents, the established complement and universal-class lifts transfer its nonintegrality to the ideal intersection graph. □

The first two middle gaps use exponent-root brackets rather than the largest-root interval. The following proof records their signs explicitly so that the classification does not rely on an unstated uniqueness claim.

Lemma 17.2. *Let $a \ge 2$ be real, $d \in \{1, 2\}$ and $b = a+d$. Put $L_1 = 2a^2-a-2$ and $L_2 = 2a^2+a-6$. As a polynomial in c, $h_C(1)$ has exactly one root above b, lying in $(L_d, L_d + 1)$. Consequently, for integer $a \ge 2$, no integer $c > b$ satisfies $h_C(1) = 0$.*

Proof. For fixed a, b, write $Q(c) = h_C(1) = Ac^3 + Bc^2 + Dc + E$, where

$$\begin{aligned}
A &= a^2 + b^2 - 1,\\
B &= a^3 - 4a^2b^2 + 2a^2b - a^2 + 2ab^2 + ab - 3a + b^3 - b^2 - 3b + 3,\\
D &= (a-1)(b-1)(2ab + 3a + 3b - 3),\\
E &= (a-1)(b-1)(a+b-1)(ab + a + b - 1).
\end{aligned}$$

Both A and E are positive. Put $L_1 = 2a^2 - a - 2$ and $L_2 = 2a^2 + a - 6$. Direct expansion gives

$$\begin{aligned}
-Q(L_1) &= 4a(a+1)(a^4 - 4a^2 - 2a + 6),\\
Q(L_1+1) &= 4a(a-1)(a+1)(a^3 - a^2 + 1),\\
-Q(L_2) &= 16a^5 + 72a^4 - 76a^3 - 433a^2 + 69a + 595,\\
Q(L_2+1) &= 4(a+2)(2a^5 - 2a^4 - 17a^3 + 25a^2 + 28a - 42),
\end{aligned}$$

where each pair uses its own $b = a + d$. The complete positive vectors at $a = 2 + m$, including those for $-Q(b)$, are in Appendix E. Thus $Q(0) > 0 > Q(b)$ and $Q(L_d) < 0 < Q(L_d + 1)$. Moreover,

$$L_1 - b = 1 + 6(a-2) + 2(a-2)^2 > 0, \qquad L_2 - b = 2(a-2)(a+2) \geq 0.$$

The second equality occurs only at $a = 2$; the final open interval still lies strictly above b there. The cubic has one root in each of $(-\infty, 0)$, $(0, b)$ and $(L_d, L_d + 1)$. These three distinct roots exhaust its degree. The unique root above b is therefore in $(L_d, L_d + 1)$. For integer a, the endpoints are consecutive integers, proving the nonvanishing conclusion without a finite base. □

Theorem 17.3 (Three-prime nonintegrality)**.** *For distinct primes p, q, r and positive integers a, b, c, the ideal intersection graph of $\mathbb{Z}_{p^a q^b r^c}$ is Laplacian nonintegral. This global conclusion combines written proofs with the established small-minimum and endpoint-two finite certificates.*

Proof. Sort the exponents. Repeated entries are covered by Theorem 8.1; minima at most seven by Corollary 12.4. Suppose $8 \leq a < b < c$ and the spectrum is integral. Theorem 13.2 supplies a positive quotient root in $(0, 3)$, necessarily one or two. Theorem 15.8 excludes the second alternative, so $h_C(1) = 0$.

For $b - a = 1$ or 2, Lemma 17.2 contradicts integer $c > b$. Otherwise $b - a \geq 3$, and Theorem 15.2 gives $b + c \geq 2a^2 - 2a + 1$. Since $c \geq b$, we have $c > a^2 - a$. Theorem 17.1 supplies a noninteger quotient root, again contradicting integrality. These cases exhaust all positive exponent triples.

The minimum-through-seven theorem retains its 27 562-triple base and earlier small-minimum dependencies; the endpoint-two theorem retains its 8 658-triple base. The new largest-root and small-gap steps have written unbounded proofs. No additional exponent scan or integer-point enumeration is used in this completion. □

18 Remaining questions

Theorem 17.3 proves nonintegrality for every positive three-prime exponent vector. Theorem 9.2 settles any higher-prime vector with a unit entry. Thus the remaining part of Q3 has at least four prime factors and every exponent at least two. By Theorem 9.3, it also fails $A \geq (m^2 - 1)B - (m - 1)$ for every choice of distinguished coordinate m, with A, B computed from the other exponents. Corollary 9.4 excludes, in particular, every fixed $m \geq 2$ with all other exponents at least $m^2(t - 1)$. Theorem 9.5 further excludes every four-prime vector with at least two exponents equal to two.

Thus a remaining four-prime vector has at most one such entry. Theorem 9.6 likewise excludes two entries equal to three. The general repeated-pair tensor lower bound does not settle the remaining repeated exponents $a \geq 4$. The endpoint-one equations retain separate arithmetic questions about exponent points, even though none of their integer quotient spectra split.

18.1 Arithmetic of endpoint-one exponent points

The spectral and coefficient reductions below remain useful descriptions of the endpoint-one surface. The new largest-root argument settles integral splitting uniformly; determining all integer exponent points on that surface is a different question and is not required by the classification.

Spectral and divisor windows. Write $h_C(x) = (x-1)F(x)$, $s = a+b+c$, $p = ab+ac+bc$ and $r = abc$. Theorem 15.2 gives four real roots of F strictly above a, and Corollary 15.3 places their smallest in $(a, \min\{c, a+b\})$. Its integer value must divide $N_4 = rs(p+s)$, while all three larger roots exceed $s = a+b+c$, without an equality or sign condition. The sign parameter $E = c+a-b(b+2)$ refines only the smallest-root window:

$$E > 0: \quad b < d < \min\{c, a+b\}, \qquad E < 0: \quad a < d < b.$$

Since the three larger roots exceed $s > b$, the sign of $F(b) = r^2E/(b-1)$ determines the side of b on which the smallest root lies. This gives the displayed refinements and, when $E = 0$, the simple smallest residual root b. The earlier regime proof retains the original interlacing argument. Proposition 15.6 also identifies b off endpoint one on the equality surface. The coefficient interval in Corollary 15.3 further requires $N_4/D < d < N_4/(D - N_4R)$, where D is minus the linear coefficient of F, $U = \min\{c, a+b\}$ and $R = 2/s + 1/(p+s-1-U)$. Both denominator and endpoints are positive.

Coefficient and exponent tests. Write $F(x) = x^4 - \alpha x^3 + \beta x^2 - \delta x + N_4$. Every integer root d satisfies

$$\begin{aligned} &d \mid N_4, && d^2 \mid N_4 - \delta d, \\ &d^3 \mid N_4 - \delta d + \beta d^2, && d^4 \mid N_4 - \delta d + \beta d^2 - \alpha d^3. \end{aligned}$$

For $e \in \{a, b, c\}$, the determinant identity $F(e) = -r^2(e^2+3e-s)/(e-1)$ also gives $d - e \mid F(e)$ when $d \neq e$; if $d = e$, the condition is $F(e) = 0$. These are necessary tests, obtained by reducing $F(d) = 0$, rather than sufficient splitting criteria. The coefficient proof and exponent tests retain the exact formulas and diagnostics. Each candidate still requires the actual root equation and the remaining factors.

An elementary equality problem. For $E = 0$, put $c = b(b+2) - a$ and $q = ac$. Let $G(a, b)$ be the plane septic in Remark 15.5, and define

$$\begin{aligned} A_3 &= q + b^3 + 5b^2 + 8b - 1, \\ B_3 &= qb(b^2+5b+5) + 2b^5 + 12b^4 + 25b^3 + 16b^2 - 8b + 1, \\ N_3 &= qb(b+3)\big[q + b(b^2+3b+3)\big], \\ P_3(x) &= x^3 - A_3x^2 + B_3x - N_3, \\ \Delta &= A_3^2B_3^2 - 4B_3^3 - 4A_3^3N_3 - 27N_3^2 + 18A_3B_3N_3. \end{aligned}$$

The generic identity is $h_C(x) = (x-1)(x-b)P_3(x) + x(x-b)G(a,b)$, so this is precisely the spectral cubic when $G(a,b) = 0$. The equality splitting system would require integers $4 \le a < b$, $d > s = b(b+3)$ and $u \ge 0$ such that

$$G(a,b) = 0, \qquad P_3(d) = 0, \qquad \Delta(a,b) = u^2.$$

Theorem 17.3 proves that this system has no admissible solutions. The following algebraic equivalence still explains the earlier diagnostic tests. The curve equation is essential to the formulation. A square discriminant alone does not imply integral splitting: $x^3 - 3x + 1$ has discriminant 81 and no rational root. With an integer root d, however, $P_3 = (x-d)[x^2 + (d - A_3)x + N_3/d]$. If $P_3'(d) \neq 0$, then

$$\Delta = P_3'(d)^2\big[(A_3 - d)^2 - 4N_3/d\big].$$

A square Δ makes the integer quadratic discriminant a rational square and hence an integer square; its parity gives integer roots. If $P_3'(d) = 0$, the repeated root is d and the third root is $A_3 - 2d$, also an integer. Thus the displayed system is equivalent to integral splitting on the equality curve. At each known point, only divisors $d \mid N_3$ above s need testing. The explicit cubic note and its certificate retain the full 61-term discriminant.

Propositions 14.6 and 14.7 supply further necessary restrictions. The finite-field table first leaves six compatible rows; the normalized discriminant excludes $(1,0)$ and $(4,0)$, leaving

$$(a,b) \bmod 5 \in \{(2,0),(3,0),(0,2),(3,2)\}.$$

In addition, $7 \nmid b$ and $11 \nmid b$. These four rows permit cubic splitting over $\mathbb{F}_5$; they assert neither integer equality points nor an integral spectrum. The complete residue table records all nine curve rows and the three excluded cubic reductions. The two further exclusions use $G = 0$ and the normalized discriminant, rather than a nonsplitting reduction over $\mathbb{F}_5$. At modulo three all six curve rows split, giving no further obstruction. Proposition 14.8 proves that at every actual equality point in the two $b \equiv 0 \pmod 5$ rows, P_3 splits modulo every power of five. The simple positive-branch roots lift by Hensel's lemma, so increasing that modulus cannot exclude those rows. For every fixed $b \in 5\mathbb{Z}_5$, the curve itself has a unique point in each of the residues $a \equiv 2, 3$, with split cubics. Thus neither row can be excluded by a finite five-power curve-and-splitting test. At odd prime divisors of b on the negative branch, the residue test for 13 is also sufficient for local splitting, except at $p = 13$. At $p = 13$, Proposition 14.9 requires $b \equiv 104 \pmod{169}$ on the negative branch for integer splitting; every other 13-divisibility class has odd discriminant valuation. These local compatibility classes no longer leave an open global integral-spectrum case: the largest-root interval excludes all of them at admissible integer exponent points. Further local analysis remains a separate arithmetic question.

Finiteness and effective classification. Remark 15.5 and Siegel's theorem show that the equality curve has finitely many integer points. They provide neither a height bound nor an exhaustive list here. Riemann–Roch spaces for specified divisors do not by themselves enumerate those points. Effective enumeration of equality exponent points remains open. Their integral cubic spectra, and integral quartic spectra away from equality, are now excluded by the global three-prime theorem. The endpoint-one surface itself is not claimed to have no integer points. The genus consequence remains a structural remark; no Jacobian rank bound or covering computation is assumed in the spectral results.

18.2 More prime factors

Which nonsquarefree exponent vectors with more than three prime factors give integral ideal intersection graphs? The present arguments do not supply the full general-exponent Q3 characterization.

The distinction between the repeated-exponent blocks remains useful: the boundary family has an integral antisymmetric block, whereas $(1, 1, 2)$ has an integral symmetric cubic and a nonintegral full spectrum. A general reduction must control all remaining blocks.

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A Exact verification and reproducibility

The infinite conclusions use the written proofs and the finite certificates to which those proofs reduce the problem. Computations use integer or rational arithmetic; no floating-point spectrum or Lean formalization is claimed for these Laplacian results. SymPy 1.14.0 and Python 3.10 or later are used, with source hashes in the saved results.

The project repository contains every checker, exact certificate and reproduction command [17]. The build guide gives the commands by filename. The complete per-checker catalogue is retained in `paper/sections/verification-details.tex`; the optional detailed build includes it without changing the proofs or saved certificates. The principal verification scopes are as follows.

A.1 Graph embeddings and repeated exponents

Direct ideal adjacency validates $LU = UB$, independently of the support construction, on 14 squarefree and $(1, 1, k)$ cases with 44 294 vertex pairs. The full 34-vertex polynomial at $(2, 2, 3)$ is checked. Three boundary graphs with $34, 174, 548$ vertices cover 165 490 pairs and verify both repeated-exponent embeddings and the quotient lifting. The 548-vertex full characteristic polynomial is not computed.

The 90 cubic diagnostics have exact rational intervals and independent Sturm/sign checks. The all-b proofs use the exhaustive divisor reduction, not extrapolation from those diagnostics. All divisors of $24, 108, 320$ for smaller-factor indices $1, 2, 3$ are exhausted; eight positive-b cubics have modular certificates checked by separate integer Horner arithmetic. Higher indices use the uniform sign proof. The separate ten-vertex graph for $(1, 1, 2)$ confirms why the symmetric cubic alone is not an integrality characterization.

A.2 Small minima and complete finite domains

The general six-support quintic agrees with the reflected source quotient and weighted symmetry. Thirty-five requested fully distinct triples and direct ideal graphs with $22, 58$ vertices check the construction. Endpoint identities and positive shifted coefficients support the written unit, minimum-two and uniform-cutoff proofs.

At minimum three, the proved cutoff leaves precisely 325 pairs $4 \le b < c < 30$. All integer quintics, endpoint signs and intervals are checked; the first endpoint has 257 positive and 68 negative values, with no zero. This complete certificate is part of the minimum-three proof. At minima four through seven, all 27 562 ordered triples in the derived region are exhausted. A separate fraction-free Bareiss implementation reconstructs every discriminant by a 9×9 integer Sylvester determinant, checking exact divisions, consecutive-square brackets and domain completeness. Together with the written cutoff, this proves Corollary 12.4; the minimum itself is not extrapolated.

A.3 Inertia and endpoint surfaces

Symbolic block, Schur-complement, determinant and positivity identities support the uniform low root and balanced region. Six secular/Sturm fixtures cover the possible nonsingular sign combinations; nine direct quotient/Sturm fixtures include equal exponents and zero-diagonal boundaries. The four bounded-gap families also have complete 38-residue modular certificates. These finite fixtures check formulas; the infinite conclusions use the written comparison and inertia arguments.

`scripts/verify_singular_schur_inertia.py` checks the zero-diagonal congruence and six fixed quotient controls, including one, two and three zero diagonal entries. It counts roots strictly below the endpoint and checks its multiplicity separately, including an exact algebraic endpoint.

The uniform bound holds for all positive real parameters. `scripts/verify_uniform_low_root_positive.py` checks the added principal-block and compression identities, including the singular minimum-three boundary, and seven fixed exact root counts. The equal-parameter quadratic also shows that the uniform constant three is optimal as its minimum root approaches three from below.

`scripts/verify_proof_structure.py` checks the balanced factorization and the fixed-gap endpoint-one expansion by a direct quotient determinant. It verifies the polynomial rescaling, nonzero implicit-function derivative and successive coefficient equations. The analytic proof in Section 3 gives $O_d(a^{-3})$ for fixed d; the classification retains its exact small-gap and largest-root arguments.

The exact factorization and worst-middle identity are checked by `scripts/verify_balanced_span_threshold.py` for the span bound $a > \rho + 4 + \sqrt{3}(\rho + 2)$, and two fixed newly covered triples. The threshold is sharp for the strict diagonal criterion, without claiming necessity for a quotient root in $(2, 3)$.

For the middle-exponent bound, all 132 cubics over 66 pairs $4 \le a < b \le 15$ are independently reconstructed. Integer trial division exhausts 8 527 positive divisors, and separate Horner evaluation checks all 7 585 above b. No upper bound on c is imposed. This is the complete certificate underlying Theorem 15.1. The linear cutoff uses the general endpoint cubic and five positive coefficient expressions. Both known repeated-exponent endpoint zeros are positive controls. The CRT scope note uses six substitution identities, two positive boundary shifts and two integer seed determinants; its all-moduli claim is the written polynomial-congruence argument.

A.4 Endpoint-two completion

Lemma 15.7 uses a full determinant identity for the middle bound and an 84-term positive residual with two nonnegative square expressions for the maximum bound. The real tail in Theorem 15.8 uses complete 44- and 112-term positive identities. All coefficient vectors are displayed in Appendix D; these written statements require no finite base.

The middle bound has the sharp real cutoff $a > (7 + \sqrt{13})/3$; at equality, its sole endpoint-zero boundary is $b = c = 2a - 2$. The separate `scripts/verify_endpoint_two_middle_cutoff.py` checks the generic cubic identity, all 36 positive coefficients of B_0, D_0, E_0 at $a = 3 + m$, the exact boundary and three fixed rational controls. The integer middle bound still starts at four and the maximum bound at eight.

The global integer conclusion uses exactly 8 658 fully distinct triples at minima eight through thirty-nine in the necessary geometric domain. Every endpoint has independent integer-Horner and 6×6 Bareiss agreement, with no zeros; the CSV and JSON reproduce byte-for-byte. This base requires no small-gap or growing-gap exclusions. Repeated exponents and minima at most seven retain their earlier proofs, including the necessary 27 562-triple minimum-through-seven certificate. The global conclusion is therefore computer-assisted. Seven selected quotient/Horner/Sturm fixtures supplement the identities; the repeated zero $(10, 10, 12)$ is preserved below the tail threshold. The pinned completion checker and its companion middle/maximum checkers reproduce the proof data. The separate `scripts/verify_endpoint_two_coefficients.py` checks the displayed real-tail coefficients and all per-minimum domain counts against a direct determinant and an arithmetic count formula; it does not reevaluate the finite base's endpoint values.

The equality cubic in the arithmetic discussion has generic determinant and independent 5×5 Sylvester-discriminant checks in its pinned checker. The square-discriminant-plus-integer-root formulation also includes the curve equation $G(a, b) = 0$. It supplies a finite splitting test at each known integer point, without a point enumeration or rank computation.

A.5 Largest-root completion and classification dependencies

Theorem 17.1 uses the complete 36- and 170-term positive identities for $a \geq 3$ and the 35-coefficient positive lower bound for $2 \leq a \leq 3$ in Appendix E. The pinned checker reconstructs the generic quotient and the independent Schur determinant, the lower sign identity, all 34 coefficient rows, and the six historical small-gap sign vectors at $a = 8 + m$. Seven specified controls, including $(3, 6, 6)$, $(4, 7, 12)$, $(8, 11, 56)$, $(9, 12, 72)$, $(9, 12, 159)$, $(9, 30, 951)$ and $(9, 150, 160)$, have 14 independent 6×6 Bareiss endpoint determinants. Exact Sturm counts, weighted by squarefree factor multiplicity, give one root in each open unit interval and none above its upper endpoint. The controls supplement written unbounded proofs; there is no new exponent scan or Lean run.

The separate `scripts/verify_largest_root_coefficients.py` reconstructs the determinant directly from the set-disjointness quotient and checks every displayed coefficient in both tables and all six small-gap vectors. The latter now use $a = 2 + m$, proving the real minimum-two brackets; the boundary $L_2 = b$ at $a = 2$ is retained explicitly in the proof. It shares the support model and SymPy backend, but does not import the original characteristic-polynomial helper.

The separate `scripts/verify_largest_root_minimum_two.py` checks the 143-term expansion at $a = 2 + m$, $b = 5 + m + u$, $c = b + v$. On $0 \leq m \leq 1$, its 39 negative terms are bounded using $m^k \leq 1$; discarding the remaining positive terms with $k \geq 1$ gives the displayed 35 positive lower-bound coefficients. The lower-sign quadratic and Schur diagonal bounds are also checked generically. Two fixed rational controls supplement the written real-domain proof; no finite base is added.

Theorem 17.3 retains the established 27 562-triple minimum-through-seven and 8 658-triple endpoint-two bases, with earlier repeated and small-minimum proof dependencies. They were not rerun for this completion. The separate 108-pair minimum-eight certificate is valid but unnecessary for this proof route. The certificate records hashes of the prior proof and result inputs. The new real unit interval and small-gap exponent brackets require no finite exponent base; the combined global theorem is computer-assisted.

A.6 Arithmetic and shifted-root cases

Scaling, all parity patterns and common residues for primes $3, 5, 7$ are checked coefficientwise, with 70 quadratic-residue evaluations and separate integer Horner arithmetic. The all-even scaling contradiction uses the written valuation proof. Complete endpoint tables modulo two and three include zero residues and zero polynomials; 70 independent integer determinant evaluations check the endpoint table.

For the cases in Appendix B, direct and reflected quintics and all exponent permutations agree. Every shifted divisibility and binary-cubic identity is checked coefficientwise for arbitrary shifts. Chosen full quintics and derivatives are independently reconstructed by integer determinants and principal cofactors, including compatible controls and both known endpoint-zero examples. The finite residue and parity tables support the identities; the infinite conclusions use the written root factorizations, rather than a scan of exponents.

Reza's higher-modulus script was run unchanged; the independent verifier checks all 28 032 residue triples with 56 064 Bareiss determinants. The coefficientwise period-eight proof explains why the fixed divisor-32 tables add no classes at larger exponent moduli. His derivative script's 384 representatives are checked by 16 380 principal-minor determinants, including the additional controls. Zero has infinite valuation, and full valuations need not be constant on an exponent residue class. The complete catalogue retains these diagnostic limits, coefficient counts, fixture counts and exact source/certificate mappings.

A.7 Endpoint-one spectral location

Theorem 15.2 and Corollary 15.3 use written proofs without a finite base. The minimum-root checker reconstructs the generic endpoint polynomial, comparison and equality factorization, the historical twelve-term expansion and the determinant identities. The current sum proof uses three positive quadratic coefficients for $a \geq 2$; `scripts/check_endpoint_one_sum_bound.py` checks its generic identities. The residual-spectrum assertions hold for the sharp real cutoff $a > 2 + \sqrt{3}$; the integer divisor corollary retains $a \geq 4$. The separate `scripts/verify_residual_spectrum_cutoff.py` checks the generic anchor identities and the repeated endpoint-one family proving sharpness, with two fixed integer controls below the cutoff and one rational control in the newly admitted range. The pair-sum checker verifies the four-support block, characteristic factor, discriminant and strict sign identity. The established controls $(8, 8, 105)$, $(9, 9, 136)$ and $(20, 20, 741)$ leave $5, 5, 13$ divisors in the smallest-root windows; all 23 integer Horner evaluations are nonzero. Rational factorization agrees, and exact Sturm counts give one quartic root in each window and none on $[0, a]$. These controls already follow from the repeated-exponent theorem. The source-hashed certificates reproduce byte-for-byte; their finite diagnostics supplement the written bounds.

The global gap in Theorem 15.2 has no endpoint or equality hypothesis. Its pinned checker verifies the shifted singleton-block determinant, symmetric total-sum identity, exact quartic coefficients and denominator-positivity identity. Six specified controls $(2, 2, 2)$, $(2, 2, 3)$, $(4, 4, 4)$, $(9, 9, 136)$, $(9, 12, 159)$ and $(9, 30, 951)$ give six direct 6×6 Bareiss checks at the total sum. The first is the sharp boundary where $\lambda_4 = 6$. Exact Sturm counts are summed over squarefree factors with their multiplicities. In particular, $(4, 4, 4)$ has characteristic polynomial $x(x - 20)(x^2 - 32x + 48)^2$: three eigenvalues lie below eight although there are only two distinct roots there. The endpoint-one control $(9, 9, 136)$ has one quartic root below 154 and three above. The reciprocal coefficient window filters the same 23 old divisors at the three endpoint-one controls to 10; 11; and $25, 26$, respectively. All four integer Horner values are nonzero, and exact Sturm counts put one quartic root in each rational interval. These fixed diagnostics supplement the written global and coefficient proofs; the controls were already settled by the repeated-exponent theorem.

Proposition 15.6 retains the written index and simplicity argument, now for real $2 \leq a \leq b$. The direct checker `scripts/verify_equality_spectrum.py` verifies the principal factorization, the positive expression at b, ordering and nonzero coupling; three fixed controls below the former minimum use exact root counts. The equality-index checker verifies the principal factors, weighted symmetry and nonzero coupling. The earlier equality pair-sum checker retains the superseded lower bound $a + b$ and its three off-endpoint controls. The present stronger bound s follows from the global theorem.

A.8 Modulo-five splitting and equality residues

Proposition 14.6 has a written finite-field proof with no exponent base. The pinned checker reconstructs the generic six-support determinant, verifies all six exponent permutations and the equality identity, and checks all 34 residue pairs at the fixed primes three and five. Separate root-multiset enumeration and polynomial factorization agree on every curve row. Independent 720-term integer determinant expansions check the endpoint residues, both global residue polynomials at seven interpolation points each, and five fixed controls, for 83 determinants. The two $E = 0$ controls are off endpoint one; the repeated endpoint-one zero $(9, 9, 136)$ is preserved. The certificate reproduces byte-for-byte. These are finite-field tables and fixed controls, with no exponent-range or integer-point search.

A.9 Normalized equality discriminant

Proposition 14.7 has a written valuation argument with no finite exponent base. The pinned checker verifies both anchor identities, all 14 coefficients of R, the curve difference factor and its unit 2, and the discriminant's generic divisibility by $q - v$. It reruns the direct six-support characteristic identity and the independent 5×5 Sylvester discriminant check. The prime examples are exactly $5, 7, 11$, and the earlier table gives the four remaining modulo-five pairs. The certificate reproduces byte-for-byte. No exponent scan, integer-point enumeration or larger prime search is used.

A.10 Local equality splitting

Proposition 14.8 uses simple-root Hensel lifting and the quadratic discriminant criterion over $\mathbb{Q}_p$. The pinned checker verifies both reduced cubics, the three positive-branch derivative identities, the curve reduction $1 - a^4$ and derivative $-4a^3$, generic polynomial division and the factor discriminant identity modulo $P_3(d) = 0$. It also reruns the direct quotient, independent Sylvester and local anchor identities. The specified residue examples at $3, 5, 7$ illustrate both branches; they are not integer equality points. The certificate reproduces byte-for-byte. The all-prime-power conclusion is a written Hensel argument, without a prime, power or exponent scan or a finite exponent base.

A.11 The exceptional prime thirteen

Proposition 14.9 is a written valuation proof with no finite exponent base. Its pinned checker verifies the refined anchor identities in $\mathbb{Z}[1/2][b]$, every coefficient of the discriminant remainder, the curve difference unit and discriminant difference divisibility. It also reconstructs the generic cubic from the six-support quotient and independently checks its discriminant by a 5×5 Sylvester determinant. The unique linear residue $u \equiv 8 \pmod{13}$ is checked exactly; the certificate reproduces byte-for-byte. There is no prime, power or exponent scan.

A.12 A unit exponent with any prime-factor count

Theorem 9.2 has a written support-counting and variational proof with no finite base. Its exact checker constructs disjoint-support adjacency for nine specified exponent vectors with three, four or five prime factors. It checks connectedness, the weighted zero sum, both energy identities and the strict rational Rayleigh bound. Two small full graphs independently check the trial energy and complement identity. These controls verify the implementation; the unbounded theorem follows from the written proof, without an exponent scan, floating-point eigenvalues or Lean verification.

A.13 Small-exponent Rayleigh criterion

Theorem 9.3 and Corollary 9.4 have written unbounded proofs and require no finite base. The checker verifies the generic energy-minus-norm factorization, a three-collection compression characteristic polynomial and the boundary vector. The compression is not equitable; its eigenvectors are trial functions, not full eigenvectors. Nine specified support controls check the exact aggregated energy matrix, connectedness, zero sums, complement identity and universal shift. They include three equality controls and two controls outside the sufficient criterion. Two small expanded graphs independently verify the trial energies. Boundary residuals confirm that the trial function is not a full eigenvector at one. These controls validate the implementation, with no exponent scan, numerical eigensolver or Lean verification.

A.14 Four prime factors with two exponents equal to two

Theorem 9.5 has a written invariant-subspace and positive-minor proof, reducing the endpoint equation to five symbolic low-b polynomials and one explicit 17-residue cubic irreducibility certificate. It requires no finite exponent base. The checker verifies all four generic principal minors, the endpoint polynomial, the complete nine-term positive tail, all five low-b identities, the two nonsquare discriminants and the exceptional characteristic polynomial and cubic residues. Eight specified support controls independently reconstruct every column of the four-dimensional invariant subspace, the weighted zero sum and the complement/join lift. Two small expanded graphs check the same embedding directly. These controls validate the implementation; no exponent-range scan, floating-point eigenvalues or Lean verification is used.

A.15 Four prime factors with two exponents equal to three

Theorem 9.6 uses a written invariant-space and tensor positivity proof. Two complete nine-term positive expansions reduce the endpoint problem to fifteen symbolic low-b polynomials: five have positive coefficients, eight have nonsquare discriminants, and two have explicit rational noninteger positive roots. This is a complete unbounded reduction with fixed arithmetic, not a finite two-parameter exponent base. The checker verifies the generic repeated-pair tensor identity and normalized two-dimensional characteristic polynomial, both endpoint polynomials, all eighteen positive coefficients and all fifteen low-b rows. Ten specified support controls reconstruct all four embedding columns, weighted zero sums and complement/join identities, including two controls at other repeated exponents. Two small expanded graphs independently check the embedding. The certificate reproduces byte-for-byte, with no numerical spectra, modular search, exponent rectangle scan or Lean verification.

B Mixed-parity case analyses and two-adic lifting

Theorem 14.5 collects the following cases. All residue-role permutations are included. Each row records either a nonintegral class or a necessary condition for an integer spectrum.

Residue roles	Modulus	Conclusion or necessary condition
Six classes listed in the main theorem	8	Nonintegral by the divisor-32 endpoint table below.
$(3,3,2)$	4	Nonintegral: endpoint and derivative requirements are incompatible.
$(3,0,0)$	4	Nonintegral: one is excluded by root distribution and $h_C(2) \equiv 4 \pmod 8$.
$(1,3,2)$	4	$a + c \equiv b(b+2) \pmod{128}$, $b \equiv 11, 15 \pmod{16}$ and the stated binary-cubic parity condition.
$(1,3,2)$, $(3,3,0)$	4	For minimum $m \geq 4$, $h_C(2) = 0$ and maximum $M < R(m)$.

The complete identities, intermediate conditions, examples and diagnostic boundaries are retained below. The variables identify residue roles rather than size order; no additional refinement or search is introduced.

B.1 Endpoint divisibility and odd-root distribution

Lemma B.1. *Let $f \in \mathbb{Z}[x]$ be monic of degree five with $f(x) \equiv x^2(x+1)^3 \pmod 2$. If all its roots are integers, then $32 \mid f(1)$ or $32 \mid f(3)$.*

Proof. The binary factorization forces precisely three odd roots and two even roots, counted with multiplicity. At least two odd roots share a residue $r \in \{1, 3\}$ modulo four. In $f(r)$, their two factors are divisible by four and the third odd-root factor is divisible by two. The even-root factors are odd. Therefore $32 \mid f(r)$, including when $f(r) = 0$. □

Proposition B.2. *Let a, b be the two odd exponents and c the even exponent of a positive three-prime exponent triple. Each row in the following table, including exchange of a, b, implies that its ideal intersection graph is nonintegral.*

a mod 8	b mod 8	c mod 8	$h_C(1)$ mod 32	$h_C(3)$ mod 32
1	*3*	*2*	*8*	*16*
1	*7*	*2*	*24*	*16*
3	*7*	*2*	*16*	*8*
3	*5*	*6*	*8*	*16*
3	*7*	*6*	*16*	*24*
5	*7*	*6*	*24*	*16*

Proof. For either mixed exponent parity, $h_C(x) \equiv x^2(x+1)^3 \pmod 2$. Substituting $(a, b, c) = (8u+r_1, 8v+r_2, 8w+r_3)$ in the general quotient quintic gives the last two columns: every nonconstant coefficient in u, v, w is divisible by 32. Both entries are nonzero modulo 32, contradicting Lemma B.1 if all quotient roots were integers. The resulting noninteger real complement eigenvalue has a noninteger graph lift. □

For example, $(10, 17, 19)$ and $(14, 19, 21)$ meet these criteria with gcd one and unequal 2-adic valuations. There is no minimum or ratio bound in the argument. The weaker divisibilities $4 \mid h_C(2t)$ and $8 \mid h_C(2t+1)$ hold identically for both mixed exponent parities; they alone do not supply the contradiction. The distribution of the three odd roots modulo four is the additional constraint.

Lemma B.3. *Let $f \in \mathbb{Z}[x]$ be monic of degree five with $f(x) \equiv x^2(x+1)^3 \pmod 2$. Suppose $F(y) = f(2y+1)/8$ belongs to $\mathbb{Z}[y]$ and $F(y) \equiv y^3 \pmod 2$. If all roots of f are integers, then its three odd roots are congruent to one modulo four, $64 \mid f(1)$ and $16 \mid f'(1)$.*

Proof. Write the two even roots as η_1, η_2 and the three odd roots as $2r_1+1, 2r_2+1, 2r_3+1$, with multiplicities. Then

$$F(y) = \prod_{j=1}^{2}(2y+1-\eta_j)\prod_{i=1}^{3}(y-r_i).$$

The first two factors reduce to one modulo two. Hence $\prod_i (y-r_i) \equiv y^3 \pmod 2$, which forces every r_i to be even. Each odd-root factor in $f(1)$ is therefore divisible by four, giving $64 \mid f(1)$. Every term in the product-rule expansion of $f'(1)$ retains at least two such factors, giving $16 \mid f'(1)$. The argument includes coincident roots and zero factors. □

Proposition B.4. *Every positive three-prime exponent triple that is a permutation of $(3, 3, 2)$ modulo four has a nonintegral ideal intersection graph.*

Proof. Relabel the odd exponents as $a = 4u + 3$, $b = 4v + 3$ and the even exponent as $c = 4w + 2$, where $u, v, w \geq 0$. Expansion of the general complement quintic gives the coefficientwise identities

$$h_C(2y+1)/8 \in \mathbb{Z}[u, v, w, y], \qquad h_C(2y+1)/8 \equiv y^3 \pmod 2,$$
$$h_C(1) \equiv 16(u+v) \pmod{32}, \qquad h'_C(1) \equiv 8(u+v+1) \pmod{16}.$$

Its binary reduction is $x^2(x+1)^3$. If all five roots were integers, Lemma B.3 would give $64 \mid h_C(1)$, so the endpoint congruence would require $u + v$ even. The derivative divisibility $16 \mid h'_C(1)$ would require $u + v$ odd, a contradiction. Thus there is a noninteger real complement eigenvalue, whose graph lift is noninteger. □

This theorem contains the two rows $(3, 7, 2)$ and $(3, 7, 6)$ of Proposition B.2 and also covers odd exponents sharing the same residue modulo eight. For example, $(10, 19, 27)$ and $(14, 23, 39)$ have odd residues $(3, 3)$ and $(7, 7)$ modulo eight, respectively. Neither theorem requires a low-root interval or an exponent-size bound.

For the fixed necessary divisor 32, increasing the exponent modulus alone gives no further exclusions. On all mixed-parity triples, the values of both $h_C(1)$ and $h_C(3)$ modulo 32 already have period eight in each exponent. This follows coefficientwise by substituting the two parity representatives $(2u+1, 2v, 2w)$ and $(2u+1, 2v+1, 2w)$ and shifting one variable by $4z$; symmetry covers their permutations. Thus the higher-modulus tables are lifts of the six classes in Proposition B.2. The additional exclusions in Proposition B.4 use the distribution of the roots and the derivative, beyond those two endpoint divisibilities.

Proposition B.5. *Let a, b, c be positive exponent residue roles with $a \equiv 1$, $b \equiv 3$ and $c \equiv 2 \pmod 4$. If their ideal intersection graph is Laplacian integral, then*

$$a + c \equiv b(b+2) \pmod{128}, \qquad b \equiv 11 \text{ or } 15 \pmod{16}.$$

Moreover, put $u = (a-1)/4$ and $t = (a + c - b(b+2))/128$. Then

$$\begin{cases} t + u + v \equiv 1 \pmod 2, & b = 16v + 11, \\ t + v \equiv 1 \pmod 2, & b = 16v + 15. \end{cases}$$

If $m = \min\{a, b, c\} \geq 4$ and $M = \max\{a, b, c\}$, then also

$$h_C(2) = 0, \qquad M < 7m - 16 + \frac{40}{m+2}.$$

Every violation under the stated hypotheses proves nonintegrality; all exponent permutations are included.

Proof. Proposition B.10 in Appendix B forces all three hypothetical integer odd quotient roots to be b modulo eight. Thus $512 \mid h_C(b)$ and $64 \mid h'_C(b)$. The exact identity

$$h_C(b) = a^2b^2c^2\big(a + c - b(b+2)\big)$$

has prefactor of 2-adic valuation two, giving the sum condition modulo 128, including when $h_C(b) = 0$. The derivative congruence and the binary cubic calculation in Appendix B give the restriction on b and the two parity conditions.

Since $b \equiv 3 \pmod 4$, one cannot be a quotient root under the integer-spectrum assumption. For $m \geq 4$, the positive root in $(0, 3)$ from Theorem 13.2 must therefore equal two. Proposition C.1 gives the strict linear upper bound. □

The arithmetic conditions imply the earlier sum congruences $a+c \equiv 15 \pmod{16}$ and $a+c+4b \equiv 27 \pmod{32}$. Appendix B gives their staged derivation and the supporting derivative thresholds. The theorem collects these arithmetic conditions together with the endpoint-two restriction. For example, $(29, 27, 754)$ and $(45, 47, 2258)$ have $h_C(b) = 0$ and $64 \mid h'_C(b)$ but fail the final parity test. The compatible triples $(25, 27, 758)$ and $(29, 27, 882)$ instead fail the linear upper bound. Satisfying the arithmetic conditions alone does not establish integrality.

Lemma B.6. *For positive exponent triples that are permutations of* $(3, 0, 0)$*,* $(1, 3, 2)$ *or* $(3, 3, 0)$ *modulo four, an integer complement quotient spectrum would force every odd root to be three modulo four. Consequently one cannot be a quotient root under this assumption.*

Proof. For each displayed residue pattern, write $(a, b, c) = (4u + r_a, 4v + r_b, 4w + r_c)$. Coefficientwise expansion gives

$$h_C(x) \equiv x^2(x+1)^3 \pmod{2},$$
$$F(y) = h_C(2y+1)/8 \in \mathbb{Z}[u, v, w, y], \qquad F(y) \equiv (y+1)^3 \pmod{2}.$$

If every root were an integer, there would be three odd roots $2r_i + 1$ and two even roots η_j, counted with multiplicity. In

$$F(y) = \prod_{j=1}^{2}(2y + 1 - \eta_j) \prod_{i=1}^{3}(y - r_i),$$

the first two factors reduce to one modulo two. Thus every r_i is odd, giving the asserted residue and exclusion of one. Symmetry of the quintic includes every permutation of the exponents. □

Proposition B.7. *Every positive three-prime exponent triple that is a permutation of* $(3, 0, 0)$ *modulo four gives a nonintegral ideal intersection graph, without a size or ratio bound.*

Proof. For residue roles $(a, b, c) = (4u+3, 4v, 4w)$, coefficientwise expansion gives $h_C(2) \equiv 4 \pmod{8}$, so two is not a root. If the minimum exponent is at least four, Theorem 13.2 provides a positive quotient root in $(0, 3)$. Under the integer-spectrum assumption it must be one or two. Lemma B.6 excludes one and the endpoint congruence excludes two, a contradiction. The only possible smaller minimum is three, already covered by Theorem 11.2. A noninteger complement root has a noninteger graph lift. □

The triples $(19, 36, 100)$ and $(35, 52, 120)$ illustrate this one-odd-exponent class beyond the minimum-through-seven and middle-through-fifteen certificates. The result uses the uniform low-root theorem together with the shifted root distribution, without enlarging either finite certificate.

The endpoint distributions above give value divisibilities. Derivatives record additional constraints on the same hypothetical integer roots.

B.2 Derivative divisibility

Lemma B.8. *Let* $f \in \mathbb{Z}[x]$ *be monic of degree five with two even and three odd integer roots. If exactly* k *of the odd roots are one modulo four, then*

$$2^{3+k} \mid f(1), \qquad 2^{\max(2,k+1)} \mid f'(1), \qquad 2^{\max(2,k)} \mid f''(1), \qquad 2 \mid f'''(1).$$

At three, the same statements hold with k *replaced by* $3 - k$*.*

Proof. At one, the odd-root factors supply k factors divisible by four and $3-k$ divisible by two; the even-root factors are odd. Each term of the jth derivative is $j!$ times a product retaining $5-j$ root factors. Removing the odd-root factors with the strongest guaranteed divisibility gives the stated bounds, including the factor two from 2! and 3!. At three, the roles of the two odd residue classes are exchanged. Repeated roots and zero evaluations obey the same divisibilities. □

The derivative thresholds are useful once the odd roots share a residue. We now establish that cluster and compare its first two shifted reductions.

B.3 Odd roots modulo four and eight

Proposition B.9. *Let a, b, c be positive exponents with $a \equiv 1$, $b \equiv 3$ and $c \equiv 2 \pmod 4$. If $a + c \not\equiv 15 \pmod{16}$, their ideal intersection graph is nonintegral. All permutations are included, without a size or ratio bound.*

Proof. Write $(a, b, c) = (4u + 1, 4v + 3, 4w + 2)$. Coefficientwise expansion gives

$$h_C(2y + 1)/8 \in \mathbb{Z}[u, v, w, y], \qquad h_C(2y + 1)/8 \equiv (y + 1)^3 \pmod 2.$$

As in Lemma B.3, integer quotient roots would force all three odd roots to be three modulo four. Consequently $64 \mid h_C(3)$. A further coefficientwise identity is

$$h_C(3)/16 \in \mathbb{Z}[u, v, w], \qquad h_C(3)/16 \equiv (2v + 1)(u + w + 1) \pmod 4.$$

Since $2v + 1$ is invertible modulo four, this forces $u + w + 1 \equiv 0 \pmod 4$, equivalently $a + c \equiv 15 \pmod{16}$. Failure of that condition gives a noninteger complement root and graph lift. □

For example, $(14, 25, 27)$ and $(10, 29, 31)$ pass the earlier divisor-32 test but have $h_C(3) \equiv 32 \pmod{64}$, contradicting the new requirement. The sum class $a+c \equiv 15 \pmod{16}$ remains compatible with this test. Complete second- and third-derivative checks at one and three give no additional exclusions with the root information modulo four used here. In particular $h_C'''(1) \equiv 2 \pmod 4$ for every mixed exponent parity. Full valuations at a chosen exponent representative are not generally constant throughout its residue class: $h_C(3) = -2592$ at $(1, 3, 6)$ but $h_C(3) = 0$ at $(9, 3, 6)$, although the exponent triples agree modulo eight.

Proposition B.10. *Let a, b, c be positive exponents with $a \equiv 1$, $b \equiv 3$ and $c \equiv 2 \pmod 4$. If $a+c+4b \not\equiv 27 \pmod{32}$, their ideal intersection graph is nonintegral. All permutations are included, without a size or ratio bound.*

Proof. Suppose all five complement quotient roots are integers. Proposition B.9 requires $a + c \equiv 15 \pmod{16}$. Write

$$(a, b, c) = (4u + 1, 4v + 3, 16t - 4u - 2).$$

Coefficientwise expansion gives

$$H(z) = h_C(4z + 3)/64 \in \mathbb{Z}[u, v, t, z], \qquad H(z) \equiv z^3 + vz^2 + vz + t + 1 \pmod 2.$$

The three odd roots are already known to be three modulo four. Writing them as $4r_i + 3$ and the two even roots as η_j gives

$$H(z) = \prod_{j=1}^{2}(4z + 3 - \eta_j) \prod_{i=1}^{3}(z - r_i).$$

The first two factors reduce to one modulo two, so the displayed cubic must split into linear factors over $\mathbb{F}_2$. For even v, it splits precisely when t is odd: otherwise it is $(z+1)(z^2+z+1)$. For odd v, it splits precisely when t is even: otherwise it is $z(z^2+z+1)$. The quadratic z^2+z+1 is irreducible. Thus $t+v$ must be odd. Since

$$a+c+4b = 16(t+v)+11,$$

this forces the asserted congruence. Its failure gives a noninteger complement root and graph lift. □

This excludes half the classes left by Proposition B.9. There are 512 role-ordered classes modulo thirty-two; the earlier condition leaves 128, and the new one leaves 64. Thus 64 classes, or 384 after all permutations, are additionally excluded. For example, the ordered triples $(13, 18, 19)$ and $(9, 31, 38)$ satisfy the old sum condition and $64 \mid h_C(3)$, but their residue-role weighted sums are eleven modulo thirty-two. The compatible binary cubics are z^3 and $(z+1)^3$, so hypothetical integer odd roots would all be b modulo eight, also requiring $512 \mid h_C(b)$. The surviving congruence classes do not assert integral spectra.

The preceding cluster makes evaluation at the odd exponent itself particularly restrictive. Its exact value and derivative lead to the next binary cubic.

B.4 Value, derivative and the third shifted cubic

Proof of Proposition B.5, arithmetic conditions. The proof of Proposition B.10 forces the three hypothetical integer odd complement roots to be b modulo eight. The two remaining roots are even. Thus $512 \mid h_C(b)$ and $64 \mid h'_C(b)$, since each first-derivative term retains at least two factors divisible by eight. The exact quintic identity

$$h_C(b) = a^2b^2c^2\big(a+c-b(b+2)\big)$$

has prefactor with 2-adic valuation exactly two. It follows that $128 \mid a+c-b(b+2)$, including when the value is zero.

Write $(a, b, c) = (4u+1, 4w+3, b(b+2)+128t-a)$. Coefficientwise expansion gives

$$h'_C(b) \equiv 16(w^2-w+2) \pmod{64}.$$

For $w \equiv 0, 1, 2, 3 \pmod 4$, the derivative residues are respectively $32, 32, 0, 0$. Its required divisibility forces $w \equiv 2$ or $3 \pmod 4$, which is the stated restriction on b.

For the two remaining residues, put $K(z) = h_C(8z+b)/512$. Coefficientwise identities give $K \in \mathbb{Z}[u, v, t, z]$ and

$$K(z) \equiv \begin{cases} z^3+z^2+(1+v+u^2)z+t, & b = 16v+11, \\ z^3+z^2+(1+v)z+t, & b = 16v+15 \end{cases} \pmod 2.$$

Writing the odd roots as $b+8r_i$ and the even roots as η_j gives

$$K(z) = \prod_{j=1}^{2}(8z+b-\eta_j)\prod_{i=1}^{3}(z-r_i).$$

The even-root factors reduce to one, so the displayed binary cubic must split completely. For $\lambda, t \in \mathbb{F}_2$, the cubic $z^3+z^2+\lambda z+t$ splits precisely when $t=\lambda$: the two possibilities are $z^2(z+1)$ and $(z+1)^3$. Otherwise it is $z(z^2+z+1)$ or the irreducible cubic z^3+z^2+1. Using $u^2 \equiv u \pmod 2$ gives the two stated parity conditions. □

C Endpoint polynomial proofs, certificates and covering scope

The two possible low integer roots have different arithmetic constraints. The endpoint two is confined to a linear region in the minimum exponent; this gives a new nonintegral tail for even exponent triples. The shifted odd-root distributions extend the same tail to two mixed-parity patterns below.

Proposition C.1. *Let $4 \le a \le b \le c$, and put*

$$R(a) = \frac{7a^2 - 2a + 8}{a + 2} = 7a - 16 + \frac{40}{a + 2}.$$

If $h_C(2) = 0$, then $c < R(a)$. In fact, $c \ge R(a)$ implies $h_C(2) > 0$.

Proof. Fix a, c and expand the symmetric endpoint polynomial in b:

$$h_C(2) = A(a,c)b^3 + B(a,c)b^2 + D(a,c)b + E(a,c),$$

where

$$\begin{aligned}
A(a,c) &= ac(a + c - 1) + 2a(a - 1) + 2c(c - 1) - 4,\\
B(a,c) &= c^2\big((a + 2)c - (7a^2 - 2a + 8)\big)\\
&\qquad + (a^3 + 2a^2 - 6a - 4)c + 2(a - 2)(a^2 - 2a - 6),\\
D(a,c) &= (a - 2)(c - 2)\big(a^2c + a^2 + ac^2 + 6ac + 4a + c^2 + 4c - 12\big),\\
E(a,c) &= 2(a - 2)(c - 2)(a + c - 2)(ac + a + c - 2).
\end{aligned}$$

For $a, c \ge 4$, the coefficients A, D, E are strictly positive. Both $a^3 + 2a^2 - 6a - 4$ and $a^2 - 2a - 6$ are positive at $a = 4$ and increasing on $[4, \infty)$. Thus $c \ge R(a)$ also gives $B > 0$, which proves the strict endpoint sign. Conversely, an endpoint zero requires $B < 0$, so $(a + 2)c < 7a^2 - 2a + 8$. □

Proposition C.2. *If $4 \le a \le b \le c$ are all even and $c \ge R(a)$, then $G_{p^a q^b r^c}$ is not Laplacian integral. For minimum exponent at least eight, $c \ge 7a - 12$ suffices.*

Proof. Every entry of C is even, so every integer eigenvalue of C is even by the rational-root theorem applied to $C/2$. In particular, one is not an eigenvalue. Proposition C.1 excludes two, and Theorem 13.2 provides a noninteger root in $(0, 3)$. Equivalently, $h_C(0) < 0 < h_C(2)$ directly yields a root in $(0, 2)$, which cannot equal one. Its graph lift is noninteger. For $a \ge 8$, $40/(a + 2) \le 4$ gives $R(a) \le 7a - 12$. □

The triples $(8, 10, 44)$, $(10, 14, 58)$ and $(40, 42, 266)$ meet this criterion with gcd two. Their largest exponents are below the earlier quadratic cutoff and their spans are outside the earlier balanced region. The already settled repeated triple $(10, 10, 12)$ has $h_C(2) = 0$ and obeys the strict bound; an integer endpoint does not establish integrality.

Proposition C.3. *Let $4 \le m \le d \le M$ be the ordered exponents of a triple that is a permutation of $(1, 3, 2)$ or $(3, 3, 0)$ modulo four. If $M \ge R(m)$, the ideal intersection graph is nonintegral. For $m \ge 8$, the simpler bound $M \ge 7m - 12$ suffices.*

Proof. Under the integer-spectrum assumption, Lemma B.6 excludes one as a quotient root. Theorem 13.2 then forces two to be a root. But Proposition C.1 gives $h_C(2) > 0$ for $M \ge R(m)$, a contradiction. The simpler bound follows from $R(m) \le 7m - 12$ for $m \ge 8$. □

Thus the linear upper bound also restricts these two mixed-parity patterns. For example, the residue-role triples $(25, 27, 758)$ and $(29, 27, 882)$ satisfy the arithmetic conditions of Proposition B.5, but their largest exponents exceed this linear bound. The root-distribution argument therefore settles these compatible controls without a further congruence lift. The triple $(19, 27, 140)$ illustrates the other pattern. The variables m, d, M identify size order, independently of residue roles.

The size bound confines one surface geometrically. The following constant-term conditions give finite candidate sets for either surface when two exponents are fixed.

Proposition C.4. *For fixed $a, b \geq 4$, any positive integer c on an endpoint-zero surface satisfies*

$$h_C(1) = 0 \implies c \mid (a-1)(b-1)(a+b-1)(ab+a+b-1),$$
$$h_C(2) = 0 \implies c \mid 2(a-2)(b-2)(a+b-2)(ab+a+b-2).$$

Proof. Both endpoints are integer cubics in c. Their constant terms are the displayed nonzero products. Reduce each endpoint equation modulo c. □

For a fixed pair this gives finite divisor candidates, which must still satisfy the endpoint equation. Reza's requested diagnostic for middle exponents through fifteen now has a complete independent certificate.

Proposition C.5. *Every positive ordered three-prime exponent triple $a \leq b \leq c$ with $b \leq 15$ gives a nonintegral ideal intersection graph. This conclusion uses the complete finite endpoint certificate below.*

Proof. Repeated exponents are covered by Theorem 8.1, and minimum exponents one, two and three by Theorems 9.1, 10.1 and 11.2, respectively. It remains to consider $4 \leq a < b \leq 15$ and $c > b$.

For each of these 66 pairs, both endpoint polynomials are integer cubics in c with the positive constant terms in Proposition C.4. The complete endpoint certificate records all 132 coefficient lists and exhausts their 8 527 positive divisors using integer trial division. Independent integer Horner evaluation at every one of the 7 585 divisors above b gives no zero. Thus neither endpoint vanishes for any integer $c > b$, with no imposed upper bound on c. Theorem 13.2 supplies a positive root in $(0, 3)$; it can equal neither one nor two, proving nonintegrality. □

The reflected quotient in this diagnostic uses $N = |V(H)| = a + b + c + ab + ac + bc$, not the full graph vertex count. The inclusive requested bound $b \leq 15$ and exact rational linear-factor extraction are cross-checked against the independent divisor certificate. The empty lists settle this pair range; they do not prove either endpoint surface globally empty.

Proposition C.6. *Every positive three-prime exponent triple congruent to $(2, 2, 2)$, or to a permutation of $(1, 2, 2)$, modulo three gives a nonintegral ideal intersection graph.*

Proof. For these two residue patterns, reduction of the general complement quintic gives, respectively,

$$h_C(x) \equiv x^5, \qquad h_C(x) \equiv (x^2+1)(x^3+x^2-x+1) \pmod 3.$$

The corresponding pairs $(h_C(1), h_C(2))$ are $(1, 2)$ and $(1, 1)$ modulo three, so neither integer endpoint is a root. Minimum exponents at most three are already covered. Otherwise Theorem 13.2 supplies a positive root in $(0, 3)$, which cannot be an integer. Its graph lift is noninteger as well. For the second pattern, the irreducible factor $x^2 + 1$ also directly rules out a completely integral quotient spectrum. □

For example, $(8, 17, 22)$ and $(8, 17, 23)$ satisfy the new conditions with gcd one and unequal 2-adic valuations. Complete endpoint tables modulo two and three, checked against independent integer determinants, are given in the residue certificate. Every pair (a, b) modulo either prime has at least one c root on each surface; retaining the full root sets, rather than only root-free pairs, is necessary to detect these exclusions.

Remark C.7. Endpoint tests at a fixed finite set of moduli cannot by themselves prove either surface empty on all fully distinct positive triples. Let M be their least common multiple. Since $(9, 9, 136)$ has $h_C(1) = 0$, the strictly ordered triples $(9 + kM, 9 + (k + 1)M, 136 + (k + 2)M)$ pass every selected endpoint-one test. The known endpoint-two zero $(10, 10, 12)$ similarly gives survivors $(10 + kM, 10 + (k + 1)M, 12 + (k + 2)M)$. For large k their middle exponents exceed fifteen. These are congruence survivors, not asserted integer zeros. These fixed-span tails are excluded by Corollary 13.4. To retain the geometric restrictions, take L divisible by M and eight and any $d \geq 32$ divisible by L. The triples $(9 + d, 9 + 2d, 136 + 4d)$ and $(10 + d, 10 + 2d, 12 + 4d)$ pass the respective endpoint tests, have gcd one and two, and satisfy both upper bounds on the maximum while failing the sufficient inertia inequality. The substitution identities are given in the accompanying scope note. This concerns endpoint-value tests alone; another-root obstruction or further arithmetic information can still exclude these candidates.

Theorem 15.8 completes the endpoint-two alternative, including the all-even triples formerly left inside this linear region. The remaining mixed-parity surface $h_C(1) = 0$ still needs another obstruction.

D Complete endpoint-two positive identities and finite base

This appendix supplies all coefficients used in Lemma 15.7 and Theorem 15.8. A vector $[v_0, \dots, v_k]$ represents $\sum_{\ell=0}^{k} v_\ell m^\ell$. The identities are coefficient identities for the generic quotient, rather than values sampled at exponent triples.

D.1 Maximum-bound residual

For $a = 8 + m$, $b = [a + (2a - 2)t]/(1 + t)$ and $c = 9a/4 - 8 + v$, define P by

$$\begin{aligned} 64(1+t)^3 h_C(2) = {} & 9m^6(1+2t)\big[14(1-t)^2 + 5t + t^2\big] \\ & + 308m^5 t(t-1)^2 + P(m,t,v). \end{aligned}$$

The complete expansion $P = \sum_{i,j=0}^{3} P_{ij}(m) t^i v^j$ is:

i	j	Coefficients in ascending powers of m
0	0	$[3014656, 4968448, 2878080, 766780, 100060, 6027]$
0	1	$[3489792, 4092928, 1603664, 277392, 21796, 632]$
0	2	$[1124352, 751424, 170816, 16144, 544]$
0	3	$[75520, 27392, 3264, 128]$
1	0	$[393216, 3104768, 2984064, 974712, 131516, 6322]$
1	1	$[7569408, 11273216, 4914464, 907536, 74792, 2252]$
1	2	$[4098048, 2845312, 670016, 65568, 2288]$
1	3	$[308736, 115456, 14208, 576]$
2	0	$[19243008, 14490880, 5200064, 995716, 79824, 1]$
2	1	$[6706176, 11799040, 5473584, 1047616, 88300, 2684]$
2	2	$[5053440, 3608768, 872192, 87472, 3120]$
2	3	$[413952, 158720, 20032, 832]$
3	0	$[44646400, 39516672, 14832512, 2952456, 316272, 15946]$
3	1	$[6137856, 7480320, 3084384, 563904, 46760, 1416]$
3	2	$[2217984, 1597824, 391424, 39840, 1440]$
3	3	$[180736, 70656, 9088, 384]$

There are 84 nonzero coefficients, all strictly positive, and the constant is $3\,014\,656$.

D.2 The sign difference at five

With $s = a + b + c$, $p = ab + ac + bc$ and $r = abc$, direct expansion gives

$$D_5 = 4h_C(2) - h_C(5) = 9s^3 + 12s^2p + 6s^2r + 3sp^2 + 9p^2 + 3pr \\ - 279s^2 - 168sp - 87sr + 1683s + 468p + 9r - 2997.$$

For $a = 8 + m, b = 8 + t, c = 8 + u$, write $D_5 = \sum_{i,j=0}^{3} V_{ij}(m)t^i u^j$. Its full coefficient vectors are:

i	j	Coefficients in ascending powers of m
0	0	$[4629843, 996483, 67377, 1353]$
0	1	$[996483, 186366, 10395, 156]$
0	2	$[67377, 10395, 417, 3]$
0	3	$[1353, 156, 3]$
1	0	$[996483, 186366, 10395, 156]$
1	1	$[186366, 29079, 1239, 12]$
1	2	$[10395, 1239, 30]$
1	3	$[156, 12]$
2	0	$[67377, 10395, 417, 3]$
2	1	$[10395, 1239, 30]$
2	2	$[417, 30]$
2	3	$[3]$
3	0	$[1353, 156, 3]$
3	1	$[156, 12]$
3	2	$[3]$
3	3	$[0]$

The zero vector is an absent component. All 44 nonzero coefficients are strictly positive, including constant $4\,629\,843$. This establishes $h_C(5) < 0$ on endpoint two for all exponents at least eight.

D.3 The sign difference at four

The corresponding generic identity is

$$D_4 = h_C(4) - 3h_C(2) = 2r^2 - 2s^2r - 2sp^2 - 2pr - 6s^2p - 4s^3 \\ + 120s^2 + 72sp + 36sr - 624s - 4p^2 - 168p - 4r + 928.$$

For $a = 40{+}m$, $b = a(1{+}2t)/(1{+}t)$ and $c = a(4{+}9u)/(4(1{+}u))$, the polynomial $64(1{+}t)^3(1{+}u)^3 D_4 = \sum_{i,j=0}^{3} U_{ij}(m) t^i u^j$ has the following complete coefficient vectors:

i	j	Coefficients in ascending powers of m
0	0	[611305472, 13438528512, 1669177344, 83080704, 2067840, 25728, 128]
0	1	[225809391616, 101554217216, 10552548352, 490948992, 11793920, 143584, 704]
0	2	[575935266816, 199453648896, 19487672672, 881946672, 20857320, 251384, 1224]
0	3	[218865180672, 94951560192, 9789841664, 453837884, 10879740, 132278, 648]
1	0	[181014296576, 89306490880, 9443545088, 442607616, 10675840, 130304, 640]
1	1	[2226833891328, 644107380480, 59528518784, 2622449728, 61045920, 728192, 3520]
1	2	[5054388092928, 1267215288320, 110831526304, 4741723008, 108398120, 1277272, 6120]
1	3	[2483032498176, 647063198720, 57495972608, 2481039396, 57022790, 674384, 3240]
2	0	[440930932736, 161760308224, 16049227264, 731519232, 17370880, 209920, 1024]
2	1	[4630697975808, 1163490925312, 101856239872, 4360110016, 99710080, 1175200, 5632]
2	2	[10376062353408, 2312188925952, 191477361952, 7938026736, 177794520, 2065240, 9792]
2	3	[5505239310336, 1225715108864, 101452569344, 4204474452, 94149320, 1093460, 5184]
3	0	[193009477632, 77502509056, 7857856000, 361629184, 8634112, 104704, 512]
3	1	[2343969284096, 585405371648, 51112650880, 2184643072, 49911296, 587872, 2816]
3	2	[5508276535296, 1195972016128, 97721281760, 4018207392, 89506456, 1035632, 4896]
3	3	[3069924728832, 652290753536, 52684826880, 2150833004, 47677022, 549714, 2592]

All 112 nonzero coefficients are strictly positive, including constant 611 305 472. On the rectangle furnished by Lemma 15.7, this establishes $h_C(4) > 0$ on endpoint two at minimum at least forty.

D.4 Complete necessary integer base

The base is exactly $8 \le a \le 39$, $a < b < 2a - 2$, $b < c < 9a/4 - 8$. Its per-minimum counts are given below; every row has zero endpoint-two zeros. The largest integer c is $\lfloor (9a - 33)/4 \rfloor$.

a	Triples	a	Triples
8	0	24	210
9	3	25	253
10	6	26	276
11	10	27	300
12	15	28	325
13	28	29	377
14	36	30	405
15	45	31	434
16	55	32	464
17	78	33	525
18	91	34	558
19	105	35	592
20	120	36	627
21	153	37	697
22	171	38	735
23	190	39	774

The total is 8 658, and the smallest absolute endpoint value is 8 208. Every row of the complete CSV records both integer-Horner $h_C(2)$ and independent 6×6 Bareiss $\det(2I - C) = 2h_C(2)$. The certificate records the CSV hash, source hashes, per-minimum counts and direct checks. The repeated control $(10, 10, 12)$ has $h_C(2) = 0$ and $h_C(4) < 0$; it is outside this fully distinct base and below the real-tail threshold. Its nonintegrality follows from the earlier repeated-exponent theorem. No $(4, 5)$ root bracket is asserted for all smaller minima.

E Positive identities for the largest-root completion

Put $s = a + b + c$, $M = bc + s$ and

$$T = (2b^2 - ab + b - a)c^2 - [(a-1)b^2 + a^2]c - ab(a+b), \qquad P = h_C(M+1).$$

For $a = 3 + m$, $b = a + 3 + u$, $c = a^2 - a + v$, each row below gives the ascending m coefficient vector of $u^i v^j$. All rows and all coefficients are displayed: 36 terms for T, 170 for P. Every coefficient is positive. Thus the identities establish the two signs on the unbounded real domain with $a \geq 3$, rather than by extrapolating from integer controls.

The independent Schur reconstruction uses $L_t = (x - s)(x - t) - xabc/t$ and the polynomial identity

$$\det(xI - C) = L_a L_b L_c + a(x-a)L_b L_c + b(x-b)L_a L_c + c(x-c)L_a L_b.$$

This specifies P directly and agrees with the support-derived determinant.

E.1 The lower sign: T

i	j	Ascending m coefficient vector
0	0	[1404, 3042, 2517, 1036, 224, 24, 1]
0	1	[603, 684, 261, 39, 2]
0	2	[57, 15, 1]
1	0	[603, 1188, 875, 305, 50, 3]
1	1	[240, 240, 72, 6]
1	2	[22, 3]
2	0	[57, 103, 67, 19, 2]
2	1	[22, 19, 4]
2	2	[2]

E.2 The upper sign: P

i	j	Ascending m coefficient vector
0	0	[513220, 2023568, 3509410, 3571354, 2381713, 1095075, 354193, 80470, 12554, 1277, 76, 2]
0	1	[461006, 1356418, 1739823, 1279064, 593441, 179723, 35391, 4351, 302, 9]
0	2	[142258, 295089, 256850, 120963, 33056, 5205, 436, 15]
0	3	[18594, 23290, 11201, 2530, 270, 11]
0	4	[882, 399, 60, 3]
1	0	[461006, 1812246, 3118532, 3129572, 2042989, 911507, 283142, 61012, 8889, 828, 44, 1]
1	1	[398473, 1159066, 1455815, 1037238, 460760, 131613, 23978, 2659, 161, 4]
1	2	[118957, 239834, 199807, 88401, 22137, 3087, 219, 6]
1	3	[15059, 17725, 7741, 1509, 131, 4]
1	4	[693, 253, 29, 1]
2	0	[142258, 547789, 915785, 883535, 547176, 227591, 64328, 12165, 1466, 101, 3]
2	1	[118957, 336158, 404388, 271110, 110621, 27997, 4258, 352, 12]
2	2	[34629, 66544, 51528, 20429, 4311, 450, 18]
2	3	[4285, 4597, 1712, 248, 12]
2	4	[193, 49, 3]
3	0	[18594, 69414, 111456, 101999, 58928, 22337, 5555, 872, 78, 3]
3	1	[15059, 40917, 46523, 28811, 10495, 2240, 257, 12]
3	2	[4285, 7749, 5460, 1864, 303, 18]
3	3	[520, 496, 147, 12]
3	4	[23, 3]
4	0	[882, 3171, 4853, 4172, 2218, 750, 158, 19, 1]
4	1	[693, 1797, 1911, 1075, 339, 57, 4]
4	2	[193, 325, 204, 57, 6]
4	3	[23, 19, 4]
4	4	[1]

E.3 The two small-gap exponent brackets

For $b = a + d$, $d = 1, 2$, put $Q(c) = h_C(1)$, $L_1 = 2a^2 - a - 2$ and $L_2 = 2a^2 + a - 6$. The following rows give the complete ascending m vectors at $a = 2 + m$ for $-Q(b)$, $-Q(L_d)$ and $Q(L_d + 1)$. Their positivity proves every sign used in Lemma 17.2.

d	Expression	Ascending m coefficient vector
1	$-Q(b)$	$[64, 480, 848, 664, 264, 52, 4]$
1	$-Q(L_d)$	$[48, 376, 768, 648, 264, 52, 4]$
1	$Q(L_d + 1)$	$[120, 412, 592, 456, 196, 44, 4]$
2	$-Q(b)$	$[57, 1689, 2513, 1516, 456, 68, 4]$
2	$-Q(L_d)$	$[57, 1009, 2119, 1140, 232, 16]$
2	$Q(L_d + 1)$	$[160, 360, 640, 892, 476, 104, 8]$

E.4 The upper sign for minima between two and three

Put $a = 2+m$, $b = 5+m+u$, $c = b+v$, with $0 \le m \le 1$ and $u, v \ge 0$, and expand $P = \sum p_{kij} m^k u^i v^j$. Define $B = \sum_{i,j} (p_{0ij} + \sum_{k \ge 1,\, p_{kij} < 0} p_{kij}) u^i v^j$. The exact identity

$$P - B = \sum_{k \ge 1,\, p_{kij} > 0} p_{kij} m^k u^i v^j + \sum_{k \ge 1,\, p_{kij} < 0} |p_{kij}| (1 - m^k) u^i v^j \ge 0$$

holds on this slab. The following complete ascending u vectors of v^j give B directly from the determinant above. All 35 coefficients are positive, so $P \ge B \ge 259876 > 0$.

j	Ascending u coefficient vector of B
0	$[259876, 480352, 371193, 157834, 41170, 6792, 693, 40, 1]$
1	$[240176, 371193, 236751, 82340, 16980, 2079, 140, 4]$
2	$[77666, 100317, 52638, 14520, 2226, 180, 6]$
3	$[10700, 11468, 4800, 987, 100, 4]$
4	$[540, 468, 147, 20, 1]$